

Ollscoil Teicneolaíochta an Atlantaigh

Atlantic Technological University

Quantifying Model, Label, and Methodological Contributions to Variance in Real-Time Sepsis Onset Prediction

By
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Masters Thesis

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Abstract

Sepsis kills about eleven million people a year, nearly one death in five worldwide, and in intensive care every hour of delayed treatment raises the risk of dying. More than ninety machine-learning systems have been built to raise the alarm earlier. The recent systematic evidence across them finds that their apparent accuracy survives the move to a new hospital while their clinical usefulness does not. This thesis asks how much of that reported performance belongs to the model at all, and how much belongs to four choices the analyst makes before any model is fitted: how sepsis is defined in the record (the *label*, which decides which patients count as septic and from which hour), how the data are split, when the prediction is timed, and which metric is reported. On 65,078 adult ICU stays from MIMIC-IV, with the models applied without retraining to a second database, three pre-registered readings of the Sepsis-3 definition were evaluated across six model specifications on the temporal test set, with selected models re-evaluated under random splitting and externally, and two metrics reported throughout. The premise that the analyst's choices would rival the model did not survive. Across the six specifications compared, the model choice moved discrimination by 0.1499 while the label moved it by 0.0123, under a tenth as much. But the two narrowest readings of the same definition agreed on which patients were septic only at $\kappa = 0.267$, so the label was deciding not how well a cohort could be ranked but which cohort existed to be ranked at all. The label's second consequence is timing: because the onset rule adopted here fires at a documented clinical action, the window before treatment contains no sepsis events by construction rather than by observation. Two post-hoc exploratory labels show what that costs: re-timing the same stays by the field's standard onset rule places 71 events in that window, and an unvalidated label built from physiology alone, which overlaps the models' own inputs and so cannot be read as a sepsis outcome, places 1,362. Every model's discrimination is of the onset hour itself rather than of a future horizon, with predictors binned to that same hour, so these are figures for detection at the recorded onset and not for advance warning. No evaluated operating point comes near the alert-burden benchmark this field sets: the best costs 51.8 false alerts for every true one against a benchmark of 1.4. Improving discrimination alone

therefore does not resolve the membership, timing and alert-burden limitations of a treatment-anchored sepsis label. Clinical actions constitute its membership, and its onset timing decides whether a prediction task defined before those actions can be posed at all.

Keywords: sepsis; MIMIC-IV; discrete-time survival analysis; label definition; variance decomposition; clinical utility; TRIPOD+AI.

Declaration

I declare that this thesis is my own work, that it has not been submitted for any other academic award, and that all sources consulted are acknowledged in the text and listed in the references. The research question, the pre-registered hypotheses, the study design and the interpretation of the findings are my own, as are the decisions recorded as Protocol Amendments 1–10 and the judgements about what the evidence does and does not support.

Use of Generative AI

During this project I applied the PAIR framework to use generative AI responsibly.

- **Problem:** building a complete pipeline for real-time sepsis onset prediction required cohort extraction and labelling under several Sepsis-3 readings, person-hour panel construction, model implementation across six model specifications, an external-validation arm, a clinical-metrics and inference layer, and the documentation of all of it.

- **AI:** I used Claude. Because "AI assistance" covers activities with very different implications for authorship, the uses are separated rather than summarised. **Coding:** discussing implementation choices, and refactoring early prototypes into the twelve-stage structure presented here. **Debugging and code auditing:** adversarial review of the pipeline against its own specification, which is how several of the defects recorded in the amendment log were found. **Thesis prose:** drafting and editing wording in this document. **Literature discovery:** locating and summarising candidate sources, each of which I then read and verified against the claim it is cited for. **Statistical reasoning:** discussing inference choices, among them the multiplicity plan and the interval-inversion reading of H6. In the last three the assistance was to the drafting and the discussion; the claims, the citations and the decisions are mine, and are the subject of the next item.

- **Interaction:** all generated output was critically reviewed, modified and adapted to fit this study's design. The research question, the pre-registered hypotheses, the label definitions, the model choices, the configuration values, the decisions recorded as Protocol Amendments 1–10 and the reported results are my own.
- **Reflection:** this approach improved development speed and documentation quality while keeping the experimental design, the analysis and the final conclusions my own work.

Generative AI was *not* used to generate data, to produce or alter any result, or to decide what the results mean. Every quantity this study estimates (every AUROC, interval, p-value, count, rate and calibration statistic) is written into this document mechanically from the pipeline's own output tables by `thesis_constants.R` (Appendix A.4), and a quantity the pipeline did not produce typesets as a visible marker rather than as a plausible number. That mechanism guarantees the *values*, and only the values. It cannot check that a sentence still describes what the value now shows, that a section's status tag matches the register, or that a claim has not outlived the analysis it was written for; those are human obligations, and the checkers described in Appendix A.4 close only part of the gap. Four classes of number are typed rather than generated, and each is identified where it occurs. Three are not results of this study: figures quoted from prior work, each carrying its citation; thresholds and design constants fixed by the pre-registration before any modelling, such as the equivalence margin and the correction levels; and, in running prose, values rounded for readability from the generated macro that states them exactly in the adjacent table. The fourth is a single deliberate exception, named as such in Appendix A.2: two *superseded* estimates from the fits that Protocol Amendment 8 corrected are typed rather than generated, because they exist to be compared against the current values and a macro would print the current values twice. The analytical claims, the hypothesis verdicts and the limitations are mine, and I take full responsibility for the content of this thesis.

Signed: Abdul Ahad **Date:** 16 October 2026

Abdul Ahad

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The open-source R and Python communities, in particular the maintainers of `data.table`, `arrow`, `nnet`, `xgboost`, `sandwich`, and `pmsampsize`, are acknowledged for the packages that made this pipeline possible.

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A.1 Repository layout. 116

B.1 Stage inputs and outputs. Which result each stage supports is recorded once, in the analysis register of Appendix C, and is not repeated here. Stage 03 is the only stage that reads the full-size source tables, and Stage 12 is the only one that reads nothing but stored predictions, which is what allows the inference layer to be re-run and checked without refitting any model. 121

C.1 Analysis register. [C] confirmatory, [E] exploratory, [D] descriptive. **H5's pre-registered estimand is not evaluable (Section 5.10.7); the slot is retained and the contrast reported in it is a post-hoc substitute, so the [C] tag marks the slot rather than the estimand actually reported. Section numbers refer to `00_preregistration.md`. *§9 pre-registered the equity analysis in scope only: neither the subgroup levels, the reference level, nor the direction of any comparison was fixed in advance, so these analyses are exploratory. †A2 = Protocol Amendment 2 (§§16–17), written after notebooks 02–12 had been run; these analyses are post-hoc and change no pre-registered estimand. ‡A4 = Protocol Amendment 4 (§18), which adds the standard onset timing as Variant F; also post-hoc, and it enlarges family E3, so every BH-adjusted p-value in that family is stricter than it would be without the amendment. §A6 = Protocol Amendment 6 (§20), the subgroup interpretability floor; post-hoc, and it *reduces* family E1, which the inference plan's own denominator rule prohibits on its face; Section 6.7 states the tension and §20.4 of the pre-registration gives the argument. ‖A9 = Protocol Amendment 9 (§23), the full-cohort external arm under Variant D; post-hoc and registered as descriptive, so it carries no p-value and is excluded from H5 on the terms that exclude Variant D from H1. #A10 = Protocol Amendment 10 (§24), which attaches intervals to three quantities otherwise reported as point estimates and tests H6's interval null at its second boundary; also post-hoc, and it changes no verdict. ¶Scored outside the inference stage; no row in `12_analysis_register`. 123

Chapter 1

Introduction

1.1 Context and Motivation

Sepsis is a life-threatening condition caused by a dysregulated host response to infection [1]. It accounts for approximately 11 million deaths annually, nearly one fifth of deaths worldwide [2], and each additional hour before antibiotic administration is associated with higher mortality [3]. An early warning system that detects sepsis before clinical recognition could allow treatment to start sooner.

This potential has motivated a large body of research, but high discrimination has not translated reliably into clinical value. Wang et al. [4] reviewed 91 published real-time sepsis prediction models and found a median internal AUROC of 0.811. Among the models that reported it, the median Utility Score, which rewards timely alerts and penalises late and false ones, fell from $+0.381$ internally to -0.164 externally; a negative score means that alerting does more harm than issuing no alerts at all. In prospective deployment, Wong et al. [5] found encounter-level AUROC between 0.82 and 0.92 for the Epic Sepsis Model v2 across 227,091 encounters at four US health systems, with positive predictive value of 0.13 to 0.26, or 21 to 35 alerts investigated per sepsis case identified. These studies show that discrimination can remain high while external utility is poor and alert burden substantial.

Study design is a further source of variation. Cohen et al. [6] showed on MIMIC-III that varying only the operationalisation of Sepsis-3 changed the identified cohort from 867 to 2,178 patients, and that the label choice affected performance as much as the model choice. Kamran et al. [7] showed that published models partly learn to predict when clinicians start treatment rather than when patients deteriorate.

These findings shift the focus from model ranking to study design. This thesis asks how much of the variation in reported performance is attributable to the

choice of model, and how much to four design choices: how sepsis is defined, when prediction is made relative to treatment, how the data are split, and which metric is reported. Each is treated as a separate source of variation and its contribution is estimated. The pre-registered term *variance decomposition*, which the title reflects, refers here to a comparison of one-factor-at-a-time performance ranges, not to a formal partition of statistical variance (Section 5.10).

1.2 Limitations of the Conventional Approach

The conventional approach to sepsis prediction selects one Sepsis-3 operationalisation, one train/test split and one headline metric, usually AUROC, and then compares models. Most of the 91 models reviewed by Wang et al. [4] follow this pattern, which has four limitations:

1. **The sepsis label is not fixed.** Different operationalisations of Sepsis-3 applied to the same database produce substantially different cohorts and performance estimates [6], which suggests the label choice may change results as much as the model choice.

2. **The data split can produce optimistic estimates.** Guo et al. [8] found that sepsis prediction degraded more than any of the other clinical tasks they evaluated when tested on a later admission era. Random splits mix admission eras and may therefore produce optimistic estimates when clinical practice or documentation changes over time.

3. **AUROC does not reflect clinical usefulness.** A model can retain a high AUROC while its clinical utility falls below that of issuing no alerts [4].

4. **The label incorporates clinician behaviour.** Because Sepsis-3 is operationalised through culture and antibiotic orders, models may learn when clinicians suspect sepsis and begin treatment rather than when the patient's condition worsens [7]. The time at which prediction is evaluated relative to treatment therefore matters.

Comparing models while holding these design choices fixed leaves their contributions unmeasured. This study varies them explicitly and estimates each contribution with an uncertainty interval, to compare their effects on reported performance with the effect of model choice.

1.3 Research Question and Pre-Registered Hypotheses

Research question. *In real-time sepsis prediction on MIMIC-IV, how much of the variation in reported performance is attributable to (a) the choice of model, and (b) the researcher's decisions: how sepsis is defined, how the data are split, when prediction is made, and how performance is measured?*

All discrimination estimates concern detection of the recorded onset hour among at-risk person-hours: the models estimate the hazard of onset in the current hour, so they measure onset detection rather than prediction over a future horizon. Because predictors and outcomes are binned hourly, the analysis does not establish that measurements preceded onset within the scored hour (Section 3.7; Limitation 3). Here, "real-time" denotes hourly dynamic estimation without retrospective information, not demonstrated advance warning.

Six hypotheses were pre-registered before any of the modelling reported here:

H1 (Label dominance) Changing the sepsis definition affects AUROC at least as much as changing the model.

H2 (Coefficient instability) At least one clinical feature changes its effect direction, or changes in magnitude by a factor of two or more, across label variants.

H3 (Treatment leakage) Restricting evaluation to before treatment begins reduces AUROC by at least 0.03.

H4 (Split optimism) Time-based testing gives at least 0.02 lower AUROC than random testing.

H5 (Metric divergence) The Utility Score declines proportionally more than AUROC when moving from internal to external validation.

H6 (Model-class null) The primary discrete-time hazard model is not outperformed by gradient-boosted trees by more than 0.02 AUROC, under a fixed label and temporal split.

H1 and H6 represent competing explanations of performance variation: confirming H1 would make the label the main source, whereas rejecting H6 would place more weight on the model.

Two hypotheses could not be evaluated in their pre-registered form. Under the pre-registered onset rule the pre-treatment window of H3 contains no onsets by construction (Section 5.10.5), and the internal Utility Score is too close to zero for

the proportional decline of H5 to be defined, so an internal-to-external AUROC contrast takes its place (Section 5.10.7). Section 5.10.8 sets out how the interval null of H6 is tested.

The formal inference and multiplicity plan, unlike the hypotheses, was specified after the analyses had been run (Protocol Amendment 1). The six hypotheses form one confirmatory family controlled by Holm–Bonferroni correction. Other analyses are classified as confirmatory, exploratory or descriptive; Section 3.14 defines these classifications and the corresponding multiplicity procedures.

1.4 Contributions

To the best of our knowledge, and within the limits of the structured narrative search described in Section 2.1, the study makes seven contributions.

1. **C1 (Label variation).** Three Sepsis-3 operationalisations are evaluated with a discrete-time competing-risks model on MIMIC-IV. Label-related and model-related AUROC spreads are estimated on the same test rows, each with an uncertainty interval, so that the two can be compared directly. Label sensitivity itself is established [6], [9]; the contribution is this combination.

2. **C2 (Onset timing and treatment anchoring).** The study proves that a conjunction-based Sepsis-3 onset rule, used here and widely in the MIMIC-IV literature, leaves the pre-treatment evaluation window structurally empty. The standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ timing rule is then evaluated empirically on the same cohort in a post-hoc analysis. C2 concerns the logical consequence of the onset-timing rule.

3. **C3 (Utility ceiling).** The adapted Utility Score is evaluated across the full threshold grid rather than at a single conventional cut-off, to test whether the low utility reported by Wang et al. [4] persists at every operating point in a controlled single-study comparison. Under the primary label no model exceeds 0.031, on a scale whose upper bound is 1.

4. **C4 (Coefficient stability).** Cross-label coefficient stability is evaluated with pre-registered sign and magnitude criteria, testing Lauritsen et al.'s [10] framing-instability finding on MIMIC-IV, followed by an exploratory comparison of each flagged change against its cluster-robust standard error.

5. **C5 (Equity by label).** Subgroup discrimination and subgroup variation in who is labelled septic are measured on the same strata of an onset-prediction cohort under common false-discovery control. Wang et al. [11] reported both

quantities, but for mortality prediction among patients already identified as septic.

6. **C6 (Reproducible pipeline).** A reproducible, seed-fixed pipeline built around a pre-registered study protocol, with every protocol amendment disclosed. Reporting was guided by TRIPOD+AI [12], with risk-of-bias considerations informed by PROBAST+AI [13].

7. **C7 (Anchor attribution).** In a post-hoc analysis, the Sepsis-3 conjunction is separated into its treatment-anchor and organ-dysfunction components, each labelled separately, to distinguish definitional label variation from empirical differences in membership, onset timing and predictability. Where C2 concerns the logic of the onset rule, C7 attributes the label effect empirically to its two components.

1.5 Thesis Structure

Chapter 2 reviews the evidence motivating the study. Chapter 3 describes the data, labels, models, validation design and statistical analysis, and Chapter 4 presents the pipeline architecture. Chapter 5 reports the results, organised around the six hypotheses; Chapter 6 interprets them and discusses their limitations; and Chapter 7 summarises the conclusions and future work. Appendix A gives the source-code repository and the procedure for reproducing every reported number, and Appendix B documents the pipeline stages.

Chapter 2

Literature Review

This chapter reviews four bodies of evidence that motivate the study design: the distinction between statistical and machine-learning models (Section 2.2), variability in sepsis labels and their dependence on clinical action (Sections 2.3 and 2.4), internal, temporal and external validation and the gap between discrimination and clinical utility (Sections 2.5 and 2.6), and methodological choices in dynamic survival modelling (Section 2.7). It then considers rule-based comparators, subgroup validity and interpretability (Sections 2.8–2.10), and Section 2.11 summarises the evidence behind each design decision taken in Chapter 3.

2.1 Search Strategy

The literature search targeted published work on real-time sepsis prediction, with particular attention to label sensitivity, validation methodology, evaluation metrics and subgroup validity. Table 2.1 documents the protocol. A companion review, *Statistical and Machine-Learning Approaches to Real-Time Sepsis Onset Prediction on MIMIC-IV*, covers the full corpus of 50 sources, including several of contextual relevance only; this chapter engages with the sources that bear on the study's design decisions.

Type of review. This is a **structured narrative review, not a PRISMA-conformant systematic review**. Searches followed a documented protocol and every included source was verified against a primary record, but a complete deduplicated search yield was not screened by title and abstract, and no second reviewer screened independently. The search covers in depth the methodological literature that constrains this study, but not the MIMIC-IV sepsis corpus exhaustively. The novelty claims in Section 1.4 are bounded accordingly, and a systematic review could identify a prior study combining the analyses in contribution C1 or C5.

Throughout this thesis, a statement that no study was identified refers to this search.

Table 2.1: Literature search strategy, supporting the novelty claims for contributions C1 and C5 (Section 1.4).

Element	**Detail**
Databases searched	PubMed/MEDLINE, Embase, Scopus, Web of Science, IEEE Xplore, ACM Digital Library, Google Scholar, Cochrane Library, ScienceDirect, SpringerLink, Wiley Online Library, Nature Portfolio, JAMA Network, BMJ Journals, arXiv/medRxiv
Searches executed	35 structured searches across the domains listed under "Search terms"
Date range	Primary sepsis-prediction literature: January 2015 – July 2026. Methodological and definitional foundations (e.g. pooled logistic regression, decision-curve analysis) were searched without a date limit, and the earliest source retained dates from 1990
Search terms	`(sepsis OR ''sepsis prediction'' OR ''sepsis detection'') AND (MIMIC OR ''critical care'' OR ''intensive care'')` combined with each of: `''label definition''`, `''label sensitivity''`, `''onset definition''`, `''utility score''`, `''clinical utility''`, `''fairness'' OR ''equity'' OR ''disparity'' OR ''bias''`, `''subgroup analysis''`, `''competing risks'' OR ''discrete-time survival''`
Inclusion criteria	(1) Peer-reviewed journal article, conference paper, or preprint on a recognised server (arXiv, medRxiv); (2) Develops or evaluates a real-time (hourly or sub-hourly) sepsis prediction model; (3) Uses EHR data from an ICU population; (4) Reports at least one quantitative performance metric
Exclusion criteria	(1) Non-English language; (2) Editorials, commentaries, or letters without original results; (3) Models for sepsis severity staging or mortality only (not onset prediction); (4) Studies on paediatric or neonatal populations exclusively

Element	Detail
Verification standard	Every included source was checked against a publisher record, PubMed or PMC record, or institutional repository entry, with journal, volume, year and DOI confirmed. Every numerical claim attributed to a source was located in that source's own abstract, results text or publisher-hosted record, never inferred from a secondary citation
Total included	50 sources in the companion review: 46 identified and verified through the searches above, and 4 retained from an earlier base review of 19 sources. 45 are cited in this thesis
Excluded at verification	4 items encountered in secondary citation lists could not be verified to the standard above and were excluded; 1 further item is retained in the companion review with an explicit note of incomplete metadata and is not cited in this thesis

Figure 2.1 summarises the identification, verification and inclusion process, following the reporting logic of PRISMA 2020 [14].

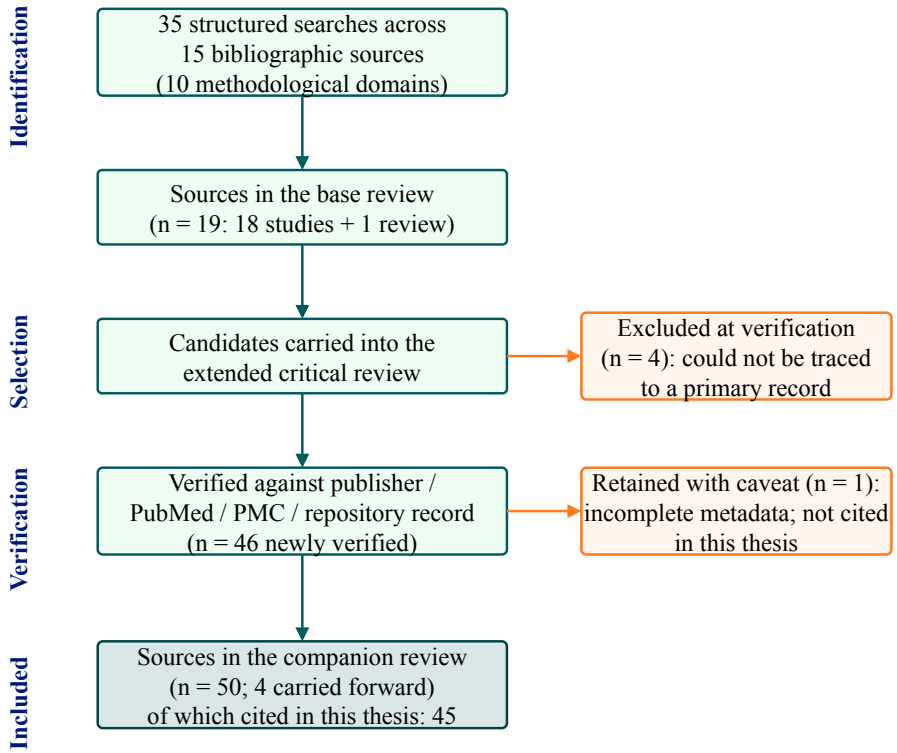


Figure 2.1: Search, verification and inclusion flow, following the reporting logic of PRISMA 2020 [14]. It is not a PRISMA flow diagram: no title-and-abstract screening of a full deduplicated yield was performed, and no second reviewer screened.

Novelty claims supported by the search. Two contributions rest on search-based evidence. Each concerns a *combination* of analyses with clear individual precedent; the search identified no study performing them together.

1. **C1 (Label variation under a dynamic survival model).** Label sensitivity is well established [6], [9], [15], and the last two studies are direct precedents for this one. Cohen et al. include a survival model among the families they compare, so the survival setting is not new either. The search identified no study applying three or more Sepsis-3 operationalisations to a *discrete-time competing-risks* model on MIMIC-IV and reporting the label spread with an interval alongside the model-class spread on the same test rows.

2. **C5 (Performance disparity and label sensitivity across the same subgroups).** Wang, Li, Naidech and Luo [11] report both subgroup performance disparity and subgroup variation in who is labelled septic, but for

mortality prediction rather than sepsis onset (Section 2.9), so their study does not provide a direct precedent for an onset model's subgroup performance. Studies of label sensitivity for onset prediction (Cohen et al., Dutta et al.) do not stratify by demographic subgroup. The contribution lies in evaluating subgroup discrimination and subgroup label sensitivity on the same strata of one onset-prediction cohort under a common multiplicity procedure.

Scope of the clinical-utility evidence. Wang et al. [4] reviewed 91 real-time sepsis prediction models. Only a minority reported the PhysioNet Utility Score [16] or an equivalent net-benefit metric; the Utility Score rewards early true alerts, penalises late alerts and false alarms, and is normalised so that issuing no alerts scores zero. The majority reported only AUROC, AUPRC or both. Among the models with a utility or net-benefit metric, the median external Utility Score was -0.164, indicating net harm. For models reporting only AUROC, clinical utility could not be assessed, and no claim about their utility is made here.

Key concepts. **MIMIC-IV** (Medical Information Mart for Intensive Care, version IV) is a freely available, de-identified database of approximately 74,000 ICU admissions at Beth Israel Deaconess Medical Center, Boston, covering 2008–2019 [17]. **Sepsis-3** is the current international consensus definition of sepsis as life-threatening organ dysfunction caused by a dysregulated host response to infection, operationalised through suspected infection (an antibiotic order plus a culture draw) together with a rise of at least 2 points in the Sequential Organ Failure Assessment (SOFA) score [1].

Review Questions

Four questions structure the review:

RQ1. How much does the choice of sepsis label definition affect model performance, and is this effect comparable in magnitude to the choice of model architecture? (Sections 2.2 and 2.3)

RQ2. Which methodological choices (validation strategy, metric selection, treatment-time leakage) most affect reported performance in the sepsis prediction literature? (Sections 2.4 to 2.6)

RQ3. What statistical framework is appropriate for modelling hourly ICU event data when proportional-hazards and independent-censoring assumptions may not hold? (Section 2.7)

RQ4. Does the existing equity literature on sepsis prediction account for label-level variation across demographic subgroups, or only model-level variation? (Section 2.9)

2.2 Reconsidering the Statistical versus Machine-Learning Distinction

Reviews of this literature conventionally divide it into a statistical tradition and a machine-learning tradition, and the distinction is often associated with interpretability on one side and predictive performance on the other. For this study, a distinction based on temporal modelling assumptions is more informative.

Several boundary cases illustrate this. The Artificial Intelligence Sepsis Expert [18], commonly classified as machine learning, uses a modified Weibull–Cox proportional-hazards formulation. TREWScore [19], a Cox model and a commonly used statistical benchmark for this task, is often described in the machine-learning literature as an early machine-learning early-warning system, and its commercial successor is marketed as one [20]. Cohen et al. [6] compare tree-based, deep-learning and survival models within a single frame, and TRIPOD+AI [12] covers "regression or machine learning" under one checklist, having concluded that the reporting failures it addresses are common to both.

A more informative distinction separates models by what they assume about time.

- **Static models** take a snapshot (baseline, day one, or the worst value in the first 24 hours) and issue one prediction. Much of the MIMIC-IV statistical corpus sits here: a candidate biomarker or composite index is regressed on mortality through a largely standard pipeline of variable selection, Cox and logistic regression, splines, a nomogram, and ROC, calibration and decision curves. These are *prognostic* instruments and cannot perform real-time onset detection.

- **Dynamic models** consume a stream and update a risk estimate as it arrives. TREWScore, the Artificial Intelligence Sepsis Expert, the PhysioNet Challenge entries [16], Moor et al.'s deep temporal model [21], and landmark and joint models all sit here. Whether the update rule is a partial likelihood, a recurrent network or a boosted hazard is secondary for this distinction.

This framing also corrects the claim that statistical methods have neglected onset prediction. Henry et al. [19] fitted a Cox model for septic-shock onset on MIMIC-II; Fagerström et al. [22] fitted one on MIMIC-III with Henry et al.'s

inputs and target as the baseline for an LSTM; and Cohen et al.'s survival arm does so on MIMIC-III again. The search identified no MIMIC-IV study combining dynamic onset modelling, explicit assumption testing, label-sensitivity analysis and temporal validation.

The evidence on model class is similarly scoped. Christodoulou et al. [23], reviewing 71 studies, found no performance advantage for machine learning over logistic regression in clinical prediction, and observed potential bias in the validation procedures of 68% of them; their evidence concerns binary outcomes on tabular data. For temporal early warning, Fagerström et al. report their deep model detecting patients up to 20 hours earlier than the Cox model at comparable sensitivity and specificity. Equivalence is therefore supported for static risk stratification but remains contested for temporal early warning. H6 tests this narrower comparison with the data, the label, the validation split and the evaluation held fixed.

The literature has two further implications for design. Moor et al.'s systematic review [24] identifies low comparability and reproducibility as the field's principal defect, and highlights case–control matching as a specific risk: when a control patient's reference time is chosen near discharge while a case is aligned near onset, the two are easily separated and apparent performance is inflated. Fleuren et al. [25], in a meta-analysis of diagnostic test accuracy, attribute the between-study heterogeneity that prevented pooling to methodological rather than clinical variation, and Cohen et al.'s results suggest that part of it is label variation.

Accordingly, this study uses all at-risk person-hours and performs no case–control matching.

2.3 Variability in Sepsis Label Definitions

MIMIC sepsis cohorts vary materially with the identification algorithm used. Johnson et al. [15] applied five retrospective sepsis identification algorithms, including the Sepsis-3 clinical criteria, to 11,791 qualifying ICU admissions drawn from 23,620, and found agreement across them that the authors characterise as only acceptable (Cronbach's $\alpha = 0.40$–0.62). MIMIC-based studies use such algorithms to construct their cohorts, so studies of sepsis in the same database may not be studying the same patients.

Cohen et al. [6] quantified the resulting effect on predictive performance, and their design is the direct precedent for this thesis. Working in MIMIC-III, they held the data fixed and varied only the reading of the Sepsis-3 onset rule across three defensible interpretations of the same consensus definition. The number of identified septic admissions ranged from 867 to 2,178, a factor of 2.5. Across representative tree-based, deep-learning and survival models, the best method gained

1–5% AUROC over its rivals when the definition was held fixed, whereas varying the definition with the method held fixed produced swings of 0–6%. The label operationalisation thus moved performance by a similar amount to the choice of modelling paradigm.

Dutta et al. [9] found the same sensitivity outside MIMIC. A locally trained gradient-boosted model generating predictions every 15 minutes across nine acute-care hospitals was evaluated prospectively on 198,494 encounters against three electronically computable definitions: Sepsis-3, the SEP-1 management bundle and the CDC Adult Sepsis Event. Performance varied by definition: expected calibration error was 0.05 for Sepsis-3, 0.06 for the Adult Sepsis Event and 0.07 for SEP-1, and decision-curve analysis showed higher standardised net benefit under Sepsis-3 and SEP-1. Dutta et al. therefore extend Cohen et al.'s retrospective MIMIC-III finding to the prospective deployment of a commercial model.

Neither study separates the part of the label effect that follows from the definitions from the part found in the data, and neither attaches an interval to the comparison between label-related and model-related variation. Cohen et al.'s "0–6% versus 1–5%" sets two ranges of point estimates side by side without uncertainty, so it cannot establish whether the two sources are of similar size. The present study extends this comparison by estimating uncertainty around both label-related and model-related performance spreads.

Figure 2.2 illustrates the mechanism: the same patient population, subjected to different but defensible Sepsis-3 interpretations, produces overlapping but distinct cohorts.

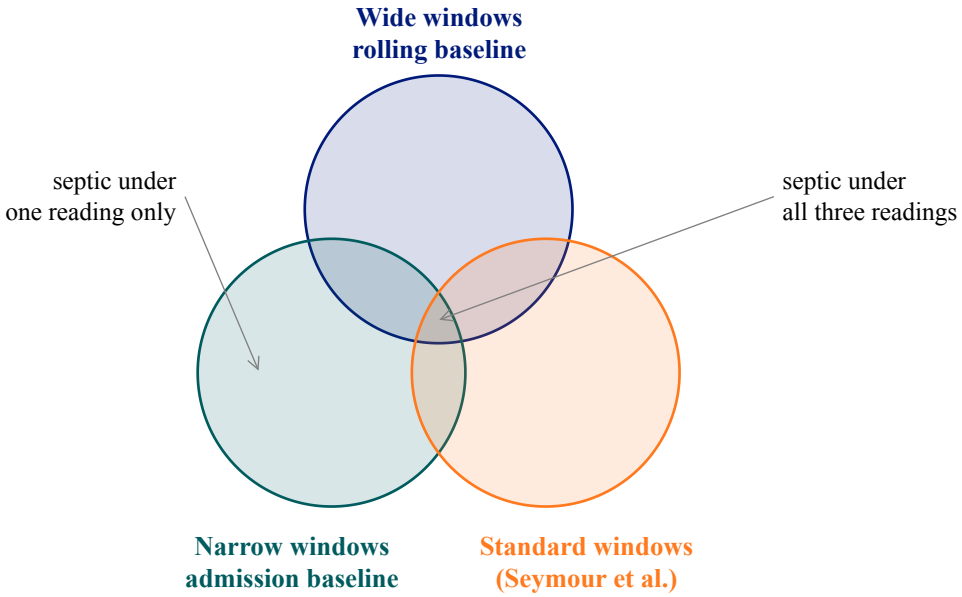


Figure 2.2: Schematic of label variation. Three defensible Sepsis-3 interpretations applied to the same ICU population produce overlapping but distinct cohorts: only the central region is septic under all three readings, and each crescent holds patients counted by one reading alone. The corresponding MIMIC-IV measurement is reported in Chapter 5, where the two narrowest readings agree at $\kappa = 0.267$.

2.4 Clinical Actions in Sepsis Label Construction

Sepsis-3 defines sepsis as organ dysfunction in a patient with suspected infection [1], and the operational criteria derived by Seymour et al. [26] represent "suspected infection" through two clinician actions: drawing a culture and administering an antibiotic. The operational label therefore depends partly on the timing of clinician actions.

Kamran et al. [7] examined this issue directly. Clinicians frequently recognise and begin treating sepsis before the formal criteria are met. A model trained to predict the moment the criteria are met is therefore partly trained to predict *when the clinician will act*, and the physiological record available at prediction time is itself shaped by the developing suspicion: tests are ordered, monitoring escalates, and observations are recorded that would not otherwise exist. Re-evaluating the Epic Sepsis Model on 77,582 hospitalisations, the authors report an AUROC of 0.62 when predictions issued after clinical recognition are retained, falling to 0.47, below chance, once they are excluded.

For a hazard model, a coefficient estimated on the interval up to Sepsis-3 onset

therefore partly reflects clinician behaviour. Lactate measurement, for example, may be prompted by clinical concern, creating dependence between the predictor, clinical suspicion and the treatment-anchored outcome. Model diagnostics such as proportional-hazards testing may not reveal this problem, because it concerns outcome construction rather than statistical fit. Addressing it requires an evaluation target that can precede the relevant clinical actions.

Lauritsen et al. [10] extend the problem beyond the label to the surrounding design. They formalise *framing* (the observation window, the prediction window, the window shift, and the event that triggers a prediction), enumerate eight possible framing structures, and evaluate four on the same data. Framing affects both discrimination and calibration and can produce opposing interpretations of the same physiological variable: in their example, the apparent relationship of SpO_2 to sepsis risk reverses direction with the framing. If framing can reverse the sign of a coefficient, the coefficient reflects the study design as well as the patient. Sign reversals are particularly consequential when coefficients are presented as clinically interpretable effects, and H2 tests whether such instability occurs on MIMIC-IV.

Treatment-anchored evaluation, as Kamran et al. apply it, restricts *when* performance is measured but leaves the treatment anchor inside the label: a Sepsis-3 case is still a case only because a culture and an antibiotic were ordered. SEP-1 and the CDC Adult Sepsis Event are also defined on antibiotic days, so a comparison confined to these definitions cannot separate definitional from empirical label variation. This motivates evaluating an additional label that does not use treatment timestamps (Section 3.6).

2.5 Internal, Temporal and External Validation

Table 2.2 summarises the published external-validation record for real-time sepsis prediction.

Study	Model	Transition	Degradation
Nemati et al. [18]	Weibull–Cox (AISE)	Emory → MIMIC-III; AUROC 0.83–0.85	Reported indistinguishable
Reyna et al. [16]	853 Challenge entries	Two seen hospital systems → one unseen; rank correlation $\rho = 0.949$ between seen systems	$\rho = -0.033$ and 0.013 to the unseen system
Moor et al. [21]	Deep temporal	Four international ICU databases, 136,478 admissions; internal AUC 0.846	External AUC **0.761** (−0.085); 0.807 with 10% fine-tuning
Wang et al. [4]	91 models (meta-level)	Internal → external, full-window	AUROC 0.811 → 0.783 (n.s.); **Utility** 0.381 → −0.164
Wong et al. [27]	Epic Sepsis Model v1	Proprietary → Michigan Medicine, 38,455 hospitalisations	AUROC **0.63**; PPV 12%
Ostermayer et al. [28]	Epic Sepsis Model v1	Proprietary → two county emergency departments	Sensitivity **14.7%**; PPV 7.6%
Wong et al. [5]	Epic Sepsis Model v2	Prospective, four US health systems, 227,091 encounters	AUROC 0.82–0.92 but **PPV 0.13–0.26**; 21–35 alerts per case
Guo et al. [8]	Feed-forward network	MIMIC-IV, within database, across admission eras	Sepsis **most degraded** of four tasks

Table 2.2: The published external-validation record for real-time sepsis prediction. Nemati et al.'s internal–external equivalence is the exception; their external set shares an era, a country and a care model with the development set, which is consistent with an external penalty that grows with the distance of the shift.

Across the studies summarised in Table 2.2, three patterns are relevant to this study.

One multi-database study reported an external AUROC reduction of 0.085. Moor et al. report 0.846 → 0.761 across four databases from the United

States, the Netherlands and Switzerland, consistent in direction, though not in significance, with Wang et al.'s meta-level 0.811 → 0.783. It is used here as a reference estimate for planning, not as an established typical value. H5 asks whether the Utility Score declines proportionally more than AUROC, so 0.085 bounds only the AUROC part of that comparison.

Model rankings may not transfer across health systems. Reyna et al. report that team Utility Scores were strongly rank-correlated between the two hospital systems on which participants trained ($\rho = 0.949$) but essentially uncorrelated with performance on the third, unseen system ($\rho = -0.033$ and 0.013). A published ranking therefore indicates little about which model would perform best at a different hospital. Every comparator in this study is refitted and rescored within the same pipeline, and published figures are used only to bound expectations.

The split design is not neutral for sepsis. Guo et al. [8] partitioned MIMIC-IV into four admission eras (2008–2010, 2011–2013, 2014–2016, 2017–2019) and evaluated four prediction tasks under temporal shift: mortality, long length of stay, sepsis and invasive ventilation. Sepsis was the most affected. In that study, the evaluated domain-generalisation and unsupervised domain-adaptation methods did not consistently outperform empirical risk minimisation (AUROC differences −0.003 to 0.050). A random train/test split on MIMIC-IV may therefore give an optimistic estimate for sepsis, more strongly than for the other ICU outcomes evaluated. The primary validation uses a temporal split, and a random split is retained to quantify any optimism (H4).

Dutta et al. [9] did not detect temporal drift over an 18-month prospective period within one health system, whereas Guo et al. examined shifts across a decade. The duration and context of the shift therefore matter, and "temporal validation" is not a single practice.

2.6 Discrimination versus Clinical Utility

A direct comparison is provided by Wang et al. [4]. Across 91 models, the median AUROC moved from 0.811 internally to 0.783 externally, a difference the authors report as not statistically significant, while the median Utility Score moved from +0.381 to −0.164. For the same transition, the two metrics disagree about whether performance changed.

The mechanism is well understood. AUROC is a ranking statistic, invariant to monotone transformation of the score, so it cannot detect miscalibration, and under low prevalence it is dominated by the large population of easily classified

true negatives. Across sites, the *location* of the score distribution relative to the alerting threshold tends to change more than the *ordering* of patients, and that location determines the false-alarm rate and hence the utility. Van Calster et al. [29] identify calibration as a central weakness of predictive analytics for this reason: a model that drives a threshold-based action depends on its calibration whatever its AUROC.

Deployment results show the same pattern. The Epic Sepsis Model v2 combined encounter-level AUROC of 0.82 to 0.92 with positive predictive value of 0.13 to 0.26 [5] (Table 2.2), and Ostermayer et al. [28], validating version 1 independently in two county emergency departments, report sensitivity of 14.7% and PPV of 7.6%. Version 1 was a logistic regression and version 2 a gradient-boosted ensemble, but version 2 also moved from global to local training, and Moor et al. show that local fine-tuning alone recovers roughly half of the external penalty ($0.761 \rightarrow 0.807$). Model class and localisation therefore changed together, and the comparison cannot isolate either effect.

Two benchmarks inform the evaluation plan. Moor et al. report their model raising 1.4 false alerts per true alert while detecting 80% of septic patients a median of 3.7 hours (95% CI 3.0–4.3) before onset. This provides a reference for alert burden and a realistic lead time for a modern system, in contrast to the 28 hours of the original TREWScore headline [19]. Seymour et al. [3], across 49,331 patients at 149 New York hospitals, estimate an odds ratio for in-hospital mortality of 1.04 per additional hour before antibiotic administration. The clinical value of a lead time of a few hours therefore depends on the alert being acted on, on the patient not otherwise being recognised in time, and on the cost of the false alerts.

These findings indicate that evaluation should include calibration, AUPRC, utility or net benefit, and alert burden in addition to AUROC, and these measures are central to this study's evaluation. The adapted Utility Score is the primary clinical metric, while AUROC is the metric on which the pre-registered comparative hypotheses are stated. Because the utility metric is threshold-dependent, it is reported across a swept grid of operating points (Section 5.4).

2.7 Methodological Challenges in Survival Modelling

Dynamic ICU prediction raises several methodological issues for conventional survival models.

1. **Proportional hazards may fail in sepsis cohorts.** Kasal et al. [30] compared a standard Cox model with Gray's flexible survival model in a

large severe-sepsis cohort and found hazards for death after sepsis frequently non-proportional. The assumption is often untested in MIMIC work.

2. **Independent censoring may also fail.** Resche-Rigon et al. [31] argue that discharge alive is not a non-informative censoring event in ICU cohorts: patients discharged alive are systematically unlike those who remain, so the assumption underlying Kaplan–Meier estimation is violated. Fine–Gray sub-distribution hazard modelling [32] is one common approach to competing risks.

3. **Competing-risks methods are themselves contested.** Schoenfeld [33] argues that survival methods, explicitly including competing-risks methods, are inappropriate for ICU outcome studies, and that studies seeking prognostic covariates should use logistic regression instead.

4. **Landmarking does not restore proportional hazards.** Landmarking, which refits the model at each prediction time on the subset still at risk, provides one practical approach to updating predictions over time [34]. It does not in general satisfy the proportional-hazards assumption; it is a different model with its own approximation error.

5. **Joint models are rigorous but computationally costly.** Modelling longitudinal biomarker trajectories and the event process through shared latent structure outperforms landmarking when the longitudinal sub-model is correctly specified [35], but the approach is computationally expensive, requires a common baseline time, and scales poorly to the heterogeneous, irregularly sampled, high-dimensional streams of an ICU record.

This study adopts a **discrete-time hazard model**, fitted as a pooled logistic regression over person-hour records with time-varying covariates and a flexible baseline hazard. Pooled logistic regression approximates time-dependent Cox regression when intervals are short and interval-specific event probabilities are low [36], and Ngwa et al. [37] confirmed the close agreement by simulation. The hourly intervals and low observed event rates place this application within the setting in which the approximation is expected to be close.

The formulation addresses the issues above as follows. It does not assume proportional hazards (item 1). A multinomial cause-specific formulation over the person-hour models discharge alive and death without sepsis as distinct competing outcomes [31], [38] (item 2). It retains a logistic-regression likelihood while modelling time explicitly (item 3). Its output is an hourly risk score, the object the Challenge utility metric consumes, so it is directly comparable with the machine-learning literature. Heyard et al. [38], [39] supply development and validation

methodology for discrete time-to-event data with competing risks, and Chapter 3 gives the full specification.

A conventional Cox model is retained as a benchmark to quantify the effect of the proportional-hazards and censoring assumptions, and its Schoenfeld test is reported.

2.8 Choice of Rule-Based Comparator

Most MIMIC-IV sepsis models benchmark against SIRS, qSOFA and NEWS. Two results constrain how that benchmark should be read.

The Surviving Sepsis Campaign guidelines [40] issue a strong recommendation *against* using qSOFA as a sole screening tool, citing poor sensitivity, so superiority over qSOFA is a weak claim. The National Early Warning Score 2 (NEWS2), which aggregates six physiological parameters (respiratory rate, oxygen saturation, systolic blood pressure, pulse rate, level of consciousness and temperature) into a bedside deterioration score, is therefore used as the principal rule-based comparator, with qSOFA and SIRS reported for completeness.

Mellhammar et al. [41] provide a direct precedent. They derived a score using LASSO on prospectively collected emergency-department data and validated it against NEWS2 and RETTS; the derived score did not outperform NEWS2 (AUC 0.73 versus 0.69, $p = 0.32$), and the authors concluded that a statistical approach did not yield better sepsis risk stratification than NEWS2. The aim here is therefore not to derive a deployable score but to compare model classes against a clinically used benchmark.

2.9 Equity and Subgroup Validity

Wang, Li, Naidech and Luo [11] was the only directly relevant source on subgroup validity in a MIMIC sepsis cohort identified in the search. The study analysed 11,791 MIMIC-III critical-care patients, of whom 5,783 met Sepsis-3 criteria, and reports two findings. First, six competing sepsis identification criteria selected systematically different sub-populations with respect to race, marital status, insurance type and language, so classification as septic in MIMIC is not demographically neutral. Second, **mortality** prediction among patients already identified as septic degraded significantly when a model trained on the full population was applied to Asian, Hispanic and Spanish-speaking patients, with pairwise testing detecting significant discrepancies between Asian and White patients and between English- and Spanish-speaking patients.

Because the second finding concerns mortality rather than onset, it does not

provide a prior estimate for an onset model's subgroup discrimination, and the two findings concern different targets and cannot be compared directly. They nevertheless show that the label is differentially sensitive by subgroup and that model performance can differ across overlapping subgroups. These findings support treating subgroup analysis as part of model validity assessment, and connect it to the label problem of Section 2.3; Chapter 3 describes how both quantities are measured here.

A methodological caution from the same source governs how the results in Section 5.9 are read. MIMIC's demographic fields are administrative records, not measured population attributes. The race field mixes broad categories with more granular ones drawn from the same population, and includes values that record the absence of an answer rather than an answer. Subgroup contrasts computed on such a field compare administrative strata and are interpreted in those terms.

2.10 Interpretability and Label Uncertainty

The main advantage of a coefficient-based hazard model in this setting is not necessarily higher performance but transparent assumptions and interpretable coefficients, and Christodoulou et al. [23] suggest that little discrimination is sacrificed for them. Cohen et al. [6] and Lauritsen et al. [10] qualify that advantage: if the identified cohort varies by a factor of 2.5 with the onset rule, and framing can reverse the sign of an estimated association, a coefficient reported without a label-sensitivity analysis reflects both biology and the recording process.

Coefficient interpretation therefore depends on a stable and clearly defined outcome. Stability across plausible label definitions strengthens the case for interpreting a coefficient, whereas sign reversal indicates sensitivity to outcome construction. The search identified no such analysis for a survival model on MIMIC-IV.

The problems reviewed in this chapter (label variation, the label's dependence on clinical action, split design and metric choice) have each been demonstrated separately, by different groups using different databases and models. The evidence therefore motivates estimating the effects of label, prediction timing, split, metric and model choices within a common dataset and pipeline, which is the research question of Section 1.3.

2.11 Summary: Design Decisions and Their Evidence

Table 2.3 maps each major design decision to the evidence supporting it.

Table 2.3: Design decisions and supporting evidence.

Decision	**Reason**	**Source**
Three label variants	Label- and model-related AUROC ranges of similar magnitude were reported	Cohen et al. [6]
Label as a studied factor	Label sensitivity was replicated prospectively at scale	Cohen; Dutta et al. [9]
Discrete-time model	Pooled logistic regression approximates time-dependent Cox regression at short intervals, without a proportional-hazards assumption	D'Agostino [36]; Ngwa [37]; Schoenfeld [33]
Competing risks	Discharge alive is informative rather than non-informative censoring	Resche-Rigon [31]; Heyard [38]
Cox benchmark	Proportional hazards may fail in sepsis cohorts	Kasal et al. [30]
No case–control matching	Control alignment can inflate apparent performance	Moor et al. [24]
Treatment-anchored evaluation	Models may learn clinician suspicion rather than deterioration	Kamran et al. [7]
Anchor-independent label	Sepsis-3, SEP-1 and CDC ASE all depend on treatment timing	Kamran [7]; Dutta [9]
Temporal split primary	Sepsis was the most affected of four MIMIC-IV tasks under temporal shift	Guo et al. [8]
Comparators refitted in-pipeline	Published rankings may not transfer across hospital systems	Reyna et al. [16]
Utility Score primary	AUROC may change little despite a substantial decline in utility	Wang et al. [4]
Utility swept, not fixed	A threshold-dependent metric at one cut-off can be dominated by calibration and threshold selection	Reyna [16]; Wong [5]
Calibration co-primary	Threshold-based decisions depend on calibration whatever the AUROC	Van Calster et al. [29]
Alert burden reported	A multi-database deep model reported 1.4 false alerts per true alert	Moor et al. [21]

Decision	Reason	Source
NEWS2 as principal benchmark	qSOFA is not recommended as a sole screening tool, and a LASSO-derived score did not outperform NEWS2	Evans [40]; Mellhammar [41]
GBT comparator	Model-class equivalence is supported for tabular prediction but contested for temporal early warning	Christodoulou [23]; Fagerström [22]
Subgroup label sensitivity	Label composition varies demographically, and no onset-model subgroup discrimination study was identified	Wang et al. [11]
TRIPOD+AI / PROBAST+AI	TRIPOD+AI is reporting guidance; PROBAST+AI is a tool for assessing model quality, risk of bias and applicability. Both cover regression and machine-learning models	Collins [12]; Moons [13]

Chapter 3

Research Methodology

3.1 Research Design

This is a retrospective, observational prediction-modelling study on de-identified critical-care data. The original protocol (§§1–12 of `00_preregistration.md`) specified the research question, the six hypotheses, the label variants, model classes, validation splits, metrics and decision thresholds before any of the modelling reported here. The uncertainty and multiplicity procedures were added after the initial analyses as Protocol Amendment 1 and are described separately (Section 3.14.4). Reporting was guided by TRIPOD+AI [12], with risk-of-bias considerations informed by PROBAST+AI [13]. The design treats the label definition, the timing of prediction relative to treatment, the validation split and the evaluation metric as study variables alongside the model. Implementation and dependency details are given in Chapter 4 and Appendix A.4.

The design decisions were informed by the structured narrative search described in Section 2.1. The search was not a systematic review, and novelty claims are interpreted within the limits stated there and in Section 6.7.

3.2 Data Sources

3.2.1 Development Data: MIMIC-IV v3.1

The development dataset is MIMIC-IV v3.1 [17], [42], [43], a de-identified critical-care database from Beth Israel Deaconess Medical Center (BIDMC), Boston, accessed through PhysioNet credentialing. Eight source tables were used (Table 3.1).

Table	Size (MB)
admissions	19.9
patients	2.8
icustays	3.3
chartevents	3,502.6
labevents	2,593.0
prescriptions	606.7
microbiologyevents	117.2
inputevents	401.3

Table 3.1: MIMIC-IV v3.1 source tables used in this study, with compressed file sizes.

3.2.2 External Validation Data: eICU-CRD v2.0

External validation uses eICU-CRD v2.0 [44], a multicentre database from more than 200 US ICUs, recorded in different electronic health record systems and covering a broader range of hospitals than the single-centre MIMIC-IV. The frozen MIMIC-IV models are applied to harmonised features without retraining.

Transporting the label as well as the model

Sepsis-3 onset is defined on documentation (an antibiotic order, a microbiology culture) whose recording conventions differ between databases more than physiology does, so the label has to be reconstructed in eICU-CRD rather than transported with the model. Three properties of eICU-CRD govern its construction.

Drugs are recorded across two tables. eICU-CRD has no item dictionary; drugs are free-text names split between `medication` (orders) and `infusionDrug` (continuous infusions), and vasopressors and much of the intravenous antibiotic exposure appear only in the latter. Antibiotic exposure is the union of both tables, and vasopressor exposure is taken from `infusionDrug` so that the SOFA cardiovascular component is not fixed at its no-vasopressor value.

Microbiology availability limits the Sepsis-3 label. The suspicion anchor requires a culture, and `microLab` covers 2,824 of 181,589 unit stays (1.55%). Culture availability is necessary but not sufficient for assigning the label: both limbs are documented for 232 stays (0.128%), and fewer remain once each variant's windows and organ-dysfunction criterion are applied (Section 5.5.1). The primary external analysis is therefore restricted to the microbiology-covered sub-cohort, in which an onset is observable, and its event rate is compared with MIMIC-IV's on that basis.

The organ-dysfunction criterion is a rise, not a level. Onset is gated on a SOFA increase of at least two points over a baseline, evaluated inside each variant's own window and against its own baseline (admission for Variants A and B, rolling 24-hour for Variant C), as in the development cohort. An absolute SOFA threshold would be a different criterion, and because missing components are scored at assumed-normal values it would reject stays for incomplete documentation rather than intact organ function, biasing towards the null in a database with sparser laboratory coverage than MIMIC-IV.

Each label variant retains its own antibiotic–culture windows and SOFA baseline in eICU-CRD, so the external label sets differ from one another as they do internally. Two further external analyses are reported separately and never pooled with the primary one. An **antibiotic-only** analysis relaxes the culture requirement across the full cohort to measure how much of the event count that requirement removes; its target is not Sepsis-3. Under Protocol Amendment 9 (post-hoc), the frozen Variant D models (Section 3.6) are applied to the full cohort, which Variant D's culture-free onset rule permits. eICU-CRD's periodic vital-sign table carries no Glasgow Coma Scale, so the external SOFA rise for Variant D rests on five components rather than MIMIC-IV's six (Section 5.5.3).

3.3 Ethical Considerations

This study uses two de-identified critical-care databases distributed through PhysioNet. Neither contains protected health information as defined under HIPAA, and no re-identification was attempted.

MIMIC-IV v3.1 is distributed by the MIT Laboratory for Computational Physiology via PhysioNet [17]. Access requires completion of the Collaborative Institutional Training Initiative (CITI) "Data or Specimens Only Research" course and signing a PhysioNet Credentialed Data Use Agreement (DUA). The DUA prohibits re-identification and restricts data sharing to credentialed collaborators. The author completed CITI training and executed the DUA before any data access; credentials are on file with PhysioNet. **eICU-CRD v2.0** [44] is distributed under the same PhysioNet credentialing framework and DUA terms.

Both databases are distributed as de-identified research datasets under the institutional approvals obtained by their custodians and PhysioNet's credentialing requirements. This study is a secondary analysis only, with no patient contact, intervention or attempt at re-identification, and the terms of the applicable data use agreements were followed throughout.

3.4 Cohort Construction

The base cohort is built by applying three inclusion criteria uniformly across all label variants:

- Age $\geq$ 18 at ICU admission.
- First ICU stay per patient only (to prevent within-patient correlation).
- ICU length of stay $\geq$ 4 hours (to permit hourly feature construction and at least one prediction).

The resulting cohort contains 65,078 **stays** in MIMIC-IV. Figure 3.1 shows the patient flow through each exclusion step.

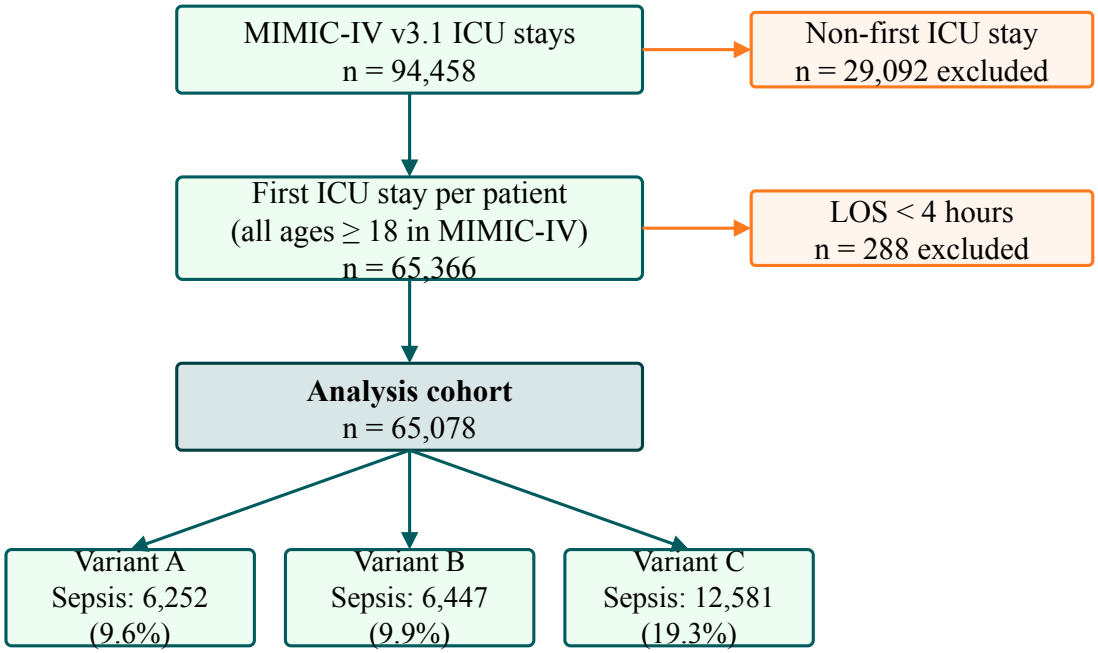


Figure 3.1: CONSORT-style patient flow diagram. All 65,078 stays enter all three label variants; only the sepsis outcome differs. Every count is generated from `02_cohort_flow`, written by script `02_cohort_labels.Rmd`. The age criterion is applied first and excludes 0 stays, which is why it appears as an annotation rather than as an exclusion box: MIMIC-IV's ICU module records adult admissions only. The filter is nevertheless run and its count generated. Sepsis counts are **cohort-wide**, at any hour of the stay; the subset falling inside the 72-hour modelling horizon, which is what the models are trained and evaluated on, is reported in the results chapter.

3.5 Outcome Definition: Three Sepsis-3 Label Variants

Following Cohen et al. [6] and Johnson et al. [15], three operationalisations of Sepsis-3 were evaluated. All three are plausible operationalisations of the same consensus definition [1]; they differ in their antibiotic–culture windows and SOFA baselines (Table 3.2).

ID	Name	ABX→cx	cx→ABX	SOFA−	SOFA+	Baseline
A	Narrow	24 h	24 h	24 h	12 h	Admission
B	Seymour-Std	72 h	24 h	48 h	24 h	Admission
C	Liberal	72 h	72 h	48 h	24 h	Rolling 24 h

Table 3.2: Three pre-registered Sepsis-3 label variants. ABX = antibiotic; cx = culture; SOFA−/SOFA+ = hours before/after suspicion time in which a SOFA rise of ≥ 2 must be observed. Baseline = SOFA reference value.

In all three variants, a patient has sepsis if there is suspected infection (antibiotic order paired with a culture within the specified windows) and the SOFA score increases by at least 2 points. Surveillance cultures (e.g. MRSA screens) are excluded.

Variant B (Seymour-Standard) is the **primary analysis**. Variants A and C are pre-specified sensitivity analyses for the variance decomposition.

3.6 Beyond Sepsis-3: An Anchor-Independent Label

Variants A–C share a culture-plus-antibiotic suspicion anchor, which records a clinician's decision to treat. Any change to the antibiotic–culture windows therefore moves the onsets, so part of the label-variant spread follows from the definition itself (**definitional** variation). Whether a label that does not use treatment timing selects different patients, at different hours, with different predictability is an **empirical** question, and it cannot be assessed within a family of labels that share the anchor.

Protocol Amendment 2 (post-hoc) therefore splits the Sepsis-3 conjunction of *suspected infection* and *acute organ dysfunction* and labels each component separately (Table 3.3): Variant E retains the suspicion anchor without requiring organ dysfunction, and Variant D retains organ dysfunction without using treatment timestamps.

ID	Name	Suspicion anchor	Organ dysf.	Uses treatment timing?
B	Seymour-Std	required	required	membership and timing
E	Anchor-Only	required	not required	yes
D	Deterioration	not required	required	**no**
F	Sepsis-3 Early	required	required	membership only

Table 3.3: The two limbs of the Sepsis-3 conjunction labelled separately (Variants E and D, Protocol Amendment 2), and the standard onset timing applied to the intact conjunction (Variant F, Protocol Amendment 4). Variant F has Variant B's membership and differs only in when the onset is placed.

Variant E (Anchor-Only) takes onset to be the suspicion time itself, using Variant B's windows, with no organ-dysfunction requirement, so its label records clinician behaviour only. The share of Variant B's above-chance discrimination that Variant E reproduces,

$$\phi_{\text{anchor}} = \frac{\text{AUROC}_E - 0.5}{\text{AUROC}_B - 0.5}, \tag{3.1}$$

is reported as the **anchor-attributable fraction**.

Variant D (Deterioration) takes onset to be the first hour after hour 1 at which a *physiology-only* SOFA score rises at least 2 points above the patient's admission baseline (the mean of each component over hours 0–3). The vasopressor and catecholamine terms are withheld, so the cardiovascular component is scored from mean arterial pressure alone, and no antibiotic order, culture draw or drug administration enters the definition. Variants D and E are fitted with the same model specification, feature set and temporal split as Variant B, so only the label changes.

Here, "anchor-independent" means that antibiotic, culture and vasopressor timestamps do not determine label membership or onset. It does not imply independence from clinical care or documentation: Variant D is built from measurements that exist because they were ordered, on patients whose physiology already reflects any treatment received.

The PhysioNet/CinC 2019 operationalisation [16] was considered as the anchor-independent label and rejected for that purpose: it sets onset to $\min(t_{\text{susp}}, t_{\text{SOFA}})$ with the same culture-plus-antibiotic trigger, so membership remains anchored to treatment. SEP-1 and the CDC Adult Sepsis Event [9] are excluded for the same reason, since both are defined on antibiotic days.

Variants D and E are used only for exploratory attribution. Variant E represents the treatment anchor rather than sepsis, so its AUROC is not a sepsis result.

Variant D is an unvalidated deterioration label derived from recorded physiology, and five of its six SOFA components are also model inputs, so its discrimination is partly self-fulfilling and its *level* is not comparable with Variant B's. The principal comparisons are therefore onset agreement (Cohen's κ) and median onset shift (Section 5.6).

3.6.1 Standard Onset Timing (Variant F, Protocol Amendment 4)

Variants A–C assign onset $= t_{\text{susp}}$ whenever the SOFA criterion is satisfied anywhere in the evaluation window, so the SOFA criterion gates membership and contributes no timing information. Under this rule no onset can precede the first antibiotic or culture, and the pre-treatment window of H3 is empty by construction (Proposition 5.1 in Chapter 5). Seymour et al. [26] and the PhysioNet/CinC 2019 Challenge [16] instead assign

$$t_{\text{onset}} = \min\big(t_{\text{susp}},\, t_{\text{SOFA}}\big), \tag{3.2}$$

with t_{SOFA} the first hour in the evaluation window at which the ≥ 2-point rise over the admission baseline is present.

Protocol Amendment 4 introduced Variant F after this limitation of the pre-registered onset rule was identified and before Variant F was fitted. **Variant F (Sepsis-3 Early-Onset)** retains Variant B's suspicion windows, SOFA threshold, evaluation window and baseline, and assigns onset by Equation (3.2). The two variants therefore label the same stays cohort-wide, and Variant F's onset is never later than Variant B's.

Here t_{SOFA} uses the *full* SOFA score, vasopressor components included, as Seymour et al. specify, so an onset placed by the cardiovascular component is free of the antibiotic–culture anchor but not of treatment in general; Variant D remains the anchor-independent comparison. Because stay-level membership is identical, the informative agreement statistic for Variant F is the onset *shift* rather than the overlap. Earlier onset assignment can also move events into the 72-hour modelling horizon when Variant B's suspicion time falls later, so within the horizon Variant F may contain more events than Variant B despite identical cohort-wide membership (Section 5.6.3).

Variants D, E and F are post-hoc and are excluded from the pre-registered variant set: they take no part in H1 or the variance decomposition, which are computed from unchanged inputs. Their contrasts against Variant B form exploratory family E3, corrected by Benjamini–Hochberg.

3.7 Prediction Framing

Following Lauritsen et al. [10] and Reyna et al. [16], the prediction problem is framed as follows:

- **Observation window:** Expanding from ICU admission, uses all data available up to the current hour.
- **Estimand:** the *cause-specific discrete-time hazard* of sepsis onset in the current hour, conditional on the stay being at risk at the start of it, $\Pr(\text{onset at hour } t \mid \text{at risk at } t, \mathbf{x}_t)$, with discharge and death as competing causes.
- **Prediction frequency:** Every hour.
- **Start time:** 1 hour after ICU admission.
- **End time:** 72 hours after ICU admission (right-censored).
- **Alerting window:** 6 hours. An alert counts as timely if it is issued in the six hours up to onset.

Estimand and scoring target. AUROC, AUPRC and calibration are computed on this hazard against an indicator that is one at the onset hour and zero at every other at-risk hour. They therefore evaluate onset-hour detection, not a binary "onset within six hours" target. The six-hour window is used only for the Utility Score, per-patient detection and lead-time metrics (Section 3.12).

Predictors and outcomes are aggregated to the same elapsed hour, so measurements recorded after the exact onset timestamp but within the same hourly bin can contribute to the onset-hour prediction. Reported discrimination should therefore be read as detection at the recorded onset hour, not as demonstrated advance warning (Limitation 3).

The hazard formulation was retained because cumulative risk over a horizon depends on both the sepsis and the competing-event hazards over that horizon. It does not yield a single measure of six-hour discrimination, a limitation recorded in Section 6.7. The alerting window and the utility scoring rule are adapted from the PhysioNet/CinC 2019 Challenge [16] rather than reproduced from it (Section 3.12), and the estimand differs from the Challenge's, which scored a binary label defined over the window itself.

3.8 Person-Hour Table and Competing Events

Each patient stay is expanded into a table with one row per person-hour at risk. Following Heyard et al. [38], each hour is assigned one of four outcomes:

- 0: No event in this hour; the stay remains at risk
- 1: Sepsis onset (the event of interest)
- 2: Discharged alive without sepsis (competing event)
- 3: Died without sepsis (competing event)

Discharge and death are treated as competing events rather than as non-informative censoring, because both remove a stay from the risk set and may be associated with sepsis risk.

3.9 Feature Construction

Features are extracted at each person-hour from chartevents, labevents and inputevents. The feature set includes:

1. **Vital signs**, 6 (last value carried forward per hour): heart rate, respiratory rate, SpO_2, MAP (invasive or noninvasive), temperature (°C), GCS total.
2. **Laboratory values**, 5 (last value carried forward): lactate, creatinine, total bilirubin, platelet count, WBC.
3. **Vasopressor indicators**, 2: any vasopressor (yes/no); any norepinephrine or epinephrine (yes/no).
4. **Six-hour trends**, 3: rolling slopes for heart rate, respiratory rate and MAP.
5. **Missingness indicators**, 6: whether each of lactate, creatinine, bilirubin, platelets, white cells and a blood gas (P/F ratio) was ordered in the current hour, one flag per test.

Temperature in Fahrenheit is converted to Celsius. FiO_2 values above 1.0 (percentage format) are divided by 100.

The P/F ratio is used in the SOFA respiratory component of the label but is not a continuous predictor, because it is recorded for a minority of person-hours and its forward-filled value reflects ventilation practice more than the current hour; only

its current-hour measurement indicator is included. No lactate slope is calculated, because lactate is not sampled densely enough to support a stable six-hour trend.

The declared set therefore holds **22 covariates**, and the fitted model carries **29 terms per outcome** once the six-column natural-spline baseline in elapsed hours and the intercept are counted. The feature vector is defined centrally in `config.R`, and the model-fitting and transport stages terminate with an error if a declared predictor is absent (Chapter 4).

3.10 Model Classes

3.10.1 Primary Model: Dynamic Discrete-Time Competing-Risks Hazard Model

The primary model is a multinomial pooled logistic regression over person-hours, following D'Agostino et al. [36] and Heyard et al. [38]:

$$\lambda_r(t \mid \mathbf{X}_{it}) = \frac{\exp\big(\alpha_r(t) + \boldsymbol{\beta}_r^\top \mathbf{X}_{it}\big)}{1 + \sum_s \exp\big(\alpha_s(t) + \boldsymbol{\beta}_s^\top \mathbf{X}_{it}\big)} \tag{3.3}$$

where $r \in \{1 = \text{sepsis}, 2 = \text{discharge}, 3 = \text{death}\}$. The baseline hazard $\alpha_r(t)$ uses a restricted natural cubic spline with 5 knots, placed at quantiles of training-set hour within each variant.

The model estimates the hazard of sepsis at every hour. With short intervals and low interval-specific event probabilities, pooled logistic regression closely approximates time-dependent Cox regression [36], while permitting explicit modelling of competing outcomes and requiring no proportional-hazards assumption.

Identification and penalisation. The unpenalised version of Equation (3.3) is **completely separated** under every label variant: under iteratively reweighted least squares run to a tight tolerance, the whole coefficient vector diverges together, with $|\beta|$ of order 10^{15}, although the design matrix is full rank. The optimiser nevertheless reports success: `nnet::multinom` returns `convergence = 0`, which records only that its stopping rule fired, not that a finite maximiser exists. A fixed ridge penalty of $\lambda = 10^{-4}$ is therefore applied to obtain finite, stable penalised estimates. The value is **not tuned**: across a 20-point ridge path the temporal-test AUROC varies by less than 0.02 for every variant. Coefficient comparisons across label variants (H2, Section 5.10.4) refer to the penalised estimates; without penalisation, a fit could stop at an arbitrary point under one label and a different one under another, producing an apparent label effect.

Coefficient uncertainty. Coefficient standard errors use a sandwich covariance clustered on the stay (implementation in Chapter 4). Because the model is ridge-penalised, these are standard errors of the *penalised* estimator, described as a ridge-regularised sandwich throughout. They do not provide uncertainty for H2's count of unstable covariates, which is a threshold rule applied across three separate fits, so H2 remains a descriptive count (Section 5.10.4).

Protocol Amendment 7 (post-hoc, §21 of `00_preregistration.md`) adds a diagnostic beside that count: for each flagged covariate, the difference between its two extreme estimates is divided by the pooled standard error of those two (Section 5.10.4). The ratio is reported alongside the pre-registered count, never in place of it. It is not a test: the three fits share stays, so the differenced estimates are correlated and a pooled standard error assuming independence is conservative by an unquantified amount; it carries no p-value and enters no corrected family. Every interval on a performance quantity comes from the stay-level cluster bootstrap of Section 3.14, and the analytic covariance enters no hypothesis verdict.

3.10.2 Comparator: Gradient-Boosted Trees (GBT)

A gradient-boosted tree (GBT) model is fitted with XGBoost for each label variant and split, on the same 22 covariates as the primary model. Its settings were fixed in advance and are applied identically to every variant, split and feature arm: logistic objective, area under the precision-recall curve as the evaluation metric, maximum depth 6, learning rate 0.05, row and column subsampling at 0.8, minimum child weight 5, and positive-class weighting by the observed class ratio. No hyperparameter search was conducted for either the GBT or the primary model, which keeps the search budget equal across the H6 comparison but may place the GBT below its attainable performance (Chapter 6).

The number of boosting rounds is chosen by early stopping on a holdout drawn from the *training* rows: 15 per cent of training stays are held out, the model is fitted on the remainder, and boosting stops after 30 rounds without improvement on the holdout. The holdout is split by stay, because each stay contributes roughly forty-five correlated person-hours and a row-level split would place the same patient on both sides. The test set is never used to select the number of rounds (Protocol Amendment 8).

3.10.3 Comparator: Cox Proportional-Hazards Model

A Cox proportional-hazards model is included as a conventional survival-model comparator. The pre-registration calls it the *static* comparator, which refers to its coefficients rather than its inputs: it is a Cox model with time-varying covariates and time-constant coefficients. It is fitted on the same training person-hours

as every other comparator, in counting-process form: each hour contributes the interval $(t, t + 1]$ with that hour's event indicator, so its covariates are the same hourly, forward-filled values the primary model sees. Each covariate has one time-constant log hazard ratio, the baseline hazard is left unspecified, and no interaction with elapsed time is estimated; Section 5.8.1 tests this assumption with Schoenfeld residuals.

The specification otherwise differs from the primary model:

- **Covariates.** Twelve of the primary model's twenty-two: heart rate, respiratory rate, oxygen saturation, mean arterial pressure, temperature, Glasgow Coma Scale, lactate, creatinine, total bilirubin, platelets, white cell count, and any vasopressor exposure. The three six-hour slope features, the second vasopressor indicator and the six order (*measured*) indicators are *not* included.
- **Time origin and risk set.** Elapsed hours since ICU admission, the same clock the primary model splines; a stay is at risk from hour one until onset, discharge, death or hour 72.
- **Event and competing events.** The event is sepsis onset in the hour. Unlike the multinomial primary model, the Cox specification treats discharge and death as censoring, which allows the consequences of the proportional-hazards and censoring assumptions to be examined.
- **Ties.** Efron's approximation, since hourly binning makes ties common.
- **Uncertainty.** Standard errors are clustered on the stay.
- **Scoring.** The linear predictor is mapped to the unit interval and scored on the same test person-hours as every other model, so the discrimination figures are comparable even though the underlying quantity is a relative hazard rather than an absolute one.

Because the Cox comparator also uses a reduced feature set and different event handling, its contrast with the primary model compares complete specifications rather than model form alone.

3.10.4 Rule-Based Comparators

NEWS2, qSOFA, and SIRS are computed at each person-hour from the same features. NEWS2 is the principal rule-based comparator (Section 2.8).

Model-class spread. The model-class spread of the variance decomposition is the AUROC range across the six evaluated specifications (primary, GBT, Cox, NEWS2, qSOFA and SIRS). Because these differ in features, event handling and whether they are fitted at all, as well as in model form, the spread measures differences between complete model specifications. The primary-versus-GBT contrast of H6, fitted on identical covariates, is the comparison closest to model class alone.

3.11 Validation Architecture

Design	How	Purpose
Internal (primary)	Time-based: train 2008–2016, test 2017–2019 [8]	Internal estimate under temporal shift
Internal (secondary)	Random 75/25 patient-level split (seed = 42)	Measure split optimism (H4)
Pre-treatment evaluation window	Person-hours before the first qualifying antibiotic or culture event [7]	Treatment leakage (H3)
External	eICU-CRD v2.0 [44], no retraining	Generalisability (H5)

Table 3.4: Validation and evaluation design. Both internal splits are patient-level, so no patient appears in both training and test sets; the pre-treatment window restricts evaluation rather than defining a split.

3.12 Evaluation Metrics

3.12.1 Primary Metrics

An **adapted PhysioNet-style Utility Score** is the primary clinical-utility metric. It rewards early true alerts and penalises late and false ones, following the design of the PhysioNet/CinC 2019 Challenge score [16]. AUROC is the metric on which the pre-registered comparative hypotheses are stated; it is not used to assess clinical utility, because it does not reflect it [4].

Departures from the Challenge score. The scoring rule is not the Challenge's own implementation, so the scores reported here are not comparable with published Challenge scores. It departs in three ways:

- Utility accumulates *per stay*, crediting the earliest qualifying alert once, rather than at every time step of every record, so holding an alarm on is not rewarded.

- The reward window is the pre-registered six-hour alerting window rather than the Challenge's wider ramp.
- A septic stay that is never alerted is penalised -1 once, rather than accumulating a penalty over the post-onset hours.

The normalisation against the inaction and optimal strategies follows the Challenge. The parameters are pre-registered constants fixed in `config.R` for every model and variant, so comparisons *within* this study are unaffected by the adaptation.

Scoring rule. Each stay contributes a raw utility:

- a septic stay first alerted at hour t within the reward window $[t_{\text{onset}} - 6,\ t_{\text{onset}}]$ earns $(t_{\text{onset}} - t)/6$, so a full-horizon warning earns 1 and an alert that coincides with onset earns nothing;
- a septic stay alerted only in $(t_{\text{onset}},\ t_{\text{onset}} + 3]$ earns -0.05 per late alert hour;
- a septic stay never alerted in either window earns -1;
- every alert hour outside a reward window, on any stay, costs -0.05.

The raw total U is normalised against the *inaction* strategy, which never alerts and earns $U_{\text{inact}} = -1$ per septic stay, and the *optimal* strategy, which earns the full reward on every septic stay with no false alarm, giving $U_{\text{opt}} = +1$ per septic stay:

$$\widetilde{U} = \frac{U - U_{\text{inact}}}{U_{\text{opt}} - U_{\text{inact}}}. \tag{3.4}$$

A score of $\widetilde{U} = 1$ corresponds to a perfect forecaster, $\widetilde{U} = 0$ to issuing no alerts, and $\widetilde{U} < 0$ to performance worse than issuing no alerts.

Operating points. The score depends on both the model and the threshold. At an hourly event rate below 0.5 per cent no model's predicted probability approaches the conventional 0.5 cut-off, so that cut-off reproduces the no-alert strategy for a well-calibrated model and an arbitrary point for a poorly calibrated one. The Utility Score is therefore reported at the 0.5 cut-off, for comparability with the literature, and at its maximum over a threshold grid. The grid is defined on the *alert rate*, the fraction of person-hours alerted, rather than on the probability scale, and spans 10^{-5} to 0.5 over sixty logarithmically spaced points, so that it adapts to each model's calibration. The maximum is taken over thresholds that raise at least one alert, and the alert rate, detection rate and false-alert burden at the utility-maximising threshold are reported with the maximum score.

Calibration is assessed using the calibration intercept and slope [39], flexible calibration curves (a restricted cubic spline of the outcome on the logit of predicted risk) summarised by the Integrated Calibration Index (ICI), E50, E90 and E_{max} [29], and the scaled Brier score. Predicted probabilities are bounded below at 10^{-6} rather than at the conventional 10^{-3}, which at this event rate would overwrite a large share of the predictions being assessed.

3.12.2 Secondary Metrics

AUPRC (important at low event rates), alert burden (false alerts per true alert; benchmark: 1.4 from Moor et al. [21]), lead time (benchmark: 3.7 hours from Moor et al.), and decision-curve net benefit [45]. Net benefit is evaluated on a threshold grid running from one tenth to ten times the observed event prevalence rather than on the conventional 0.01–0.20 grid, which at this prevalence lies far above the event rate.

3.12.3 Deployment-Facing Metrics

AUROC, AUPRC and the Utility Score characterise a model but not the alert load on a ward. The following are therefore reported at a threshold set to the largest cut-off achieving a fixed per-hour sensitivity of 80 per cent:

- **Positive predictive value** and its reciprocal, the number needed to evaluate (NNE), at that fixed sensitivity.
- **False alerts per patient-day**, which normalises alarm volume by exposure time rather than by the number of true alerts.
- **Per-patient (event-level) detection**: the fraction of septic stays alerted within the actionable window $[\text{onset} - 6\,\text{h},\ \text{onset}]$, the fraction alerted at any point before onset, patient-level PPV, and the fraction of all stays alerted. Per-hour and per-patient metrics are reported separately because they use different denominators and answer different operational questions.
- **False-alarm episodes per patient-day**, counting a contiguous run of alert hours once rather than once per hour.
- **Lead time** from first alert to onset: median, interquartile range, and the fraction of septic stays alerted at least 3 and 6 hours ahead.

The same quantities are computed on eICU-CRD with thresholds re-derived on the external cohort.

3.13 Sample Size

Sample-size adequacy is assessed with the Riley et al. [46] method, with an anticipated C-statistic of 0.846 from Moor et al. [21], a target shrinkage factor of at least 0.9, and 29 parameters: the 22 covariates, the six-column spline baseline and the intercept of the sepsis-specific linear predictor (Section 3.9). Each label variant is evaluated separately, and an underpowered variant is reported as such.

Because person-hours are clustered within stays (roughly 45 rows per stay, with one stay per patient) and Riley et al.'s criteria assume independent observations, the calculation uses training **stays** as the independent unit, with the variant-specific proportion of training stays labelled septic as the outcome proportion. The person-hour count is reported only as the size of the design matrix.

Riley et al.'s criteria are derived for a binary outcome, so applying them to the sepsis-versus-at-risk contrast is an approximation: the multinomial model estimates 29 terms for each of three non-reference outcomes. The minimum sample size scales approximately linearly with the parameter count, and the margins reported in Section 5.2 exceed even the all-outcomes count by a factor of ten, so the adequacy verdict does not depend on this choice.

3.14 Statistical Inference and Multiplicity

3.14.1 Uncertainty: the Stay-Level Cluster Bootstrap

Person-hours are nested within ICU stays, a single stay contributing on average about 45 correlated hourly rows, so treating rows as independent would understate standard errors. Every interval in this thesis therefore comes from a **stay-level cluster bootstrap**, in which whole ICU stays are drawn with replacement together with all their person-hours. Intervals are 95% percentile intervals from $B = 2{,}000$ replicates for internal and subgroup analyses and $B = 1{,}000$ for eICU-CRD, which contains 7.9 million person-hours. The seed is fixed at 42.

Contrasts between quantities measured on the *same* person-hours are evaluated within replicate, so the interval accounts for their correlation. This applies to the label-variant comparison (all three variants are built on the same test stays) and to the model-class comparison (all models score the same hours). Contrasts across *different* evaluation sets (temporal versus random split, and MIMIC-IV versus eICU-CRD) use independent bootstrap streams and propagate both variances. For subgroup analyses the bootstrap is stratified, resampling stays within each subgroup so that observed subgroup sizes are held fixed.

P-values are obtained by recentring the bootstrap distribution on the null boundary and reading off the tail area, using the $(1 + \text{count})/(B + 1)$ conven-

tion, so the smallest reportable p-value is $1/(B+1)$.

3.14.2 Confirmatory and Exploratory Families

The six pre-registered hypotheses form a single **confirmatory** family, adjusted by Holm–Bonferroni at a family-wise error rate of 0.05. The denominator remains $m = 6$ even though H2 and H3 do not yield a formal p-value (Section 5.10).

All subgroup analyses are **exploratory**. The pre-registration fixed their *scope* (that race, language and insurance would be examined), but not the subgroup levels, the reference level, or the direction of any comparison. Exploratory contrasts are corrected by Benjamini–Hochberg at $q = 0.05$ within three families: E1, all subgroup AUROC contrasts against the reference level across every equity variable and both models; E2, all subgroup label-sensitivity contrasts; and E3, the alternative-label contrasts of Protocol Amendments 2 and 4, in which the AUROC under each alternative label (Section 3.6) is compared with Variant B for both models. The anchor-attributable fraction of Equation 3.1 is a ratio with no meaningful null at zero and is reported as an interval only, outside the corrected family. The correction denominator is the number of contrasts *computed*, not the smaller number selected for presentation in Chapter 5, and Bonferroni-adjusted p-values are also reported for a reader who prefers family-wise control. Each subgroup is contrasted against the largest level of its own variable, so the reference is chosen by sample size alone.

Event floors (Protocol Amendment 6). `MIN_SUBGROUP_EVENTS` = 5, together with a 50 person-hour minimum, decides whether a subgroup AUROC is *computed*. `MIN_SUBGROUP_EVENTS_INTERPRET` = 100, equal to the external-validation floor, decides whether a computed AUROC is *interpreted* and whether its contrast enters family E1. Estimates below the interpretation floor are tabulated and flagged but excluded from statements about spread and from E1. The floor does not apply to E2, whose stay-level proportions are estimable at any stratum size.

The interpretation floor is **Protocol Amendment 6** (post-hoc, 6 August 2026, §20 of the pre-registration). It reduced E1 from thirty computed contrasts to 6, a departure from the rule that the correction denominator is the number of contrasts computed; the departure and its consequences are set out in Section 6.7.

3.14.3 Analysis Status Tagging

Every result in Chapter 5 carries a status tag:

- **Confirmatory [C]**: the estimand, threshold and direction were fixed in `00_preregistration.md` before any modelling.
- **Exploratory [E]**: not pre-specified as a specific contrast (specified only in scope, added by a protocol amendment, or absent from the protocol); generates rather than tests hypotheses.
- **Descriptive [D]**: estimates or counts with no inferential claim attached.

[C] marks a pre-specified hypothesis or estimand. It is not synonymous with a formal confirmatory test, and where a test or interval is reported, its uncertainty and multiplicity procedure follows Protocol Amendment 1, which was specified after the initial analyses. Alongside H1–H6, the analyses tagged [C] are sample-size adequacy, internal temporal performance, calibration, decision-curve net benefit, per-patient detection, external validation, proportional-hazards testing and the variance decomposition table; most of these are estimation rather than testing and so take no multiplicity correction. Analyses introduced by a protocol amendment are tagged [E] or [D], with one exception: the contrast substituted for H5 occupies the confirmatory slot reserved for H5. The tags are held in a machine-readable analysis register (`12_analysis_register.parquet`) produced by the pipeline, and are reproduced in Table C.1.

3.14.4 Status of the Inference Plan Itself

The uncertainty and multiplicity plan was introduced as **Protocol Amendment 1** on 30 July 2026 (§§13–15 of `00_preregistration.md`), after the initial analyses had been run and inspected. It did not alter the six hypotheses, their estimands, thresholds or directions, or any label, split, subgroup or model. It added bootstrap intervals, formalised the multiplicity procedures, and where the correction changes a verdict recorded in the original protocol, the corrected verdict is reported and the change is stated. The resulting inference should therefore be read as a post-hoc error-control plan applied to a pre-specified hypothesis set.

3.14.5 Feature Ablation: Action-Derived Predictors (Protocol Amendment 5)

Two kinds of predictor record clinician behaviour rather than patient physiology, in the same way that the label does (Section 2.4): vasopressor exposure, a treatment, and the per-test order indicators, which record a decision to test. **Protocol Amendment 5** (post-hoc, 6 August 2026, §19 of `00_preregistration.md`) adds an exploratory ablation of these 8 action-derived predictors: the two vasopressor

indicators and the six order indicators. The primary model and the GBT are refitted on the same training rows and splits with unchanged hyperparameters. Forward-filled laboratory values are retained, even though a value exists only because the test was ordered, because removing them would confound the ablation with a loss of physiological information. Both arms are scored by the same metric code on the same test rows, and the change in discrimination and in cross-label coefficient stability is reported in Section 5.7. The ablation enters no hypothesis, confirmatory family or variance decomposition.

3.15 Pre-Registration and Reporting

The original protocol specified the six hypotheses, the pre-registered label variants, model classes, validation splits, metrics and decision thresholds in `00_preregistration.md`; its label variants were fixed before cohort extraction and the protocol was locked before any of the modelling reported here. Analyses added later are tagged exploratory or descriptive, and every amendment is dated and disclosed in the protocol. Because the uncertainty and multiplicity procedures were specified after the initial analyses, Section 3.14.4 reports their status separately. Reporting follows TRIPOD+AI [12]; code availability and the constraints the data use agreement places on it are set out in Section 4.7.2.

Chapter 4

System Design and Pipeline Architecture

The analysis pipeline consists of twelve numbered scripts in `sepsis/project/`, executed in sequence after the pre-registration document `00_preregistration.md`. Each stage reads defined upstream outputs and writes stage-specific results to `sepsis/project/output/`, leaving the source data unchanged. This chapter describes the architecture, stage interfaces, validation checks and reproducibility controls; the statistical rationale for each step is given in Chapter 3. Figure 4.1 shows the data flow, and Appendix B lists each stage's inputs and outputs.

4.1 Design Principles

Read-only, streamed source data. Large source tables (`chartevents`, 3.5 GB compressed; `labevents`, 2.6 GB) are queried through DuckDB with the cohort and item-identifier filters pushed into the scan, so the full tables are never loaded into memory. Source directories are read-only, and all derived data are written to `output/`, so a run never alters the inputs of the next.

Centralised configuration. `config.R` defines every path, identifier, window, threshold, seed, family size, event floor and feature list used by more than one stage. Because the study varies label definitions, a constant duplicated across scripts could diverge without warning and be indistinguishable from a finding. The model-fitting and transport stages read the same declared feature vector and verify names and order before scoring.

Fail-fast interface validation. Interface violations terminate the pipeline with an error rather than letting it continue on inferred or incomplete inputs. Selecting

predictors by intersecting a declared list with the columns present would silently omit any predictor whose name did not match while the model still fitted, so `require_features()` stops a stage when a declared predictor is absent. Runtime assertions also check key invariants, including the bounds of the Utility Score (a threshold above every prediction must score zero, and no operating point may exceed one) and that the label variants produce distinct external onset sets.

Independent stages and isolated outputs. Each stage corresponds to a distinct analytical operation and writes reusable intermediate outputs in Parquet, so individual stages can be rerun and validated independently. Stage 12 operates only on stored predictions, so intervals and multiplicity corrections can be recomputed without refitting any model. Variant-specific and ablation analyses are written to separate outputs, so additional analyses can be run without overwriting pre-registered fits.

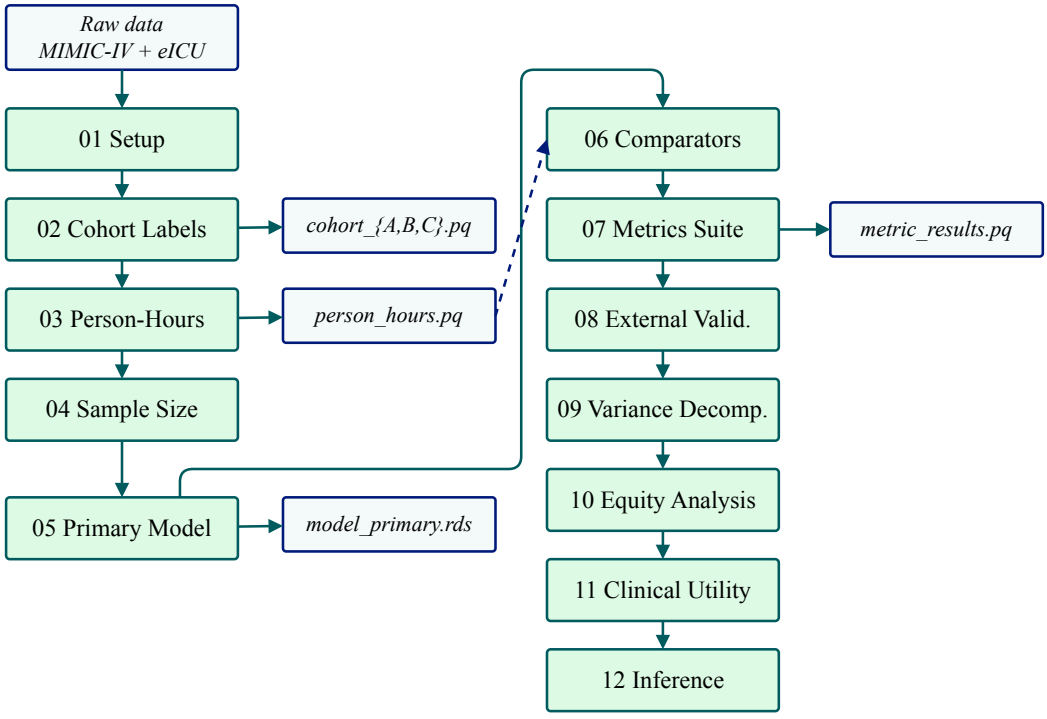


Figure 4.1: Pipeline data flow. Each box is a single script; italic nodes are output files. Raw MIMIC-IV and eICU-CRD data are read-only.

4.2 Configuration and Utilities

`config.R` defines the shared pipeline parameters, including data paths, cohort criteria, antibiotic name lists, MIMIC-IV item identifiers, label variant definitions, split boundaries, prediction framing constants, equity variable names and Utility Score parameters.

`utils.R` provides common input/output, data-processing, validation and diagnostic functions: a DuckDB-backed streaming reader, Parquet wrappers, datetime parsing, last-observation-carried-forward (LOCF) imputation, feature checks and Schoenfeld test wrappers. It also supplies the score function that the `sandwich` package lacks for `multinom` fits: for row i and non-reference category k, the score for the coefficient on covariate j is $x_{ij}(y_{ik} - p_{ik})$, which together with the retained Hessian yields the stay-clustered covariance of Section 3.10.1.

4.3 Data Preparation (Stages 01–04)

4.3.1 Stage 01: Setup Validation

`01_setup.Rmd` verifies that the required MIMIC-IV source tables are present and terminates if any is unavailable.

4.3.2 Stage 02: Cohort Extraction and Labelling

`02_cohort_labels.Rmd` constructs the base cohort, attaches the equity variables, and generates cohort-level labels for Variants A–C and the post-hoc alternative labels described in Sections 3.5 and 3.6. Variant D's onsets and Variant F's retimed onsets need the hourly SOFA trajectory and are derived in Stage 03. The post-hoc variants are excluded from the pre-registered variant set in `config.R` and written to separate files, so they cannot enter H1 or the decomposition table.

4.3.3 Stage 03: Person-Hour Table Construction

`03_person_hours.Rmd` expands each stay into hourly rows and is the only analytical stage that queries the full-size event tables directly. For each person-hour:

- Vitals and labs are LOCF-imputed from the most recent measurement.
- Six-hour rolling slopes are computed for heart rate, respiratory rate and MAP.
- Vasopressor indicators are set from concurrent infusions.

- Missingness flags are set for each lab test.
- The competing-event code (0/1/2/3) is assigned based on the hour's relationship to onset, discharge, or death.
- The temporal split group is assigned from the stay's admission year.
- Treatment anchor timestamps (first antibiotic, first culture) are recorded.

4.3.4 Stage 04: Sample Size Calculation

`04_sample_size.Rmd` implements the stay-level sample-size assessment of Section 3.13 and also produces a person-hour reference calculation, labelled as such because person-hours are not independent.

4.4 Modelling (Stages 05–06)

4.4.1 Stage 05: Primary Model

`05_primary_model.Rmd` fits the primary model of Section 3.10.1 for each requested label variant under the temporal and random splits. It validates the declared feature set, applies the fixed ridge penalty, and logs the convergence flag alongside the largest fitted coefficient, flagging any fit in which $\max|\beta|$ exceeds 20. It writes the fitted models, coefficients, coefficient covariances and test predictions; the full and ablated specifications (Section 3.14.5) are written to separate outputs.

4.4.2 Stage 06: Comparator Models

`06_comparators.Rmd` fits the GBT and the Cox comparator of Sections 3.10.2 and 3.10.3, runs the Schoenfeld tests, and computes NEWS2, qSOFA and SIRS at each person-hour. The GBT is fitted under both splits and on the ablated feature set with identical hyperparameters, and its early-stopping holdout is drawn from training stays only; GBT feature importance is exported for each variant. The rule-based scores never use the ablated features, and the Cox comparator is not ablated.

4.5 Evaluation (Stages 07–11)

4.5.1 Stage 07: Metrics Suite

`07_metrics_suite.Rmd` evaluates stored predictions using the metrics of Section 3.12 and writes split-specific, label-specific and ablation result tables. The

utility threshold grid is evaluated in a single sorted pass, and the Utility Score invariants are asserted automatically. A stand-alone Python implementation of the same scoring rule, `utility_score.py`, is kept in step with the stage so that the two implementations can be cross-checked.

4.5.2 Stage 08: External Validation on eICU-CRD

`08_external_validation.Rmd` applies the frozen MIMIC-IV models to harmonised eICU-CRD features without retraining. It produces three separate analyses: the Sepsis-3 variants on the microbiology-covered sub-cohort, the antibiotic-only sensitivity analysis on the full cohort, and the post-hoc Variant D analysis on the full cohort (Section 3.2.2). `check_model_features()` rebuilds each design matrix from the declared feature vector and verifies compatibility before scoring. Runtime guards require the three Sepsis-3 variants to produce distinct onset sets, and keep the post-hoc Variant D output and the pre-registered external output in separate files, so a post-hoc label cannot enter H5.

4.5.3 Stage 09: Variance Decomposition

`09_variance_decomposition.Rmd` assembles the one-factor-at-a-time AUROC contrasts defined in the pre-registration and generates the label-anchor attribution and onset-agreement summaries. It records the pre-registered H5 contrast as undefined and stores the substituted AUROC transport contrast separately (Section 5.10.7).

4.5.4 Stage 10: Equity Analysis

`10_equity_analysis.Rmd` estimates subgroup discrimination and label sensitivity for the equity variables available in MIMIC-IV: race, language and insurance. It applies the estimation and interpretation floors defined in Section 3.14.

4.5.5 Stage 11: Clinical Utility Metrics

`11_clinical_metrics.Rmd` generates the calibration, decision-curve, fixed-sensitivity, per-patient alert, alarm-episode and lead-time outputs defined in Section 3.12, for both MIMIC-IV and eICU-CRD.

4.6 Inference and Multiplicity (Stage 12)

`12_inference.Rmd` reads stored predictions and produces stay-level bootstrap intervals, the multiplicity-adjusted confirmatory results, the exploratory-family cor-

rections and the analysis register (Section 3.14). It resamples ICU stays rather than person-hours. Reduced-replicate settings for smoke tests are documented in Appendix A.4; reported results always use the replicate counts fixed in `config.R`.

4.7 Reproducibility

The pipeline is orchestrated by `run_pipeline.R` (or the equivalent `run_pipeline.sh`), which executes all stages sequentially or by phase, renders each stage to PDF in `output/pdf/`, and writes per-stage and combined run logs to `output/logs/`. Intermediate outputs are Parquet, and small result tables are also written as CSV. The end-to-end run that produced the reported results took about two and a half hours on an Apple silicon workstation, dominated by Stage 05. `thesis_constants.R` generates the numerical values used in this thesis from the pipeline result tables (Appendix A.4).

4.7.1 Fixed Random Seed

The fixed seed 42 is defined centrally and applied to stochastic splitting, the sample-size procedure, model fitting and bootstrap resampling, to support repeatability within the analysis environment. The temporal partition is deterministic, because it is defined by admission year. Seeding does not by itself guarantee bitwise reproducibility across environments: GBT results were monitored when comparing runs, because they are the component most likely to vary slightly across library builds or architectures (Appendix A.4). The pipeline can be rerun on a credentialed MIMIC-IV v3.1 installation by setting `MIMIC_RESEARCH_ROOT` to the directory containing the `data/` folder.

4.7.2 Code Availability

The pipeline is held in a public version-controlled repository at `https://github.com/ahadarif007/sepsis-variance-decomposition`, released under the Apache License 2.0. The repository contains the source code but no MIMIC-IV or eICU-CRD records, derived person-hour tables or fitted model objects, because the PhysioNet data use agreement (Section 3.3) prohibits redistribution of the underlying data. Reproduction therefore requires independent PhysioNet credentialing for both databases. Appendix A describes the repository layout, the reproduction procedure and the scope of the licence.

Chapter 5

Results

This chapter reports cohort characteristics, sample-size adequacy, model performance and calibration, clinical utility, external validation, label-anchor analyses, the feature ablation, model diagnostics, equity analyses and, last, the variance decomposition and pre-registered hypothesis verdicts (Section 5.10), which are assembled from the earlier estimates. Results are marked as confirmatory [C], exploratory [E] or descriptive [D] (Section 3.14); the status of every analysis is recorded in the analysis register (Appendix C). Unless otherwise stated, intervals are 95% stay-level cluster-bootstrap intervals. Chapter 6 interprets the results and sets out the limitations.

5.1 Cohort Description and Label Variation [D]

Table 5.1 summarises the three cohorts. The base extraction identifies 65,078 adult first ICU stays. The database and patients are the same across all three variants; only the sepsis definition differs.

Var	Name	Stays	Onsets (cohort)	%	Deaths, no sepsis	Onset h (IQR)
A	Narrow	65,078	6,252	9.6	5,934	10[4–26]
B	Seymour-Std	65,078	6,447	9.9	5,892	11[4–30]
C	Liberal	65,078	12,581	19.3	4,956	10[4–30]

Table 5.1: Cohort description by label variant. "Deaths, no sepsis" counts stays ending in death with no labelled sepsis onset before it, so it varies with the label. The Liberal variant's wider antibiotic-culture windows and rolling SOFA baseline identify about twice as many cases. Onsets are counted **cohort-wide, at any hour of the stay**; the subset inside the 72-hour modelling horizon (5,662, 5,887 and 11,531), on which the models are trained and evaluated, is reported in Table 5.16.

The twofold difference in sepsis prevalence between Variant A (9.6%) and Variant C (19.3%) is consistent with the label sensitivity previously reported by Johnson et al. [15].

5.2 Sample Size Adequacy [C]

Because Riley et al.'s criteria [46] assume independent observations, adequacy is assessed on the stay rather than the person-hour (Section 3.13; Table 5.2).

Var	Name	Train stays	Sepsis stays	Stay prev.	Riley min-N (stays)	EPV
A	Narrow	52,946	4,647	0.0878	1,729	160.2
B	Seymour-Std	52,946	4,847	0.0916	1,666	167.1
C	Liberal	52,946	9,403	0.1776	991	324.2

Table 5.2: Sample size adequacy on the independent unit. Every row is a count of *stays*: the training split contains 52,946 stays, of which the given number are labelled septic under that variant. Riley min-N is the minimum number of independent observations required at that variant's stay-level prevalence, an anticipated C-statistic of 0.846 [21], and 29 parameters, which is the number of terms in the sepsis-specific linear predictor (22 covariates, a six-column spline baseline and an intercept). The criterion is applied as an approximate binary-outcome assessment of the sepsis-versus-at-risk contrast (Section 3.13); EPV is events per parameter on that reading. All three variants exceed the requirement by a wide margin (30.62×, 31.78× and 53.43× respectively).

The corresponding design matrices contain 2,304,923, 2,301,842 and 2,120,376 training person-hours, at per-hour event rates of 0.00202, 0.00211 and 0.00443. These figures govern the computational cost of the fit but do not multiply the effective sample size.

Sample size is therefore not a binding constraint: under the primary label the requirement is 1,666 independent observations against 52,946 available, a margin of 31.78×. Precision is limited instead in the external cohort (Section 6.3) and in the smaller demographic strata (Section 5.9).

5.3 Model Performance: Internal Temporal Evaluation [C]

Table 5.3 reports performance on the temporal test set (2017–2019) under the unrestricted evaluation window.

Var	Model	AUROC	AUPRC	Utility (0.5)	Utility (best)	Alert burden
A	Primary	0.7713	0.0093	0.000	0.026	∞
A	GBT	0.7455	0.0088	−1.614	0.038	192.1
A	NEWS2	0.6413	0.0032	−0.509	−0.001	236.4
A	Cox	0.6589	0.0035	−4.790	0.000	356.4
A	qSOFA	0.6012	0.0027	−2.293	−0.263	343.1
A	SIRS	0.6157	0.0032	−3.209	−0.035	346.9
B	Primary	0.7590	0.0096	0.000	0.025	∞
B	GBT	0.7424	0.0087	−1.781	0.031	196.5
B	NEWS2	0.6415	0.0032	−0.491	0.000	246.7
B	Cox	0.6444	0.0033	−4.720	0.000	361.6
B	qSOFA	0.6091	0.0028	−2.193	−0.252	316.8
B	SIRS	0.6114	0.0030	−3.086	−0.035	351.7
C	Primary	0.7695	0.0201	0.000	0.095	∞
C	GBT	0.7550	0.0188	−0.557	0.111	89.2
C	NEWS2	0.6483	0.0074	−0.099	0.001	102.2
C	Cox	0.6733	0.0085	−1.613	0.003	148.7
C	qSOFA	0.6012	0.0060	−0.705	−0.048	150.9
C	SIRS	0.6242	0.0070	−1.065	0.000	148.4

Table 5.3: Internal temporal test performance (2017–2019, all variants use temporal split). N test rows: 548,827 (B). **Utility (0.5)** is the normalised adapted PhysioNet-style Utility Score (Section 3.12) at the default cut-off; **Utility (best)** is its maximum over the threshold grid of Section 3.12. Zero is the no-alert strategy under both. Alert burden is at the 0.5 cut-off (∞ = no true alerts there); the benchmark is 1.4 [21]. Per-threshold curves are in Figure 5.1 and the attaining operating points in Table 5.5. Confidence intervals for Variant B are in Table 5.4. **[C] confirmatory.**

Model	AUROC	95% CI	AUPRC (95% CI)
Primary	0.7590	0.7434–0.7749	0.0096 [0.0082, 0.0123]
GBT	0.7424	0.7265–0.7583	0.0087 [0.0075, 0.0110]
NEWS2	0.6415	0.6248–0.6589	0.0032 [0.0029, 0.0037]
Cox	0.6444	0.6271–0.6614	0.0033 [0.0030, 0.0038]
qSOFA	0.6091	0.5934–0.6254	0.0028 [0.0026, 0.0033]
SIRS	0.6114	0.5950–0.6281	0.0030 [0.0028, 0.0039]

Table 5.4: Variant B temporal test performance with 95% stay-level cluster-bootstrap intervals ($B = 2{,}000$ replicates over 548,827 test person-hours carrying 1,040 sepsis onsets). These are *marginal* intervals; paired differences are reported separately where claimed (H6, Section 5.10.8). The primary model and GBT overlap almost completely, and both are separated from the other four models. **[C] confirmatory.**

5.3.1 Utility and Alert Burden Across Thresholds

At the default 0.5 cut-off no model does better than issuing no alerts, but this threshold is uninformative. The primary model scores 0.000 under Variant B because its largest predicted hourly probability is far below 0.5, so it generates no alert. The models that do cross 0.5 (the rule-based scores, rescaled onto the unit interval, and the Cox linear predictor) do so so often that they score well below zero: -4.720 for Cox and -2.193 for qSOFA under Variant B. At this cut-off the models are therefore separated by calibration rather than clinical value.

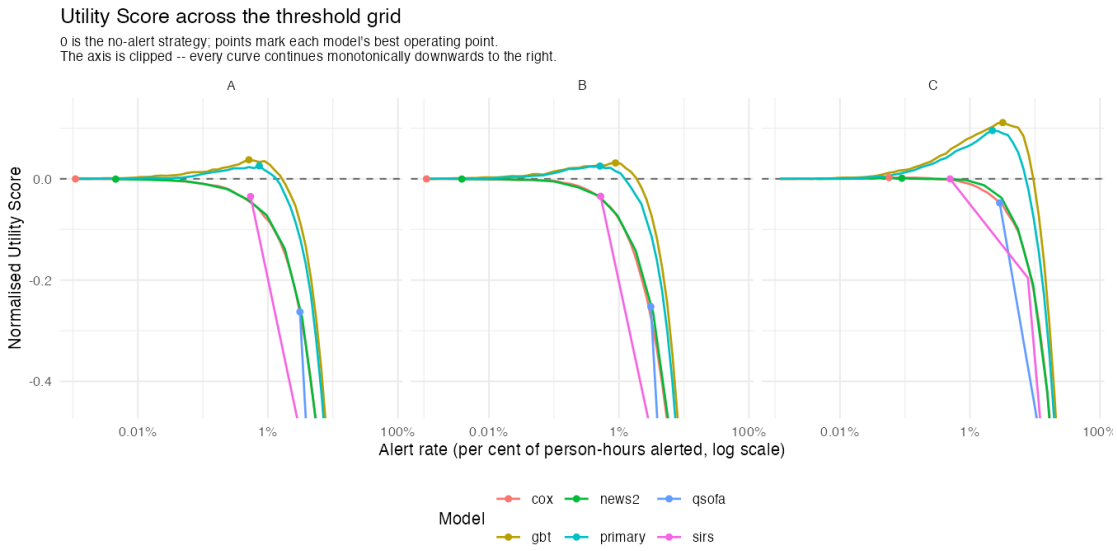


Figure 5.1: Normalised Utility Score against alert rate for every model and pre-registered label variant on the temporal test set. The dashed line at zero is the no-alert strategy, and points mark each model's best operating point. Only the primary model and GBT rise above zero, peaking between 0.5 and 3 per cent of person-hours alerted. The vertical axis is clipped; every curve falls to -5 or below at a 50 per cent alert rate.

Figure 5.1 traces the whole threshold grid, and Table 5.5 reports each model's best attainable score together with the operating point that attains it.

Var	Model	Max utility	Alert rate (% p-h)	Stays detected (%)	Alert burden
A	Primary	0.026	0.737	15.6	44.9
A	GBT	0.038	0.511	13.0	40.2
A	NEWS2	−0.001	0.005	0.0	∞
A	Cox	0.000	0.001	0.0	∞
A	qSOFA	−0.263	3.114	13.9	254.1
A	SIRS	−0.035	0.542	4.1	122.9
B	Primary	0.025	0.511	11.2	47.3
B	GBT	0.031	0.885	17.0	51.8
B	NEWS2	0.000	0.004	0.0	∞
B	Cox	0.000	0.001	0.0	∞
B	qSOFA	−0.252	3.095	13.5	248.8
B	SIRS	−0.035	0.523	3.5	135.8
C	Primary	0.095	2.213	28.5	29.3
C	GBT	0.111	3.194	35.5	34.0
C	NEWS2	0.001	0.090	0.8	49.2
C	Cox	0.003	0.057	0.6	46.7
C	qSOFA	−0.048	2.868	13.0	111.2
C	SIRS	0.000	0.493	3.4	59.7

Table 5.5: Maximum achievable Utility Score and the operating point that attains it, over the sixty-point threshold grid, on the temporal test set. "Alert rate" is the percentage of test person-hours alerted; "stays detected" is the percentage of septic stays alerted within the reward window [onset − 6, onset]; alert burden is false alerts per true alert at that same point, against the benchmark of 1.4 [21]; ∞ with a maximum at or below zero means the best strategy is to raise no alert. **[E] exploratory:** thresholds are chosen on the test set, so these are upper bounds on achievable utility.

After threshold sweeping, the two fitted models reach positive utility under every variant. Under Variant B their maximum scores are 0.025 (primary) and 0.031 (GBT), on a scale bounded above by 1. These optima are attained at 47.3 and 51.8 false alerts per true alert, while detecting 11.2 and 17.0 per cent of septic stays within the actionable window. Under Variants A and B, NEWS2, Cox, qSOFA and SIRS have no threshold at which alerting beats not alerting, and under Variant C their maxima do not exceed 0.003. Across all variants, every positive maximum is attained at between 29.3 and 51.8 false alerts per true alert, twenty to thirty-seven times the benchmark of 1.4 [21]. The thresholds are selected on the test set, so these values are optimistic upper bounds. AUROC values in the mid-to-high 0.7s (Table 5.3) therefore did not translate into an operating point

with a clinically plausible alert burden.

Decision-curve net benefit prices false alarms differently: at a threshold matched to the event prevalence, the primary model achieves positive net benefit over both default strategies (Table 5.7, Section 5.4). This is an internal contrast between two metrics, not a test of H5's internal-to-external estimand. Under both metrics, no configuration tested approaches the alert-burden benchmark.

5.3.2 Calibration [C]

Calibration intercept and slope are two summary numbers describing a whole curve, so both the curve itself and a scalar summary of its distance from the diagonal are reported. Following Van Calster et al. [29], the flexible calibration curve is a restricted cubic spline of the outcome on the logit of predicted risk, and it is summarised by the Integrated Calibration Index (ICI, the mean absolute distance between predicted risk and that curve) and the scaled Brier score, $1 - \text{Brier}/\text{Brier}_{\text{null}}$, where the null model predicts the prevalence for every person-hour.

Var	Model	Intercept	Slope	ICI	$E_{\max}$	Scaled Brier
A	Primary	−0.077	1.013	0.0002	0.019	0.0037
A	GBT	−6.049	1.143	0.3789	0.893	−89.1792
A	Cox	−6.418	1.034	0.4986	0.978	−140.9298
B	Primary	−0.106	1.055	0.0002	0.033	0.0038
B	GBT	−6.059	1.306	0.3993	0.864	−93.5482
B	Cox	−6.399	0.957	0.5016	0.975	−138.8289
C	Primary	−0.035	1.084	0.0003	0.006	0.0082
C	GBT	−5.295	1.253	0.3971	0.836	−42.3814
C	Cox	−5.564	1.168	0.4933	0.984	−60.3299

Table 5.6: Calibration on the temporal test set. Perfect calibration is intercept = 0, slope = 1, ICI = 0. A scaled Brier score at or below zero means the model is no better than predicting the prevalence for everyone. NEWS2, qSOFA and SIRS are omitted: their score-to-probability lookups take only 21, 4 and 5 distinct values respectively, too few to support a flexible calibration curve.

The primary model is the only evaluated model whose predicted probabilities stay close to observed risks, under every label variant: its intercept sits within 0.11 of zero, its slope within 0.08 of one (1.013, 1.055, 1.084), and its largest ICI across the three variants is 0.0003, under a fifth of the 0.2 per cent event rate. Its scaled Brier score is positive under all three, though barely (0.0037 to 0.0082), meaning it improves on the null model by less than one per cent.

Intervals on intercept and slope. Protocol Amendment 10 attaches intervals to these two quantities, which at 548,827 person-hours are narrow. The slope covers one under the Narrow and Standard variants ([0.953, 1.073] and [0.993, 1.119]) but not under the Liberal one ([1.040, 1.127]); the intercept covers zero under the Liberal variant ([−0.077, 0.008]) but not under the other two ([−0.140, −0.018] and [−0.166, −0.042]). Under each variant one of the two absolute targets is therefore missed at this confidence: the miscalibration is statistically detectable but small. An intercept of −0.106 on the logit scale at an event rate of 0.00189 moves the predicted hourly risk by about a tenth of itself and moves no operating point on the threshold grid of Section 5.4.

GBT and Cox are severely miscalibrated, with intercepts near −6 and strongly negative scaled Brier scores. The Variant B GBT scaled Brier score of −93.5482, for example, indicates a squared error 94.5 times that of predicting the prevalence for every person-hour. GBT emits predicted risks between 0.2 and 0.9 for person-hours whose observed event rate never exceeds 0.04 (Figure 5.2), so its best operating point, near 0.8157 under Variant B (Table 5.5), can be located only empirically and has no interpretation as a risk.

Across variants, slopes of 1.013, 1.055 and 1.084 sit within 0.071 of one another, and the paired cross-label contrasts against the primary label are −0.042 [−0.114, 0.027] for the Narrow variant and 0.029 [−0.031, 0.085] for the Liberal one: both intervals cover zero, so the label is not shown to change the calibration slope. The Liberal-against-Standard *intercept* contrast, by contrast, is 0.071 [0.009, 0.133], which excludes zero; the displacement is small on the probability scale at a 0.00189 event rate. The label definition therefore shifts the calibration intercept slightly but is not shown to change the slope.

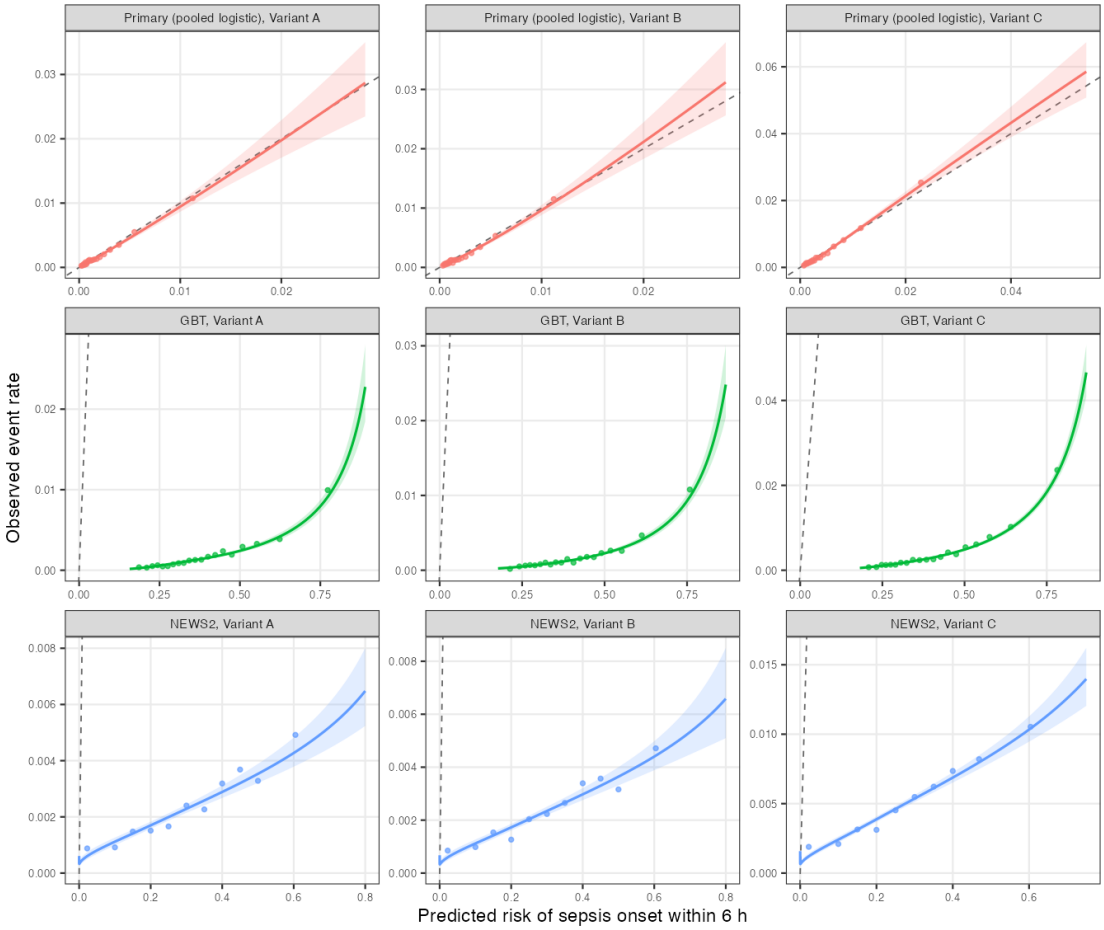


Figure 5.2: Flexible calibration curves on the temporal test set. Dashed line = perfect calibration; points = twenty equal-sized bins of predicted risk; shaded band = 95% confidence interval. Each panel has its own scale because the predicted-risk ranges differ by orders of magnitude.

Sensitivity to the Probability Clamp

Calibration is often computed after clamping predicted probabilities to $[0.001, 0.999]$. That guard is standard for models whose predictions approach 0 or 1, but at a 0.2 per cent hourly event rate it distorts the estimate: 46.8% (Variant A), 39.6% (Variant B) and 8.8% (Variant C) of the primary model's predicted probabilities fall *below* the clamp floor, so the clamp overwrites a large share of the quantity being measured. No predicted probability exceeds 0.2342 under any variant, so only the floor binds. The values in Table 5.6 use a floor of 10^{-6}. Under the conventional clamp the primary model's Variant A slope is 1.137, against

1.013 on the 10^{-6} floor, so a calibration guard chosen without reference to the event rate itself changes the reported estimate.

5.4 Clinical Utility: Net Benefit, Operating Points, and Event-Level Detection

This section reports four operational quantities: net benefit across a plausible threshold range, the positive predictive value and false-alarm rate at a fixed sensitivity, detection expressed per patient rather than per person-hour, and the lead time gained when detection occurs.

Under the discrete-time competing-risks specification, event = 1 marks the onset hour itself, so each septic stay contributes one event hour out of a mean of 45. Per-hour sensitivity requires the alert to fire *in the onset hour*; per-patient sensitivity asks only whether the patient was alerted before onset. Per-hour figures alone understate detection and do not show the associated alarm volume.

5.4.1 Decision-Curve Analysis [C]

Decision-curve analysis requires a threshold grid, and at this prevalence the choice of grid determines the result. A threshold probability of t encodes a willingness to accept $(1 - t)/t$ false alarms per true detection. The hourly event rate here is 0.19–0.42 per cent, so the conventional grid of 0.01–0.20 sits between five and one hundred times the prevalence: at $t = 0.01$ the analyst is asserting a tolerance of only 99 false alarms per detection, far below what an hourly sepsis alert must absorb, and every curve falls trivially below "treat none". Net benefit is therefore evaluated on a grid from one tenth to ten times the prevalence, following Vickers and Elkin [45].

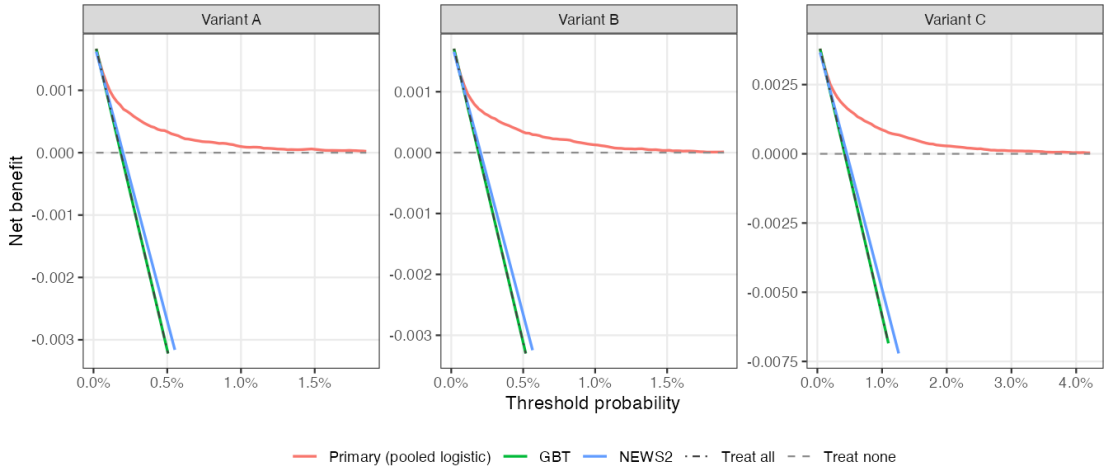


Figure 5.3: Decision curves on the temporal test set. Net benefit against threshold probability, with both default strategies shown. Curves are clipped below $-1.75\times$ prevalence for legibility.

Var	Model	Net benefit	Std NB	Alert rate	Alert-rate span
A	Primary	7.53×10^{-4}	0.407	0.305	0.987
A	GBT	0	0.000	1.000	0.000
A	Cox	0	0.000	1.000	0.000
A	NEWS2	1.33×10^{-4}	0.072	0.894	0.000
A	qSOFA	2.70×10^{-4}	0.146	0.656	0.000
A	SIRS	2.51×10^{-4}	0.136	0.676	0.000
B	Primary	7.19×10^{-4}	0.380	0.318	0.995
B	GBT	0	0.000	1.000	0.000
B	Cox	0	0.000	1.000	0.000
B	NEWS2	1.31×10^{-4}	0.069	0.894	0.000
B	qSOFA	2.84×10^{-4}	0.150	0.656	0.000
B	SIRS	2.83×10^{-4}	0.149	0.676	0.000
C	Primary	1.73×10^{-3}	0.411	0.289	0.997
C	GBT	0	0.000	1.000	0.000
C	Cox	0	0.000	1.000	0.000
C	NEWS2	2.98×10^{-4}	0.071	0.888	0.000
C	qSOFA	6.45×10^{-4}	0.153	0.644	0.000
C	SIRS	6.91×10^{-4}	0.164	0.665	0.000

Table 5.7: Net benefit at a threshold probability equal to the event prevalence. At that threshold both defaults have net benefit exactly zero and the maximum attainable net benefit is the prevalence itself, so standardised net benefit (Std NB = net benefit / prevalence) is the fraction of the attainable benefit captured. "Alert-rate span" is the change in alert rate across the threshold grid.

Decision-curve analysis produced positive but modest net benefit for the primary model. At a threshold equal to the prevalence, the primary model captures 40.7% (Variant A), 38.0% (Variant B) and 41.1% (Variant C) of the attainable net benefit. The three variants span barely three percentage points, so, unlike discrimination, this share is insensitive to the reading of Sepsis-3. The qualitative result matches the swept Utility Score (Table 5.5): a small positive margin for the fitted models and none for the rule-based comparators. The magnitudes differ because the metrics price errors differently: net benefit weights a false alarm by $t/(1-t)$, which at $t = 0.00189$ is very small, whereas the Utility Score imposes a fixed -0.05 penalty per false-alert hour. Net benefit is therefore the more generous metric here, and the gap widens as the alert rate rises. The preferred weighting of false alerts depends on the intended clinical decision and should be specified before deployment. This within-cohort disagreement is not an instance of H5's internal-to-external estimand.

For every model except the primary one, the alert-rate span is zero across

the plausible threshold range. GBT and Cox alert on *100% of person-hours* at every threshold in this range, so their decision curves coincide with treating everyone (Figure 5.3); with predicted probabilities in the 0.2–0.9 range (Table 5.6), no threshold below 4 per cent excludes anything. NEWS2, qSOFA and SIRS are non-tunable for the opposite reason: their integer scores map to a few fixed probabilities, all above the grid, and they alert on a constant 58–86 per cent of person-hours. Within the clinically plausible range, the primary model is therefore the only one whose alert rate responds to the threshold; GBT's best operating point lies far above that range (Table 5.5).

5.4.2 Operating Points at Fixed Sensitivity [E]

Alerting systems are usually commissioned by choosing a sensitivity target rather than a fixed cut-off. Table 5.8 sets the threshold to the largest cut-off achieving 80 per cent per-hour sensitivity and reads off the resulting operating characteristics.

Var	Model	Specificity	PPV	NNE	Alerts/pt-day	FA/pt-day	FP:TP
A	Primary	0.562	0.0034	296	10.52	10.48	295
A	GBT	0.540	0.0032	311	11.05	11.02	310
A	NEWS2	0.301	0.0023	442	16.79	16.75	441
A	Cox	0.404	0.0025	403	14.32	14.28	402
B	Primary	0.522	0.0032	315	11.49	11.45	314
B	GBT	0.511	0.0031	322	11.74	11.70	321
B	NEWS2	0.301	0.0023	430	16.79	16.75	429
B	Cox	0.380	0.0024	409	14.88	14.85	408
C	Primary	0.565	0.0077	129	10.48	10.40	128
C	GBT	0.541	0.0073	136	11.04	10.96	135
C	NEWS2	0.315	0.0052	191	16.47	16.38	190
C	Cox	0.423	0.0058	171	13.87	13.78	170

Table 5.8: Operating characteristics at a fixed per-hour sensitivity of 80%. NNE = number needed to evaluate (1/PPV), the alerted person-hours required per true event hour. FA/pt-day = false-alert hours per patient-day. qSOFA and SIRS are reported separately in the text because they degenerate at this target.

At 80 per cent sensitivity, PPV never exceeds 0.0077. The best case, the primary model under the Liberal label, still requires 129 alerted person-hours per true event hour; under the Seymour-Standard label the same model requires 315. Every configuration generates between 10.40 and 16.75 false-alert hours per patient-day; even the best configuration generates 10.40 false-alert hours per patient-day.

This ordering differs from the AUROC ordering. The primary model and GBT are separated by less than 0.02 AUROC (H6), but at a fixed sensitivity target the

primary model needs fewer alerts per true event under all three labels.

A sensitivity target also exposes the coarse rule-based scores. qSOFA cannot be tuned to 80 per cent sensitivity under the Liberal label at all: with only 4 distinct score values, the lowest cut-off reaching the target is the one that alerts on every person-hour (specificity 0.000, 23.90 false-alert hours per patient-day). It reaches the target under Variants A and B only at specificity 0.344 and 15.72 false-alert hours per patient-day.

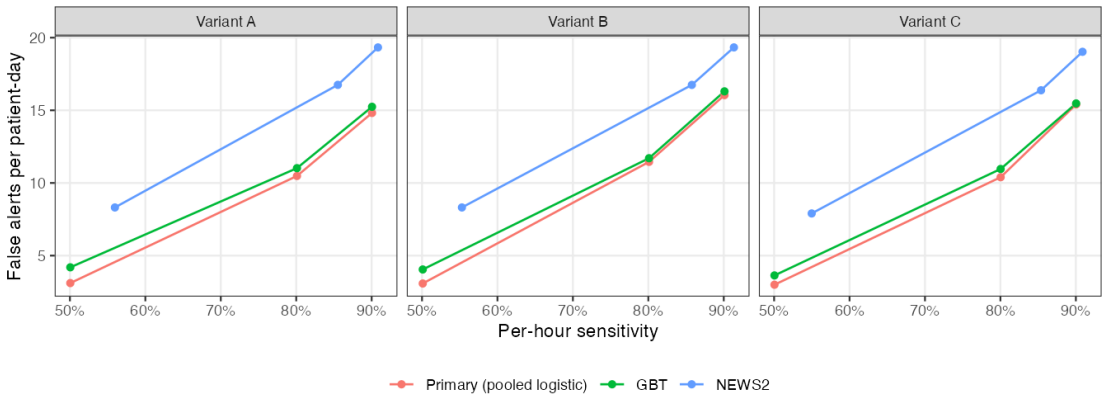


Figure 5.4: False alerts per patient-day against per-hour sensitivity, at sensitivity targets of 50%, 80% and 90%. The relationship is close to linear across the range.

5.4.3 Per-Patient (Event-Level) Detection [C]

Table 5.9 re-expresses the same operating points per patient. Episode sensitivity within the actionable window counts a septic stay as detected if any alert fired between six hours before onset and onset itself. A false-alarm episode is a contiguous run of alert hours that does not overlap the window from six hours before to three hours after onset, so a five-hour alert run counts once rather than five times.

Var	Model	Ep. sens (6 h)	Ep. sens (any pre)	Patient PPV	% stays alerted	FA ep./pt-day
A	Primary	0.880	1.000	0.084	100.0	0.92
A	GBT	0.914	0.989	0.085	97.0	1.37
A	NEWS2	0.968	0.987	0.084	98.2	2.02
A	Cox	0.895	0.924	0.083	93.6	1.94
B	Primary	0.880	0.996	0.085	100.0	1.03
B	GBT	0.904	0.987	0.088	95.8	1.19
B	NEWS2	0.967	0.991	0.086	98.3	2.02
B	Cox	0.902	0.923	0.084	93.8	1.89
C	Primary	0.864	1.000	0.175	100.0	0.95
C	GBT	0.902	0.988	0.181	95.7	1.03
C	NEWS2	0.964	0.988	0.177	98.1	2.03
C	Cox	0.896	0.925	0.177	91.9	1.77

Table 5.9: Per-patient detection at the 80% per-hour sensitivity operating point, on 12,132 test stays of which 1,015 (A), 1,040 (B) and 2,128 (C) are septic. Ep. sens (6 h): fraction of septic stays alerted within [onset − 6 h, onset]. Ep. sens (any pre): alerted at any point before onset. FA ep./pt-day: false-alarm episodes, not alert-hours, per patient-day.

Under the primary label, 0.880 of septic patients are alerted inside the six-hour window and 0.996 at some point before onset. Interpreted alone, patient-level detection obscures the associated alert burden: to achieve it, the models alert on 93.8–100.0 per cent of *all* stays. Patient-level PPV falls to 0.085 under Variant B and 0.175 under Variant C, which are, to three decimal places, the septic fractions of the cohort themselves (1,040/12,132 and 2,128/12,132). Because nearly every stay is alerted, patient-level alerts provide little discrimination beyond the cohort prevalence; under Variants A and C the primary model alerts on 100.0 per cent of the 12,132 test stays.

Per-Hour and Per-Patient, Side by Side

Var	Model	Sens (hr)	Sens (pt)	PPV (hr)	PPV (pt)	FA hr/pt-day	FA ep./pt-day
A	Primary	0.801	1.000	0.0034	0.084	10.48	0.92
A	GBT	0.801	0.989	0.0032	0.085	11.02	1.37
A	NEWS2	0.855	0.987	0.0023	0.084	16.75	2.02
B	Primary	0.801	0.996	0.0032	0.085	11.45	1.03
B	GBT	0.801	0.987	0.0031	0.088	11.70	1.19
B	NEWS2	0.858	0.991	0.0023	0.086	16.75	2.02
C	Primary	0.800	1.000	0.0077	0.175	10.40	0.95
C	GBT	0.800	0.988	0.0073	0.181	10.96	1.03
C	NEWS2	0.854	0.988	0.0052	0.177	16.38	2.03

Table 5.10: The same operating point viewed per person-hour and per patient. Across these rows the two views differ by more than twentyfold in PPV and by roughly tenfold in alarm count.

Per-hour PPV of 0.0032 and per-patient PPV of 0.085 describe one model at one threshold on one test set. The choice of denominator changes the reported PPV by a factor of twenty-six, more than the spread attributed to label definition (H1) or model class (H6), so it is an analyst decision that requires pre-specification.

5.4.4 Lead Time to Onset [C]

Lead time is the interval between the first alert issued at or before onset and the onset hour, over septic stays that were alerted at all.

Var	Model	Median (h)	IQR (h)	≥ 3 h	≥ 6 h
A	Primary	8.0	3–19	0.850	0.593
A	GBT	8.0	3–19	0.832	0.591
A	NEWS2	8.0	4–20	0.868	0.618
A	Cox	8.0	3–19	0.749	0.539
B	Primary	8.0	3–21	0.827	0.585
B	GBT	8.0	3–21	0.819	0.579
B	NEWS2	8.0	4–21	0.866	0.605
B	Cox	7.0	3–21	0.744	0.531
C	Primary	7.0	3–21	0.822	0.574
C	GBT	7.0	3–21	0.822	0.577
C	NEWS2	8.0	4–21	0.865	0.605
C	Cox	7.0	3–21	0.722	0.511

Table 5.11: Lead time at the 80% sensitivity operating point. The ≥ 3 h and ≥ 6 h columns are fractions of *all* septic stays, not only detected ones.

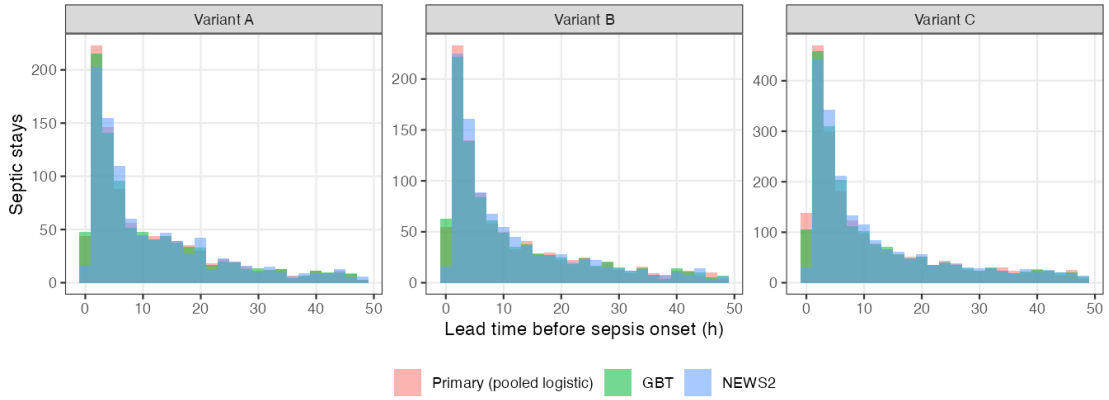


Figure 5.5: Distribution of lead time before onset at the 80% sensitivity operating point, truncated at 48 h. The mode lies in the first four hours, with a long right tail. Because almost every stay is alerted at this operating point, the distribution is not evidence of advance warning.

Median lead time is 7.0–8.0 hours for the primary model, with 0.585 of septic stays alerted at least six hours ahead under the primary label, but nearly every stay is alerted at this operating point. The long lead-time tail (Figure 5.5) therefore includes alerts unrelated to the subsequent onset, and these estimates are not comparable with the 4–12 hour horizons in the literature [19], [21]. They should not be interpreted as evidence of actionable advance warning without controlling alert burden.

5.5 External Validation on eICU-CRD [C]

External validation asks whether a frozen model ranks person-hours as well in a different health system as in the one it was fitted on. Because the Sepsis-3 label must be transported as well as the model, this section reports the labelling inputs first, then discrimination, then operating characteristics.

5.5.1 Coverage of the Sepsis-3 Anchor in eICU-CRD

In eICU-CRD the culture limb of the Sepsis-3 anchor is documented for 1.55% of the analysis cohort (2,824 of 181,589 unit stays), whereas antibiotic exposure, pooled across the `medication` and `infusionDrug` tables, reaches 30.44% of stays. Both limbs are documented at any offset for 232 stays (0.128%); they co-occur inside each variant's antibiotic–culture window for 186, 196 and 211 stays under Variants A, B and C; and adding the organ-dysfunction criterion leaves 27, 31 and 73 labelled stays.

Under Variant B, 31 stays are labelled but 29 events are scored: an onset after a stay's last observed hour is treated as right-censoring, the horizon rule the development cohort applies at 72 hours. Two of Variant B's onsets fall this way and none of Variant A's or C's; every other labelled onset hour receives a prediction from all three models.

Because no onset is observable in a stay without a culture record, the primary external analysis is restricted to the 163,458 person-hours of the microbiology-covered sub-cohort rather than all 7,875,057. Table 5.12 reconciles the resulting event rates with the development cohort's.

Var	Cohort	Person-hours	Events	Events / 1,000 person-h
A	MIMIC-IV temporal test	548,854	1,015	1.849
A	eICU, microbiology-covered	162,070	27	0.167
A	eICU, antibiotic-only anchor	7,656,521	5,589	0.730
B	MIMIC-IV temporal test	548,827	1,040	1.895
B	eICU, microbiology-covered	161,992	29	0.179
B	eICU, antibiotic-only anchor	7,627,689	6,140	0.805
C	MIMIC-IV temporal test	504,828	2,128	4.215
C	eICU, microbiology-covered	159,580	73	0.458
C	eICU, antibiotic-only anchor	7,450,975	9,886	1.327

Table 5.12: Event-rate reconciliation for the development cohort and both external anchors. The antibiotic-only anchor is not Sepsis-3 and its rate is not interchangeable with the other two.

On the restricted cohort the external event rate is 0.179 per thousand person-

hours against MIMIC-IV's 1.895 under the primary label, a ratio of 0.094. Relaxing the culture requirement raises it to 0.805 per thousand (ratio 0.425), suggesting that much of the gap lies in the culture limb. The residual includes case mix, charting frequency and differences in illness severity; Section 5.5.3 narrows it further by removing the anchor altogether. A gap of two or more orders of magnitude between two large adult ICU populations would be far more consistent with missing labelling inputs than with a difference in incidence, and would call for an audit of those inputs before the rate is interpreted.

5.5.2 Discrimination Under Transport

Var	Model	N rows	N events	AUROC (eICU)	AUROC (MIMIC)
A	Primary	162,070	27	0.6913	0.7713
A	GBT	162,070	27	0.7106	0.7455
A	NEWS2	162,070	27	0.5866	0.6413
B	Primary	161,992	29	0.6960	0.7590
B	GBT	161,992	29	0.6815	0.7424
B	NEWS2	161,992	29	0.5259	0.6415
C	Primary	159,580	73	0.7370	0.7695
C	GBT	159,580	73	0.6877	0.7550
C	NEWS2	159,580	73	0.5074	0.6483

Table 5.13: External validation on eICU-CRD (frozen models, no retraining), Sepsis-3 anchor on the microbiology-covered sub-cohort. Each variant carries its own external label set, so the row counts and event counts differ across variants as they do internally.

The point estimate falls under every label. The primary model falls from 0.7713 to 0.6913 (Narrow), from 0.7590 to 0.6960 (Seymour-Standard) and from 0.7695 to 0.7370 (Liberal). Only the Seymour-Standard contrast, the tested contrast of H5, carries an interval, and it covers zero (Section 5.10.7); the Narrow and Liberal figures are point estimates on 27 and 73 events, so the three indicate a direction rather than three established degradations. The Seymour-Standard drop of 0.0630 is about three-quarters of the 0.085 reference estimate reported by Moor et al. [21]. The point estimates show the same cross-label ordering as the internal results, although the low event counts preclude interpreting that ordering.

NEWS2 falls to 0.5259 under the primary label, close to chance. On this arm the fitted models therefore retain more of their advantage over a bedside score, although under the antibiotic-only anchor, with two orders of magnitude

more events, the ordering between the boosted model and NEWS2 does not hold (Section 6.3).

The differing event counts and NEWS2 AUROCs across variants are consistent with variant-specific external labels: NEWS2's score is label-invariant, but its AUROC depends on the outcome it is ranked against.

Every external event count above falls below the interpretability floor (Section 6.3).

5.5.3 The Anchor-Independent Label on the Full External Cohort [D]

The estimates above are computed on the 2,824 stays (1.55%) with documented microbiology. Variant D carries no treatment timestamp, so it can be constructed on the whole cohort. This subsection reports the frozen Variant D models applied to all 181,589 eICU-CRD stays under Protocol Amendment 9.

Model	Internal AUROC	External AUROC	Drop (95% CI)	Events
Primary	0.8493	0.7701	0.0792 [0.074, 0.085]	52,535
GBT	0.9147	0.8400	0.0747 [0.070, 0.078]	52,535
NEWS2	n/a	0.6012	n/a	52,535

Table 5.14: Variant D applied to all 181,589 eICU-CRD stays (5,866,697 person-hours). The target is acute organ dysfunction, not Sepsis-3, so these values are not pooled with the Sepsis-3 arm. eICU-CRD's periodic vitals carry no Glasgow Coma Scale, so the external SOFA delta rests on five components against MIMIC-IV's six. The drop's interval resamples the two databases independently and differences the streams, as for H5 (Protocol Amendment 10). Post-hoc under Protocol Amendment 9; excluded from H5.

The arm scores 52,535 events over 5,866,697 person-hours, against 29 under Variant B, and is the only external estimate that clears the 100-event interpretability floor (Section 6.3).

AUROC drop. The primary model falls from 0.8493 internally to 0.7701 externally, a difference of 0.0792 (95% CI [0.074, 0.085]), and the gradient-boosted comparator from 0.9147 to 0.8400, a difference of 0.0747 ([0.070, 0.078]). Both drops are of the order of the 0.085 degradation Moor et al. [21] report across four databases, and of the same order as the H5 estimate. The event count affects the width rather than the location: both intervals exclude zero and are narrower than a hundredth of a point, whereas H5's substituted contrast, 0.0630 [−0.0215, 0.1563] on 29 events, is compatible with both no degradation and a degradation twice as large. For the deterioration target, transport between these databases

therefore reduces discrimination; this is not evidence about the Sepsis-3 target, a different estimand.

Event rate. Under the Sepsis-3 anchor the external event rate is 0.094 of the development cohort's, a gap of roughly an order of magnitude located largely in the culture limb (Section 5.5.1). Under the anchor-independent label the two databases give 12.007 and 8.955 events per 1,000 person-hours, a ratio of 0.746. The reduction in the event-rate gap after removing the treatment anchor suggests that label transport accounts for a substantial part of the original discrepancy (Section 6.4).

5.5.4 External Clinical Utility on eICU-CRD [C]

Operating characteristics are reported on the same microbiology-covered subcohort and inherit its coverage limitation (Section 5.5.1).

Var	Model	Specificity	PPV	NNE	FA/pt-day	Patient PPV
A	Primary	0.459	25.10×10^{-5}	3,984	12.98	0.00956
B	Primary	0.500	29.65×10^{-5}	3,372	11.99	0.01027
C	Primary	0.484	71.65×10^{-5}	1,395	12.38	0.02516
A–C	NEWS2	0.110	16.65×10^{-5}	6,004	21.35	0.00889

Table 5.15: eICU-CRD operating characteristics at the same 80% target sensitivity, thresholds re-derived on the external cohort. All rows are drawn from the same 2,824 microbiology-covered stays, but each variant carries its own label and therefore its own denominator: 162,070, 161,992 and 159,580 person-hours on 27, 29 and 73 event hours under A, B and C (Table 5.13). The NEWS2 row is scored under Variant B. Re-deriving thresholds externally excludes the further loss a threshold transported from MIMIC-IV would incur.

Operating characteristics on eICU-CRD are substantially worse than internal ones at the same target sensitivity. Reaching 80 per cent sensitivity requires 1,395 to 3,984 alerted person-hours per true event hour, at 11.99–12.98 false-alert hours per patient-day, roughly one alert every two hours per patient. NEWS2 cannot reach the target without alerting on nearly every person-hour (specificity 0.110, 21.35 false-alert hours per patient-day). Patient-level PPV of 0.01027 is again the cohort septic fraction (29/2,824), because the model alerts on essentially every stay. At this event count the contributions of model and label cannot be separated, since PPV, NNE and false-alarm rate all scale with the event rate; these figures are therefore reported as an upper bound on transferability and support no separate conclusion.

5.6 Label-Anchor Attribution [E]

Variants A, B and C vary the label within Sepsis-3 and share the culture-plus-antibiotic anchor, so H1 cannot distinguish an effect of reading the definition differently (definitional) from an effect of choosing a different construct (empirical). This section separates the two by labelling each limb of the Sepsis-3 conjunction separately (Section 3.6; post-hoc, Protocol Amendment 2, exploratory family E3).

5.6.1 The Anchor Determines Every Onset Time

Under Variants A, B and C the onset time is the suspicion time t_{susp}; the SOFA criterion determines only *whether* a stay is labelled, not *when*. Variant E, which removes the SOFA criterion, is therefore a strict superset of Variant B: every one of B's 5,887 labelled stays is also labelled by E, and in 100.0% **of them the onset hour is identical**. Adding the organ-dysfunction requirement discards 11,364 of E's 17,251 labelled stays, two thirds, and moves not one onset by a single hour.

Var	Label	Onsets (≤72 h)	%	κ vs B	Sens. to B	Exact onset match
A	Narrow	5,662	8.7	0.267	0.325	97.5%
B	Seymour-Std	5,887	9.0	n/a	n/a	n/a
C	Liberal	11,531	17.7	0.344	0.625	98.6%
E	Anchor-Only	17,251	26.5	0.432	1.000	100.0%
D	Deterioration	27,209	41.8	0.057	0.555	6.5%
F	Sepsis-3 Early	6,074	9.3	0.983	1.000	73.7%

Table 5.16: Onset agreement with Variant B across all five comparison variants, on the 65,078 shared stays. Counts are restricted to onsets inside the 72 h modelling horizon so that Variant D, whose onsets exist only within the panel, is comparable. "Sens. to B" is the fraction of B's labelled stays that the variant also labels; "exact onset match" is computed among stays labelled by both.

Within Sepsis-3 (Table 5.16, top block), variants that agree a stay is septic almost always agree on the hour: 97.5% for A, 98.6% for C. The pre-registered label variance therefore concerns *which patients* rather than *when*, since timing is fixed by the treatment anchor in all three. That disagreement is substantial: A and B share only 0.325 of B's septic stays ($\kappa = 0.267$) despite near-identical prevalence.

A label built from physiology alone (Variant D) agrees with Sepsis-3 at $\kappa = 0.057$, barely distinguishable from chance, and matches B's exact onset hour in only 6.5% of the stays both label, with a median absolute shift of 9.0 hours (IQR −11 to +7). The anchor-independent label thus identifies largely different patients at different times.

Variant F keeps the construct and the patients but changes the onset timing. It labels every stay B labels ($\kappa = 0.983$, the shortfall being the 187 stays discussed above whose onset only enters the horizon under the earlier rule) yet agrees with B on the exact hour in just 73.7% of them: two defensible readings of the same definition place about a quarter of Sepsis-3 onsets at a different hour on the same patients.

5.6.2 Discrimination Under Each Limb

	AUROC				AUPRC			
Model	**B**	**E**	**D**	**F**	**B**	**E**	**D**	**F**
Primary	0.7590	0.7697	0.8493	0.7951	0.0096	0.0321	0.0767	0.0131
GBT	0.7424	0.7665	0.9147	0.7458	0.0087	0.0334	0.1362	0.0090
NEWS2	0.6415	0.6650	0.6810	0.6409	0.0032	0.0129	0.0243	0.0034
Cox	0.6444	0.6854	0.7868	0.6528	0.0033	0.0147	0.0382	0.0035
qSOFA	0.6091	0.6243	0.7308	0.6167	0.0028	0.0105	0.0312	0.0031
SIRS	0.6114	0.6407	0.5549	0.5983	0.0030	0.0127	0.0146	0.0029
Events	1,040	3,073	4,774	1,082				

Table 5.17: Discrimination under each limb of the Sepsis-3 conjunction and under standard onset timing (temporal test set, unrestricted window). B = Sepsis-3 (anchor **and** organ dysfunction); E = suspicion anchor only; D = organ dysfunction only, anchor-independent; F = Sepsis-3 with onset at $\min(t_{\text{susp}}, t_{\text{SOFA}})$. AUPRC is reported alongside AUROC because the four labels have very different prevalences.

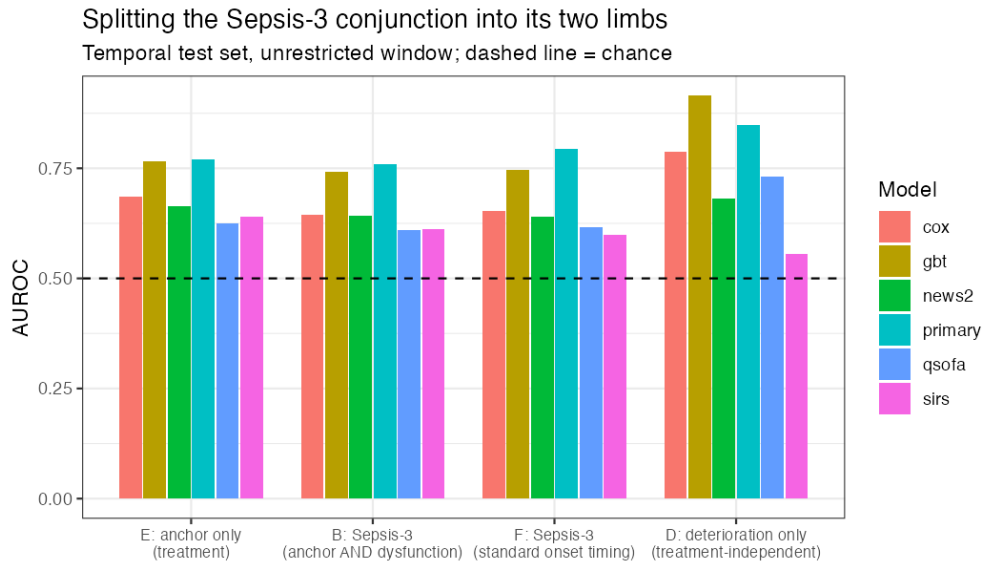


Figure 5.6: Splitting the Sepsis-3 conjunction. The two limbs and the intact conjunction sit close together; only the anchor-independent deterioration label separates from them, and its level is not comparable with the others (Section 6.4). The dashed line is chance.

Anchor-attributable fraction. The share of Sepsis-3's above-chance discrimination that the treatment anchor reproduces on its own (Equation 3.1), with the ratio bootstrapped end to end, is

$$\phi_{\text{anchor}} = 1.041 \;\; [0.990,\, 1.097] \quad \text{(primary)}, \qquad 1.099 \;\; [1.037,\, 1.164] \quad \text{(GBT)}.$$

For the primary model the interval **includes 1**, and the underlying contrast is not significant: E against B is $\Delta = 0.0107$ $[-0.0029,\, 0.0238]$, $p_{\text{BH}} = 0.139$ in family E3. For GBT the interval sits just above 1 ($\Delta = 0.0241$, $p_{\text{BH}} = 0.003$), a small but detectable effect.

On this cohort the treatment anchor alone is therefore about as predictable as Sepsis-3 itself. Dropping the organ-dysfunction requirement nearly triples the event count (3,073 against 1,040) and leaves discrimination essentially unchanged: the SOFA gate selects which stays are labelled without selecting the more separable ones.

Anchor-independent label. Variant D differs substantially: $\Delta = 0.0904$ [0.0733, 0.1075] for the primary model and 0.1723 for GBT, both significant in family E3. Its AUPRC gap is larger still (0.0767 against 0.0096), which matters more at this prevalence. Because five of D's six SOFA components are model inputs, part of this separation is self-fulfilling (Section 3.6).

Onset timing. Variant F, whose label is Sepsis-3 unmodified in membership, is also significantly more separable than B ($\Delta = 0.0361$ [0.0257, 0.0461], $p_{\mathrm{BH}} = 0.002$). Its AUPRC is *higher* than B's despite a comparable event count, so this is not a prevalence artefact. Onsets placed when organ dysfunction is first detectable are more separable from surrounding at-risk hours than the same onsets placed at the clinician's action. This is consistent with, but does not demonstrate, the physiological signal preceding the clinical response, since predictors are binned to the onset hour (Section 6.7).

Summary. Variant E produces discrimination similar to Variant B, and Variant F, Sepsis-3 under standard onset timing, produces higher discrimination for the primary model. Variant D produces substantially higher discrimination, but its AUROC is not directly comparable, because the label is derived from variables that overlap with the model inputs. Within Sepsis-3, reduced discrimination is therefore associated more with an onset placed at the clinician's action than with the conjunction itself.

5.6.3 The Empty Pre-Treatment Window Is a Property of the Onset Rule

Under the onset rule of Variants A–C the pre-treatment window is empty by construction (Section 5.10.5). Table 5.18 reports its contents under alternative rules.

Var	Label	Pre-treatment person-h	Onsets	AUROC (primary / GBT)
B	Seymour-Std	179,003	0	undefined
E	Anchor-Only	179,003	0	undefined
F	Sepsis-3 Early-Onset	178,303	71	0.8081 / 0.8018
D	Deterioration	147,299	1,362	0.8498 / 0.9360

Table 5.18: Events in the treatment-anchored (pre-treatment) evaluation window. E's and B's zeros are entailed by Proposition 5.1. Variant F holds the Sepsis-3 conjunction and the labelled stays fixed and changes only the onset timestamp to the standard $\min(t_{\mathrm{susp}}, t_{\mathrm{SOFA}})$; Variant D removes the infection anchor altogether.

The zeros for E and B carry no information: E's onset is the anchor, and B is a subset of E with identical onset times.

Variant F. Variant F shares Variant B's windows, SOFA threshold, evaluation window and all 6,447 septic stays; only the timestamp differs. With that change, 1,738 onsets move earlier (27.0%; median −14.0 h, largest −48.0 h), 419 land before the first antibiotic or culture across the cohort, and 71 of those fall in the

temporal test set. Sepsis-3 onsets therefore do precede clinical action under the standard timing, and the primary model identifies those hours at 0.8081 AUROC on the onset-hour estimand; whether they are predictable *in advance* cannot be assessed, because predictors are binned to the onset hour (Limitation 3). Because F's membership is unmodified Sepsis-3, this result does not depend on the Variant D caveat.

Event count. Agreement between F and B is 99.7% ($\kappa = 0.983$), and the discrepancy is one-directional. Within the 72-hour prediction horizon F labels 6,074 stays against B's 5,887, and every one of B's is among them: $F \supseteq B$. The 187 additional stays are those whose t_{susp} falls after hour 72, so no model here ever sees a labelled event for them, but whose SOFA rise occurred inside the window. The onset rule therefore determines how many events a real-time model can be trained and tested on, as well as when they are recorded.

Variant D. Variant D records 1,362 pre-treatment onsets, more than F, as expected of a label without the infection requirement, identified at 0.8498 AUROC (0.9360 for GBT). The caveat of Section 3.6 applies to that level but not to the event count.

5.6.4 Exploratory Family E3: Intervals and Correction

Var	Model	AUROC	95% CI	Δ vs B	95% CI (Δ), q_{BH}
E	Primary	0.7697	[0.761, 0.779]	0.0107	[−0.0029, 0.0238], $q = 0.139$
E	GBT	0.7665	[0.757, 0.775]	0.0241	[0.0096, 0.0375], $q = 0.003$
D	Primary	0.8493	[0.844, 0.855]	0.0904	[0.0733, 0.1075], $q = 0.002$
D	GBT	0.9147	[0.911, 0.918]	0.1723	[0.1545, 0.1885], $q = 0.002$
F	Primary	0.7951	[0.780, 0.810]	0.0361	[0.0257, 0.0461], $q = 0.002$
F	GBT	0.7458	[0.730, 0.762]	0.0033	[−0.0087, 0.0148], $q = 0.559$
B	Primary	0.7590	[0.7434, 0.7749]	n/a	n/a
B	GBT	0.7424	[0.7265, 0.7583]	n/a	n/a

Table 5.19: Exploratory family E3: AUROC under each limb of the Sepsis-3 conjunction with 95% stay-level cluster-bootstrap intervals, and the paired difference against Variant B corrected by Benjamini–Hochberg across the family of $m = 6$ contrasts. All variants are labelled on the same test stays, so each difference is evaluated within replicate.

Every contrast is positive, and two of the six do not survive correction. The first, E against B for the primary model ($q = 0.139$), is the basis of the anchor-attribution

result above; the corresponding GBT contrast is significant but small. The second is F against B for GBT ($\Delta = 0.0033$, 95% CI -0.0087–0.0148, $q = 0.559$): re-timing the onset separates the labels for the primary model but not for the boosted one, so the gain in separability from re-timing is model-specific. The pre-treatment event counts of Section 5.6.3 do not depend on this contrast.

These intervals quantify sampling uncertainty only. Prevalence differs by a factor of 3.0 (E) and 4.6 (D) relative to B, and Variant D's label overlaps the model's inputs, so the intervals establish which orderings are stable, not that the labels measure comparable constructs.

5.7 Action-Derived Features: An Ablation [E]

8 of the 22 covariates record clinician actions: vasopressor exposure and the per-test indicators of whether a laboratory value was obtained in a given hour. Cross-label coefficient instability is concentrated in these features (Section 5.10.4). This section tests whether the models depend on them by removing them and refitting, with the specification otherwise unchanged (Section 3.14.5).

Var	Model	AUROC full	AUROC ablated	Δ	$\text{Utility}_{\max}$ full	$\text{Utility}_{\max}$ ablated
A	Primary	0.7713	0.7633	−0.0080	0.026	0.013
A	GBT	0.7455	0.7418	−0.0037	0.038	0.037
B	Primary	0.7590	0.7514	−0.0076	0.025	0.014
B	GBT	0.7424	0.7359	−0.0065	0.031	0.032
C	Primary	0.7695	0.7602	−0.0093	0.095	0.092
C	GBT	0.7550	0.7516	−0.0034	0.111	0.095

Table 5.20: Discrimination and best-threshold Utility Score with and without the 8 action-derived features. Δ is a paired difference; Table 5.22 carries its interval. **[E] exploratory.**

5.7.1 Coefficient Stability After Ablation

Table 5.21 compares cross-label coefficient stability before and after ablation.

Specification	Covariates	Unstable	% unstable
Full	22	6	27.3
Ablated	14	1	7.1

Table 5.21: Cross-label coefficient instability before and after removing the action-derived features, under the identical stability rule (sign reversal or magnitude change $\geq 2\times$ across Variants A, B and C). **[E] exploratory.**

Removing 8 of the 22 covariates changes AUROC by −0.0076 for the primary model (95% CI [−0.0131, −0.0022]) and −0.0065 for GBT ([−0.0130, −0.0001]) under the primary label, and by no more than −0.0093 for either model under any variant (Table 5.22). 5 of the six paired contrasts exclude zero, so the cost is small, below 0.01 AUROC everywhere, but not absent; no practical-equivalence margin was pre-registered.

Var	Model	AUROC full	Δ (ablated − full)	95% CI
A	Primary	0.7713	−0.0080	[−0.0123, −0.0034]
A	GBT	0.7455	−0.0037	[−0.0086, 0.0012]
B	Primary	0.7590	−0.0076	[−0.0131, −0.0022]
B	GBT	0.7424	−0.0065	[−0.0130, −0.0001]
C	Primary	0.7695	−0.0093	[−0.0129, −0.0059]
C	GBT	0.7550	−0.0034	[−0.0061, −0.0007]

Table 5.22: Paired ablated-minus-full AUROC contrasts with 95% stay-level cluster-bootstrap intervals (Protocol Amendment 10), differenced within replicate. **[E] exploratory; enters no corrected family.**

The cross-label unstable count falls from 6 of 22 covariates to 1 of 14 (7.1%). This is not a subset relation: the single remaining flag falls on `lactate`, which the full fit does not flag, because the refit re-estimates the remaining coefficients. Together with the by-kind split of Section 5.10.4 (50.0% unstable among action-derived features against 14.3% among physiological ones), these results indicate that the features recording clinician behaviour contribute little to discrimination and carry most of the cross-label instability (Section 6.5).

5.8 Model Diagnostics

Two checks on the comparator models are collected here: a test of the Cox comparator's proportional-hazards assumption, and the predictors the gradient-boosted trees used, which bear on Sections 5.7 and 5.10.4.

5.8.1 Proportional-Hazards Testing [C]

The Schoenfeld residual test on the Cox comparator (Variant B) rejects proportional hazards ($\chi^2 = 126.2$, df $= 12$, $p = 3.6 \times 10^{-21}$). Because the comparator uses hourly covariates but time-constant coefficients, the test rejects the hazard-ratio structure (Section 3.10.3). 8 of the 12 covariates violate the assumption individually at $\alpha = 0.05$, led by GCS, bilirubin, respiratory rate and creatinine. These

are per-covariate tests reported without multiplicity adjustment; the global test alone is sufficient for the pre-registered check, and the smallest p-values survive Bonferroni correction over the covariate set by many orders of magnitude. The Cox coefficients therefore cannot be read as time-constant hazard ratios on these data.

5.8.2 GBT Feature Importance [E]

Feature	Gain	Cover	Frequency
Temperature (°C)	0.145	0.144	0.083
Resp. rate slope (6 h)	0.138	0.085	0.075
MAP slope (6 h)	0.087	0.098	0.068
GCS	0.072	0.069	0.055
Heart-rate slope (6 h)	0.083	0.069	0.084
Creatinine	0.058	0.076	0.065
WBC	0.065	0.086	0.078
Heart rate (level)	0.054	0.061	0.085
Lactate	0.044	0.047	0.054
MAP (level)	0.050	0.038	0.084

Table 5.23: Top-10 GBT features by information gain (Variant B). Three of the five most informative predictors are six-hour trend features.

The prominence of six-hour slope features is consistent with Lauritsen et al.'s [10] concern: the GBT model draws much of its signal from trends, so its performance depends on the length of the history window used for feature construction, a choice rarely reported or varied in the comparable literature. Heart rate contributes through both its level and its slope.

5.9 Equity Analysis [E]

5.9.1 What the Subgroup Strata Are

Two properties of MIMIC-IV's administrative record govern how this section is read.

Strata are documentation categories. `race` in MIMIC-IV is a single code assigned at registration. Its levels mix broad categories with more granular ones drawn from the same population: "White" and "White – Other European" are

separate levels, as are "Asian" and "Asian – Chinese". A stay carries exactly one code, so the levels partition the cohort, but they do not partition it into disjoint *populations*: the reference level "White" is a residual category comprising stays not assigned a more specific European code, and a contrast against it is a contrast against that residual. "Unknown" and "Unable to obtain" record the *absence* of an answer; they are retained because the completeness of the record is informative (Section 5.9.3) but are not demographic groups.

Event floor. The interpretation floor is 100 sepsis onsets, the same as the external-validation floor (Section 6.3; [12], [46]). Strata below it are estimated and tabulated but **not interpreted**, and are excluded from statements about spread and from family E1, which therefore contains 6 contrasts. Of the racial strata, 2 clear the floor.

5.9.2 Performance Disparities by Subgroup [E]

Racial stratum	Rows	Sepsis	GBT AUROC (95% CI)	Primary AUROC (95% CI)
White	317,688	597	0.739 (0.719–0.761)	0.756 (0.736–0.777)
Unknown	107,158	235	0.773 (0.742–0.803)	0.773 (0.740–0.804)
Black / African American†	31,716	58	0.749 (0.676–0.817)	0.759 (0.685–0.825)
Other†	22,825	35	0.669 (0.571–0.756)	0.724 (0.631–0.806)
White – Other European†	11,768	18	0.672 (0.566–0.780)	0.730 (0.600–0.846)
Asian†	6,543	16	0.765 (0.625–0.886)	0.841 (0.726–0.929)
Hispanic / Latino – Puerto Rican†	6,462	16	0.624 (0.448–0.792)	0.702 (0.546–0.845)
Asian – Chinese†	5,386	13	0.601 (0.398–0.802)	0.701 (0.511–0.869)

Table 5.24: AUROC by racial stratum (Variant B, temporal test) with 95% stay-level cluster-bootstrap intervals, stratified within stratum, $B = 2{,}000$, ordered by exposure. † marks strata with fewer than 100 sepsis onsets, which are neither interpreted nor entered into family E1 (Section 5.9.1). **[E] exploratory.**

The spread of subgroup point estimates is not commensurable with the model-class and label spreads of Section 5.10: those are whole-cohort contrasts with one factor varied, whereas a subgroup range is the extreme order statistic of separate estimates on disjoint slices with few events, and is biased upward by construction even when every underlying value is identical. The apparent spread largely reflects how few events the small strata contain. Of the racial strata, 2 carry at least 100 onsets; the remaining 6 are estimated on counts between 13 and 58, and their intervals are correspondingly wide.

Table 5.25 therefore reports contrasts against the reference level on the same bootstrap replicates.

Variable	Stratum	Sepsis	Model	ΔAUROC vs. reference (95% CI)	p	q_{BH}
Insurance	Private	251	GBT	+0.050 (+0.012–+0.088)	0.011	0.066
Insurance	Private	251	Primary	+0.039 (+0.003–+0.077)	0.033	0.099
Race	Unknown	235	GBT	+0.033 (−0.003–+0.069)	0.067	0.134
Race	Unknown	235	Primary	+0.017 (−0.022–+0.055)	0.387	0.580
Insurance	Medicaid	183	Primary	+0.008 (−0.036–+0.052)	0.683	0.819
Insurance	Medicaid	183	GBT	−0.003 (−0.049–+0.046)	0.888	0.888

Table 5.25: Exploratory family E1 in full: all 6 subgroup AUROC contrasts clearing the 100-event floor, across race, language and insurance and for both fitted models, ordered by adjusted p-value. q_{BH} is the Benjamini–Hochberg adjusted value over the whole family. **No contrast is significant**; two unadjusted intervals exclude zero. The reference level is the largest stratum of its variable (Section 5.9.1). **[E] exploratory.**

Applying the floor leaves 6 contrasts in family E1 across all three equity variables and both models, and **none survives false-discovery correction**. The smallest adjusted value in the family belongs to a contrast on insurance rather than on race: Private insurance against the Medicare reference, 0.050 AUROC for the gradient-boosted model, with a bootstrap interval of [0.012, 0.088] and $q_{\mathrm{BH}} = 0.066$.

Its interval excludes zero but its adjusted p-value does not clear the threshold. The corrected verdict is the one read here, because the protocol (§9) fixed the scope of the equity analysis but not the levels, reference or direction of any contrast; the contrast is reported but not claimed.

Of the racial strata, only two carry enough events to be interpreted, and one of them is *Unknown*, a missingness category (Section 5.9.1). **The study therefore contains no adequately powered comparison between two racial strata that both record a demographic answer.** The study is **underpowered to detect subgroup disparities**: even above the event floor, the intervals are compatible both with a stratum being served materially worse than the reference and with its being served slightly better.

Wang et al.'s [11] finding of degraded performance for Asian, Hispanic and Spanish-speaking patients concerns mortality prediction (Section 2.9), and the corresponding strata here fall below the event floor, so it is neither confirmed nor contradicted. The point estimates for these strata lie below the reference under GBT, in the direction Wang et al. report, but are not interpreted.

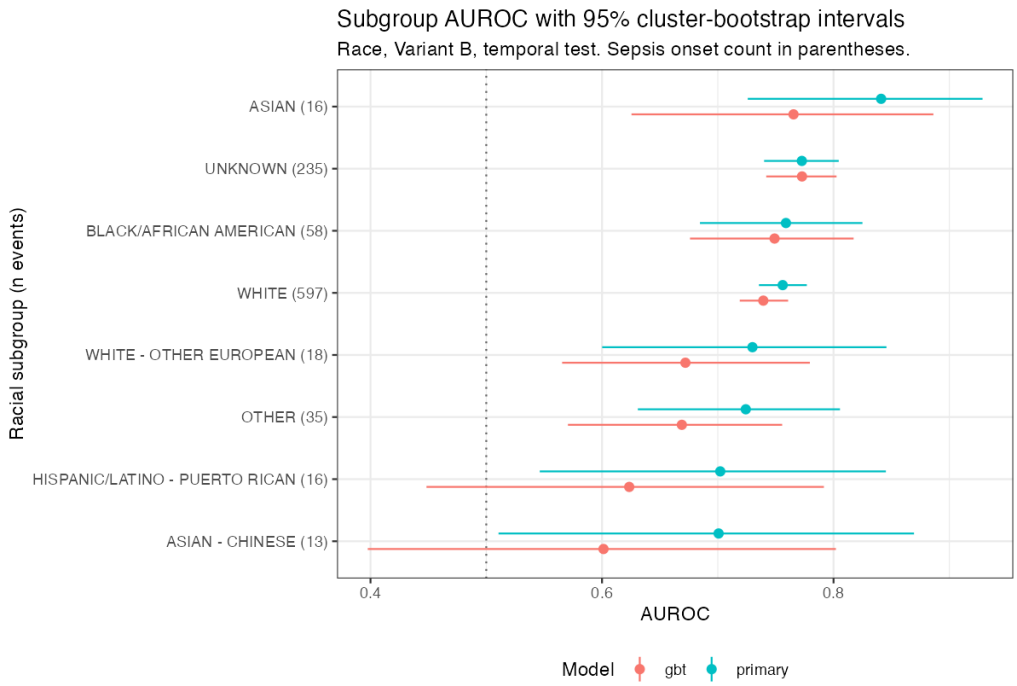


Figure 5.7: AUROC by racial subgroup with 95% stay-level cluster-bootstrap intervals (Variant B, temporal test); sepsis onset count in parentheses. Interval width is governed almost entirely by the stratum event count, and several intervals reach the 0.5 chance line.

5.9.3 Label Sensitivity by Subgroup [E]

Racial group	Stays	A (%)	B (%)	C (%)	Range (pp, 95% CI)
White	40,212	9.4	9.6	18.6	9.3 (8.9–9.7)
Unknown	7,264	11.6	12.8	23.9	12.3 (11.3–13.3)
Black / African American	4,929	9.3	9.1	19.0	9.9 (9.0–11.1)
Other	2,254	8.8	9.8	17.5	8.7 (7.1–10.2)
Unable to obtain	1,629	9.7	11.1	22.5	12.8 (10.7–14.9)
White – Other European	1,589	9.4	7.7	16.6	8.9 (7.2–10.7)
Asian	799	10.5	11.5	21.3	10.8 (8.3–13.8)
Asian – Chinese	721	10.1	11.1	19.0	8.9 (6.4–11.9)

Table 5.26: Percentage of stays labelled as septic by racial subgroup and label variant, with a 95% stratified cluster-bootstrap interval on the A–C range. The fraction roughly doubles from Variant A to C across all groups. Per-variant intervals are in `12_subgroup_labelsens_ci.csv`. **[E] exploratory.**

The fraction of patients labelled as septic changes by 8.7–12.8 percentage points across label variants within each racial stratum. Unlike discrimination, this contrast survives correction, because it is a stay-level proportion computed on the full 65,078-stay cohort rather than on the few hundred test-window onsets. No event floor is therefore applied; the construct limitation of Section 5.9.1 still applies.

Racial group	**Stays**	**Δ range vs. White (pp, 95% CI)**	p	q_{BH}
Unknown	7,264	+3.01 (+1.93–+4.06)	0.001	**0.006**
Unable to obtain	1,629	+3.49 (+1.38–+5.55)	0.002	**0.009**
Black / African American	4,929	+0.62 (−0.35–+1.87)	0.186	0.347
Asian	799	+1.48 (−1.04–+4.50)	0.249	0.407
Other	2,254	−0.63 (−2.26–+0.98)	0.425	0.510
White – Other European	1,589	−0.41 (−2.10–+1.41)	0.635	0.714
Asian – Chinese	721	−0.40 (−3.02–+2.61)	0.675	0.714

Table 5.27: Difference in label sensitivity (A–C range) against the reference group (White). q_{BH} is adjusted over exploratory family E2, all 18 label-sensitivity contrasts across race, language and insurance. Two racial contrasts survive at $q < 0.05$; across the full family, 4 do (adding Spanish-speaking patients, +2.61 pp, $q = 0.006$, and privately insured patients, −1.76 pp, $q = 0.006$). **[E] exploratory.**

2 racial strata are significantly more label-sensitive than the reference after false-discovery correction, and *both of them are missingness categories*: Unknown (+3.01 pp, $q = 0.006$) and Unable to obtain (+3.49 pp, $q = 0.009$). No stratum recording an actual demographic answer differs from the reference. Across the full family, Spanish-speaking patients are more label-sensitive than English-speaking patients (+2.61 pp, $q = 0.006$), and privately insured patients are *less* label-sensitive than Medicare patients (−1.76 pp, $q = 0.006$).

The significant contrasts involve documentation-related categories and strata of language and insurance, and may reflect differences in record completeness; this mechanism was not tested (Limitation 9). Of the two equity analyses, only this one is precise enough to support a claim, and reporting either without the other would invert which conclusion the data support.

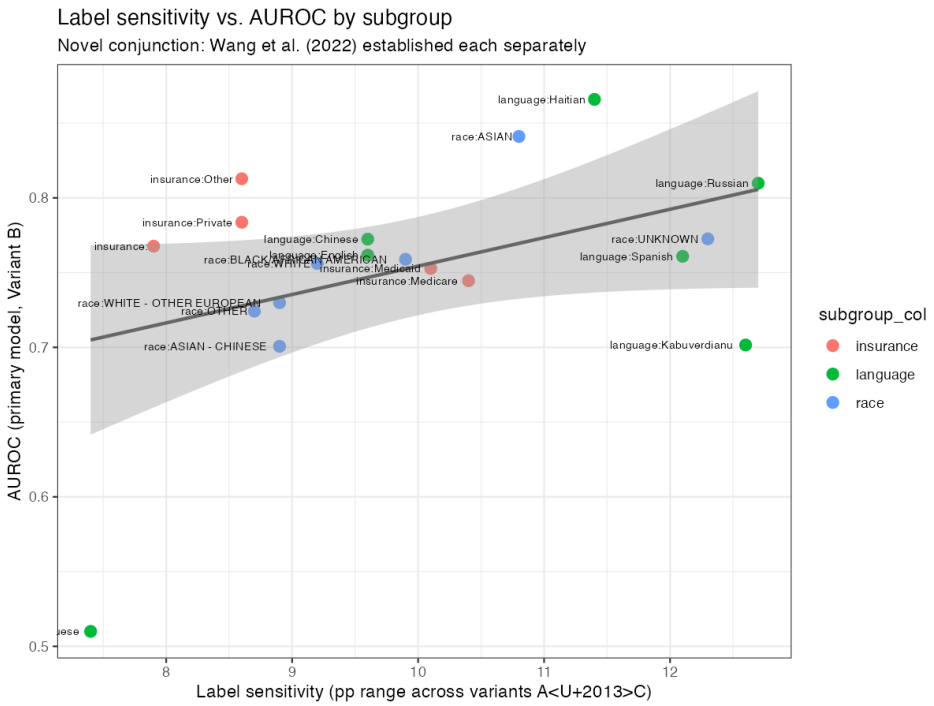


Figure 5.8: Label sensitivity (fraction labelled septic) vs. AUROC by racial subgroup, across all three label variants. No association is tested, and the AUROC uncertainty of Figure 5.7 is not shown. **[E] exploratory, visual only.**

5.10 Variance Decomposition and Hypothesis Verdicts [C]

This section assembles the preceding estimates into the variance decomposition and the six hypothesis verdicts. The label spread is read across the three variants of Section 5.3, the model-class spread across the model rows of the same table, the split contrast against the random-split arm, and the transport contrast against the external estimates of Section 5.5. Each is an AUROC contrast with an interval from the same stay-level cluster bootstrap.

Spreads as ranges. Each spread is a *range*: the largest AUROC in a set minus the smallest. The label spread is a range over three variants and the model-class spread over six specifications. Because a range grows with the number of values even when the underlying quantities are identical, a range over six is expected to exceed one over three by roughly half again, a bias present in every bootstrap replicate (the same reason the subgroup range is not treated as an effect size in

Section 5.9).

The ordering does not depend on this statistic. The model-class range spans a structural gap rather than sampling scatter, between the two fitted models and the four rule-based and Cox comparators, and *any* single contrast across that gap is larger than the whole label range. The narrowest of them, GBT against the Cox model, is 0.7424 against 0.6444, which exceeds the label range by a factor of eight and involves no order statistic. An asymmetry of about half again cannot account for 0.1499 against 0.0123.

A variance-components partition is not used, for two reasons: the AUROCs are computed on the same test rows, so the design supplies no suitable error term, and §7 of the pre-registration fixed the estimand as the *spread*. The range is reported as a descriptive summary with a bootstrap interval.

Two of the six hypotheses admit no test statistic, for reasons of design; both still occupy a slot in the correction (Section 5.10.2).

5.10.1 The Decomposition Table

Factor (others fixed)	**ΔAUROC**	**95% CI**	**H**
Model class (label B, temporal)	0.1499	0.134–0.171	n/a
Label definition (primary, temporal)	0.0123	0.003–0.032	H1
Split method (primary, label B)	−0.0165	−0.0372–0.0040	H4
External transport (primary, label B)	0.0630	−0.0215–0.1563	H5
GBT − primary (label B, temporal)	−0.0165	−0.0308–−0.0012	H6

Table 5.28: Variance decomposition with 95% stay-level cluster-bootstrap intervals. The name is the pre-registration's (§1) and the quantities are ranges, not variance components: they neither estimate nor partition a variance, and they are not on a common denominator, so they cannot be summed or expressed as shares of a total. Each row varies one factor with the others held fixed, over the alternatives named. The first row tests no hypothesis; H1 contrasts it with the label range. H6, below the rule, is the contrast between the two *flexible* estimators alone. **[C] confirmatory.**

The four factors differ widely in effect. **The spread across the six evaluated model specifications is the largest** (0.1499). That set runs from a 29-term

dynamic hazard model to qSOFA and differs in features, event handling and fitting as well as model form (Section 3.10), so the claim concerns this comparison, not model architecture in general. **External transport has the second-largest point estimate** (0.0630), but its interval is the widest in the table and covers zero, so its rank is not established. The label moves discrimination by 0.0123 and the split by −0.0165, both small, and the split's interval covers zero. The pre-registered expectation that the four factors would be comparable is not supported. Three qualifications apply:

- Between the two *flexible* estimators, fitted on identical covariates, the gap is 0.0165, an order of magnitude below the six-specification range and of the same order as the label's (H6; Section 5.10.8).
- The label factor is small *in discrimination* only. The variants disagree about which patients are septic ($\kappa = 0.267$, A vs B), so they define different cohorts, prevalences and denominators for every downstream clinical claim.
- The split interval [−0.0372, 0.0040] excludes the pre-registered +0.02 optimism threshold while covering zero. The transport row rests on 29 events, and its interval [−0.0215, 0.1563] contains both zero and a degradation twice the size of the label effect.

5.10.2 Corrected Confirmatory Verdicts

Table 5.29 gives the verdict for each of the six pre-registered hypotheses, with its estimate and, for the four tested contrasts, the interval and Holm-adjusted p-value.

H	Contrast θ	Est.	95% CI	p_{Holm}	Verdict
H1	Label spread − model spread	−0.1375	−0.1623–−0.1083	1.000	NOT confirmed
H2	Unstable covariates (count)	6	n/a	n/a	Descriptive
H3	Pre-treatment onsets (count)	0	n/a	n/a	Definitional
H4	Random − temporal AUROC	−0.0165	−0.0372–0.0040	1.000	NOT confirmed
H5†	Internal − external AUROC†	0.0630	−0.0215–0.1563	0.551	Not evaluable as pre-registered; substituted AUROC contrast inconclusive
H6	GBT − primary AUROC	−0.0165	−0.0308–−0.0012	1.000	H6 not rejected

Table 5.29: The confirmatory family. One-sided cluster-bootstrap tests against each hypothesis's pre-registered threshold ($\theta_0 = 0$ for H1, and for the amended AUROC transport contrast shown in H5's reserved slot; 0.02 for H4 and H6), Holm–Bonferroni corrected at family size $m = 6$ and FWER 0.05. H2 and H3 reserve a slot without contributing a p-value, which makes the correction on the other four stricter. H3's count is entailed by the onset rule (Proposition 5.1). **The Verdict column reports the pre-registered test only**: H6's non-rejection is not evidence of equivalence, and its one-sided non-inferiority is read off the interval (Section 5.10.8). The smallest reportable p-value is $1/(B+1)$. **[C] confirmatory, except** †: H5's pre-registered estimand is **not evaluable**, and the row shows an **[A] amended, post-hoc** substitute contrast occupying its correction slot (Section 5.10.7).

H1 and H4 are not confirmed: their adjusted p-values are 1.000 and 1.000, and their point estimates lie on the opposite side of zero from the one predicted. H6 is not rejected and its interval supports one-sided non-inferiority but not two-sided equivalence. H2 is descriptive, and H3 and H5 were not evaluable as pre-registered; the contrast substituted for H5 is inconclusive. The following subsections give each verdict with its estimate; the underlying results are in the sections cited.

5.10.3 H1: Label Dominance (Not Confirmed)

H1 [C]: Spread in AUROC across label variants ≥ spread across model classes.
Point estimates: label spread = 0.0123; model spread = 0.1499.
Contrast: $\theta = -0.1375$ (95% CI −0.1623–−0.1083), Holm-adjusted $p = 1.000$. **H1 NOT CONFIRMED; the estimate lies in the direction opposite to the one hypothesised.**

Across the six evaluated specifications, model choice affected AUROC more than the three Sepsis-3 variants did: the difference of spreads excludes zero. The specification spread runs from rule-based scores (qSOFA 0.6091, SIRS 0.6114, NEWS2 0.6415) through the Cox comparator (0.6444) to the two flexible esti-

mators (primary 0.7590, GBT 0.7424). The variants nevertheless changed cohort membership substantially: A and B agree on only 88.1% of stays ($\kappa = 0.267$) despite near-identical prevalence (Section 5.6).

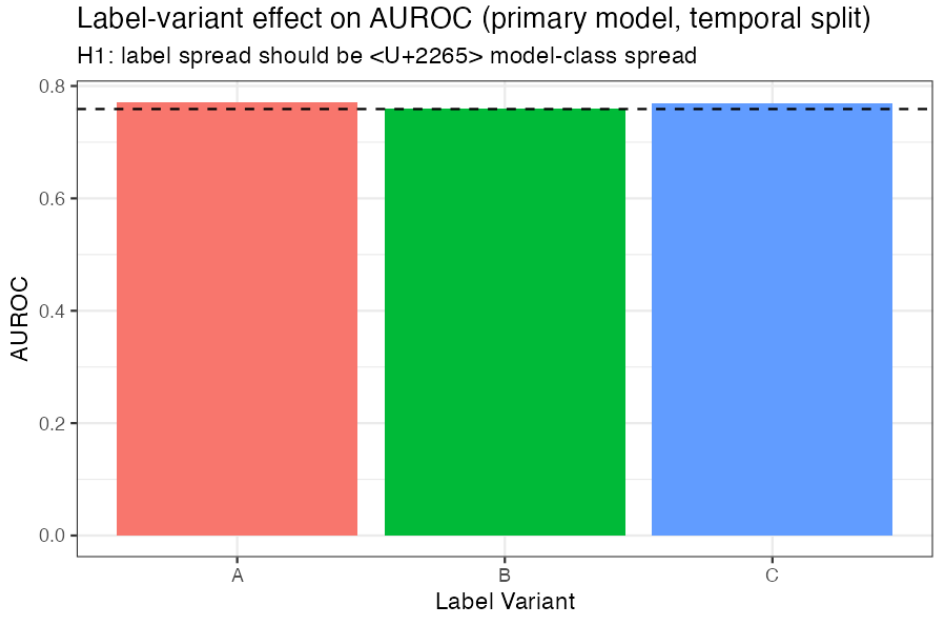


Figure 5.9: AUROC by label variant for each model (temporal split). Both quantities are *ranges*, largest minus smallest: the range within each model group is the label contribution (0.0123, over three variants) and the range across groups is the spread across the six model specifications (0.1499), larger by roughly an order of magnitude (Section 5.10).

5.10.4 H2: Coefficient Instability (Met, Descriptive)

H2 [C]: At least one feature changes sign or magnitude $\geq 2\times$ across label variants.

Result: of the 22 covariates shared by all three variants, 2 reverse sign, 5 change magnitude by $\geq 2\times$, and 6 do at least one of the two. **H2's criterion is met (descriptive; no test statistic).**

Multiplicity: H2 reserves a slot in the confirmatory family of six but contributes no p-value, which makes the Holm correction on the remaining hypotheses stricter rather than weaker.

The count is the pre-registered answer, reported unmodified, and depends on the penalised fit of Section 3.10.1. Because a threshold rule on point estimates

cannot separate genuine movement from imprecision, it is an upper bound on the instability present.

Feature	What changed
Bilirubin	Sign reversal across variants
Any vasopressor	Sign reversal *and* magnitude $\geq 2\times$
Platelets	Magnitude $\geq 2\times$
Creatinine measured	Magnitude $\geq 2\times$
Platelets measured	Magnitude $\geq 2\times$
WBC measured	Magnitude $\geq 2\times$

Table 5.30: Features with sign reversal or magnitude instability across label variants. 2 of the 22 shared covariates reverse sign and 6 are unstable on either criterion. Full per-covariate flags are in `12_h2_coefficient_stability.csv`.

Does the instability exceed its own estimation noise? (Protocol Amendment 7)

Using the cluster-robust covariance (Section 3.10.1), the difference between each flagged covariate's two extreme estimates is referred to their pooled standard error (Table 5.31).

Covariate	β_A	β_B	β_C	Max diff.	Pooled SE	Ratio	Kind
`wbc_measured`	0.0970	0.5374	0.8989	0.8020	0.0517	15.51	action
`platelets_measured`	-0.2564	-0.5614	-0.8173	0.5610	0.0529	10.60	action
`creatinine_measured`	0.2877	0.2303	0.0343	0.2534	0.1024	2.48	action
`platelets`	0.0002	0.0001	0.0003	0.0002	0.0002	1.07	physiol.
`vaso_anyTRUE`	-0.0374	0.0271	0.0025	0.0645	0.0864	0.75	action
`bilirubin_total`	0.0014	-0.0016	-0.0020	0.0034	0.0056	0.61	physiol.

Table 5.31: Each covariate flagged by H2, with its coefficient under each label variant, the difference between the two extremes, the pooled cluster-robust standard error of those two estimates, and their ratio. 3 of the 6 flagged covariates move by more than roughly twice their pooled standard error; the remaining 3 move by less, the largest such ratio being 1.07; no covariate falls between 1.07 and 2.48. "Kind" marks clinician action or patient physiology (Section 5.10.4). **A diagnostic, not a test**: the fits share stays, and the covariance is that of the penalised estimator (Section 3.10.1).

The pre-registered count stands; the post-hoc diagnostic (Protocol Amendment 7) is reported beside it, not in its place. The covariates whose movement exceeds their standard errors are 3 of 3 action-derived, with ratios from 2.48 to 15.51, and

all are indicators of whether a test was ordered. Both sign reversals, bilirubin and vasopressor exposure, carry the two smallest ratios, so no claim rests on either individually. This agrees with the by-kind split (Section 5.10.4) and the ablation (Section 5.7), but all three are computed from the same three fitted models and are not independent corroboration.

The instability is concentrated in the action-derived features

The covariates fall into two kinds. **Action-derived** features record clinician behaviour (vasopressor exposure and the per-test order indicators); **physiological** features record the patient (vital signs, laboratory values and their rolling slopes). Table 5.32 splits the instability by kind.

Feature kind	Covariates	Unstable	% unstable
Action-derived	8	4	50.0
Physiological	14	2	14.3

Table 5.32: H2 instability by feature kind. Action-derived covariates are a minority of the specification but a majority of the unstable terms. **[E] exploratory.**

Instability is concentrated in the features that encode clinician behaviour, which a change of label affects in a way that a heart rate is not. With 22 covariates the counts are small and the difference between the two rates is not tested. The result supports Lauritsen et al.'s [10] framing-instability concern for specific effect estimates, not instability of the fitted model as a whole.

5.10.5 H3: Treatment Leakage, Definitional and Empirical

H3 [C]: Restricting to pre-treatment hours reduces AUROC by ≥ 0.03.

Definitional result: 0 **sepsis onset events** in 179,003 pre-treatment person-hours. AUROC is undefined, so the pre-registered contrast cannot be formed. This zero is *proved* below (Proposition 5.1) to follow from the onset rule used by variants A–C; it is a statement about the implementation, not a measurement.

Empirical result (Protocol Amendment 4): relabelling the same stays with the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ onset (Variant F) moves 1,738 of 6,447 onsets earlier (27.0%, median −14.0 h) and places 71 **onsets before treatment**, identified at 0.8081 AUROC (0.8018 GBT). **H3 is not evaluable as pre-registered; the definitional and empirical statements replace it.**

Multiplicity: like H2, H3 reserves a slot in the confirmatory family with-

out contributing a p-value. Variant F's contrast against B is exploratory (family E3).

Definitional result. In the hours before the first antibiotic or culture there are zero onset events for variants A–C and E. This follows from how the onset timestamp is assigned:

Proposition 5.1. *Let t_{anchor} be the earlier of the first qualifying antibiotic order and the first culture draw of an admission, and let t_{onset} be the onset assigned by variants A, B, C or E. Then $t_{\text{onset}} \geq t_{\text{anchor}}$ for every labelled stay, so the window $[\cdot, t_{\text{anchor}})$ contains no onset.*

Proof. Under E, $t_{\text{onset}} = t_{\text{susp}}$, and t_{susp} is defined as the *later* element of a qualifying antibiotic-culture pair; both elements are at or after t_{anchor}, so $t_{\text{susp}} \geq t_{\text{anchor}}$. Under A, B and C the SOFA criterion is a *membership* gate: a stay is labelled only if the $\geq$ 2-point rise occurs somewhere in the evaluation window, but the timestamp written is still t_{susp} (Section 3.6). The inequality therefore carries over unchanged. □

The exact one-sided 95% upper bound on the pre-treatment onset rate in the inference tables therefore bounds a quantity the definition fixes at zero.

Empirical result. Under the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ onset (Equation 3.2), an onset precedes the anchor whenever the SOFA rise is detected before the clinician acts, and Proposition 5.1 does not apply. Variant F, with the same septic stays as Variant B, records 71 pre-treatment onsets in the temporal test set, identified at 0.8081 AUROC (Section 5.6.3). The conjunction-timing rule of the pre-registered variants is therefore what prevents pre-treatment onsets from being recorded; Section 6.1 interprets this result.

5.10.6 H4: Split Optimism (Not Confirmed)

H4 [C]: Temporal-split AUROC is at least 0.02 lower than random-split AUROC.

Result: primary model, random AUROC 0.7424 against temporal 0.7590. Difference = −0.0165 (95% CI −0.0372–0.0040): the temporal split is the *higher* of the two. Test against the pre-registered threshold $\theta_0 = 0.02$: Holm-adjusted $p = 1.000$. **H4 NOT CONFIRMED.**

Model	Temporal	Random	Random − temporal
Primary	0.7590	0.7424	−0.017
GBT	0.7424	0.7132	−0.029
Cox	0.6444	0.6310	−0.013
NEWS2	0.6415	0.6088	−0.033

Table 5.33: Temporal vs. random split comparison (Variant B, AUROC). Every model scores lower on the random split.

The interval includes zero and excludes +0.02, and all four models fitted under both splits score lower on the random split, by 0.013 to 0.033. The temporal test set is the 2017–2019 cohort whereas the random test set spans all years, so the contrast also confounds split design with cohort composition, and the random split cannot be assumed to be the more permissive design (Section 6.7).

5.10.7 H5: Metric Divergence (Not Evaluable as Pre-Registered)

H5 [C]: Utility Score declines proportionally more than AUROC from internal to external validation.

Testable arm: AUROC degradation internal → external (primary, Variant B) = 0.0630 (95% CI −0.0215–0.1563); internal 0.7590, eICU-CRD 0.6960 on 29 external events. Holm-adjusted $p = 0.551$. **The point estimate falls, but the interval does not exclude no change.**

Non-testable arm: the internal Utility Score is 0.000 at the default cut-off and 0.025 at its best threshold, so a proportional decline cannot be formed. **H5 AS PRE-REGISTERED IS NOT EVALUABLE; the substituted AUROC contrast is not supported.**

H5's proportional contrast requires a non-zero internal Utility Score, and the internal score under the primary label is 0.000 at the default cut-off and 0.025 at its best threshold, so the contrast cannot be formed. Its slot is occupied by the AUROC limb alone (§13.6 of `00_preregistration.md`), to which the Holm-adjusted 0.551 applies. That contrast, 0.0630, is in the predicted direction and about three-quarters of the 0.085 anticipated from Moor et al. [21], but its interval is consistent with both no loss and a loss twice as large, because it rests on 29 external events (Section 5.5.1). The Narrow label loses most and the Liberal least, and the antibiotic-only arm, a different target on 6,140 events, shows a degradation of the same sign (0.7590 to 0.6644); neither observation is a test.

5.10.8 H6: Model-Class Null (Not Rejected; One-Sided Non-Inferiority)

H6 [C]: $|\Delta_{\text{AUROC}}(\text{GBT} - \text{primary})| \leq 0.02$.
Result: $\Delta = -0.0165$ (GBT 0.7424, primary 0.7590), 95% CI -0.0308–-0.0012. The *upper* limit lies inside the bound $\theta_0 = 0.02$; the lower limit does not. **H6 NOT REJECTED. The interval supports one-sided non-inferiority of the primary model (GBT is not better by more than the margin), not two-sided equivalence.**

Both models are scored on the same person-hours, so the contrast is paired, and the primary model is also ahead under the other two labels (Table 5.3). The pre-registered form (§7) places equivalence in the *null*, H_0: $|\theta| \leq 0.02$; **this is not the two one-sided tests (TOST) procedure**, and failure to reject the interval null is not evidence of equivalence. Protocol Amendment 10 tests both boundaries: at $\theta = +0.02$ $p = 1.000$, and at $\theta = -0.02$ $p = 0.668$. Neither boundary is cleared, so H6 is not rejected.

The conclusion rests on the interval. Its upper limit, -0.0012, lies below $+0.02$, so GBT is not supported as outperforming the primary model by the pre-registered margin: a one-sided non-inferiority conclusion, the only form claimed. Its lower limit, -0.0308, falls outside -0.02, so two-sided equivalence is not established. The interval lies entirely below zero, so the primary model is ahead by -0.0165 AUROC, of the same order as the label spread. The results therefore do not support choosing GBT over the primary model on discrimination. A caution on the comparator's fitting procedure is stated in Section 6.7.

The AUROC difference is not accompanied by a clinically meaningful utility advantage for either model: at each model's best threshold GBT is marginally ahead, 0.031 against 0.025, and both maxima lie a few per cent above the no-alert strategy.

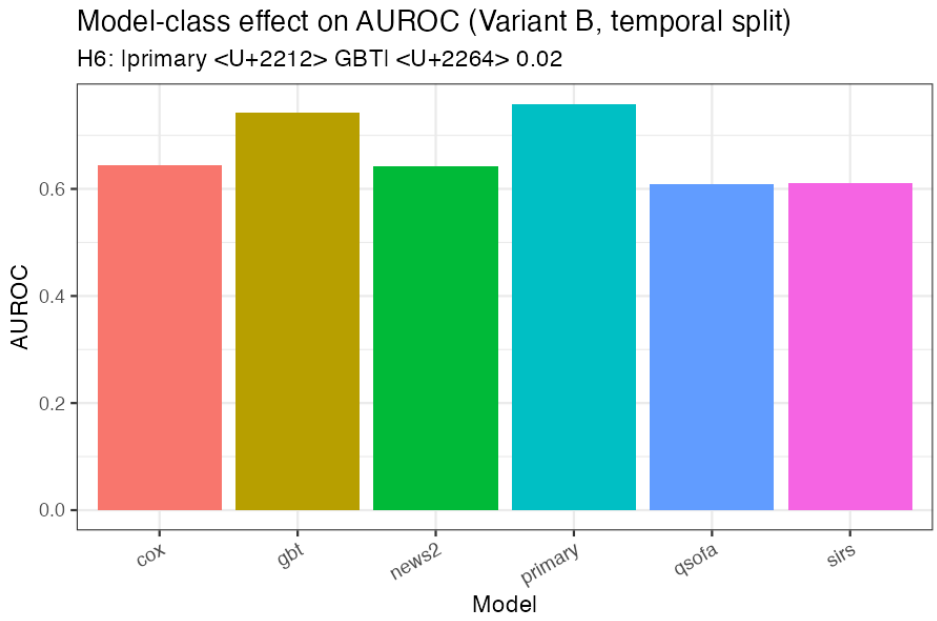


Figure 5.10: AUROC by model class (Variant B, temporal split). The two flexible estimators are close to one another and clearly ahead of the rule-based scores and the Cox comparator; on the paired contrast the primary model leads GBT by 0.0165 (Section 5.10.8).

Chapter 6

Discussion

This chapter interprets the results of Chapter 5. Section 6.1 states the central finding and Section 6.2 reads H1 and H6 together. Sections 6.3 to 6.5 set out the scope of the external estimate, the anchor-independent label and the feature ablation. Section 6.6 draws out implications for how studies of this kind are reported, and Section 6.7 sets out the limitations.

6.1 The Core Finding

The central finding concerns outcome construction. Under the pre-registered conjunction-based timing rule, no Sepsis-3 onset can occur before the treatment anchor (Proposition 5.1). Applying the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ timing to the same cohort, whose septic membership is cohort-wide identical, recovered 71 pre-treatment onsets, separable from their surrounding at-risk hours at 0.8081 AUROC on the onset-hour estimand. Whether those onsets are predictable in advance cannot be assessed here, because predictors are binned to the onset hour (Limitation 3).

This finding has an analytic and a measured part. Analytically, any study that uses a conjunction-based onset rule and reports pre-treatment performance is reporting on an empty set, and a study that does not restrict the window is, to that extent, scoring discrimination around a clinician-defined decision. Empirically, the operationalisation rather than the data removes the pre-treatment onsets. Neither part supports the proposition that no patient develops sepsis before clinicians act, which external evidence contradicts: the Epic Sepsis Model v2 evaluation [5] reports median lead times of 1.4–7.1 hours ahead of clinician recognition across four sites. Sepsis-3 *membership*, however, depends on clinical action under every timing rule, since without a qualifying antibiotic–culture pair there is no label, so Kamran et al.'s [7] concern applies to which patients can enter the positive class.

The Utility Score stays close to the no-alert strategy for every model. This reflects both the low hourly event rate and the scoring rule's cumulative penalty for false-alert hours, which is charged for every alerting hour while only the first timely alert is credited; the study does not separate their individual contributions, nor test whether the temporal split contributes through era-related differences in recording. AUROC remains in the high 0.7s under the same conditions because the ranking of person-hours is preserved even when the achievable alert volume is not clinically usable. Wang et al. [4] documented this divergence across 91 models moving from internal to external validation (Section 2.6); here it is already present internally, under a time-based split, where utility is close to the no-alert baseline before any external validation.

6.2 H1 and H6 Together

H1 and H6 address different levels of comparison. H1 compares the label spread with the spread across six complete model specifications, which captures the large gap between flexible fitted models and simpler comparators: the specification spread 0.1499 exceeds the label spread 0.0123, and the interval on the difference excludes zero. H6 compares two flexible models fitted on the same predictors and shows a much smaller difference, 0.0165 AUROC in the primary model's favour, with GBT not better by more than the pre-registered margin. In this study, therefore, the difference between the two flexible estimators was small relative to the difference between fitted models and simpler comparators, while the label changed which patients were counted far more than it changed discrimination.

The H6 result also bears on the statistics-versus-machine-learning question of Section 2.2. Christodoulou et al.'s [23] equivalence finding concerns binary outcomes on tabular data, and Fagerström et al. [22] reported a deep model detecting septic-shock onset up to 20 hours earlier than a Cox baseline. Here, with the label, the split, the features and the evaluation held fixed, the primary discrete-time hazard model is not outperformed by GBT. This does not contradict Fagerström et al.: their comparator was a static Cox specification, the same family as the Cox benchmark retained here, which is also well behind the fitted models in this study (0.6444 against 0.7590).

The intervals add one thing the point estimates could not. The label spread's interval [0.003, 0.032] and the specification spread's [0.134, 0.171] do not overlap, so H1's reversal is not attributable to resampling noise. The label effect is real, since its interval excludes zero, but its main consequence is on the cohort, the event count and every denominator computed on it, which AUROC does not capture.

6.3 Interpreting the External Validation

The interpretability floor adopted here is 100 events, following the external-validation sample-size guidance of Riley et al. [46] and Collins et al. [12]: below it the confidence interval on an external AUROC is wider than the degradation the study is trying to detect. All three Sepsis-3 variants fall below it, with 29 events under the primary label, so the external Sepsis-3 degradations are stated as directions rather than magnitudes, and the H5 interval is the formal expression of that limit. External AUPRC and Utility Score are not interpreted at this event count.

The microbiology-covered sub-cohort is also selected by documentation: a culture is ordered when infection is already suspected, so the covered stays are enriched for patients a clinician had begun to investigate, and the external estimate describes transport into that sub-population rather than into eICU-CRD as a whole (Limitation 2).

The antibiotic-only arm clears the floor with two orders of magnitude more events, but it labels a broader target, treated suspicion of infection without the culture limb, and is never pooled with the Sepsis-3 arm. Its degradation has the same sign and label ordering as the Sepsis-3 arm's and a somewhat larger magnitude. Because the two arms label different targets, that agreement is descriptive. Accordingly, the external analysis supports a direction of effect, but not an estimate for Sepsis-3 alone. The same arm shows that the internal ordering of comparators does not transport: GBT scores 0.5316, 0.5486 and 0.6106 under Variants A, B and C, close to chance under the two narrowest labels, while NEWS2 scores 0.6121, 0.6071 and 0.6169, so a bedside score ranks ahead of GBT under every label in this arm. The two flexible estimators are close internally, but only the primary model stays ahead of NEWS2 externally. The arm's event count supports this ordering but not the size of any gap.

The Variant D arm is the one external estimate above the floor for its own target (Section 6.4); it provides precise transport estimates for deterioration but does not improve inference for Sepsis-3. The Sepsis-3 point estimate, 0.0630, is about three-quarters of the 0.085 penalty Moor et al. [21] report across four databases. Unlike Moor et al., this study cannot attribute the loss, because their harmonised databases shared labelling inputs and eICU-CRD does not.

6.4 Scope of the Variant D Results

Variant D's label is a threshold function of recorded physiology, and five of the six SOFA components are model inputs, so its discrimination is partly self-fulfilling (Section 3.6). Its AUROC *level*, internal or external, is therefore not comparable

with Variant B's and is not evidence about sepsis prediction. This caveat applies throughout this section. Within it, Variant D supports four conclusions.

1. **The label sensitivity is not wholly definitional.** At $\kappa = 0.057$ and a 6.5% onset match, an anchor-independent label disagrees with Sepsis-3 about both membership and timing, so the H1 label result is not only an artefact of how one definition is read.

2. **The alert burden remains high under every label; the utility ceiling does not.** The best achievable Utility Score rises from 0.031 under Variant B to 0.198 under E and 0.271 under D, with the primary model following the same pattern. Even D's best configuration carries 12.7 false alerts per true alert, several times the 1.4 benchmark of Moor et al. [21]. Contribution C3 therefore holds across labels in its alarm-volume form but not as a utility ceiling near zero; the rise is itself expected for a label built from recorded physiology.

3. **The two metrics order the models differently, and the ordering is stable across labels.** At its own best threshold GBT leads on the Utility Score under B, D and E alike, while on AUROC the primary model leads under B and E and GBT only under D. Changing the label does not reverse either ordering; changing the metric does. This illustrates the concern behind H5, that the verdict depends on the metric reported, without testing H5's internal-to-external contrast.

4. **The external event-rate gap is largely attributable to the anchor.** The Sepsis-3 arm yields an external event rate 0.094 times the development cohort's. Applied to all 181,589 eICU-CRD stays (Protocol Amendment 9), Variant D gives a ratio of 0.746. Because the two labels are applied to the same two cohorts, a population difference large enough to produce the anchored ratio would also have produced a comparable shortfall under the anchor-independent label, and it does not. The reduction therefore suggests that label transport accounts for a substantial part of the original discrepancy. It bounds the anchor's contribution rather than isolating it: roughly a quarter of the development cohort's rate remains unaccounted for, and case mix, charting frequency, the five-component external SOFA rise and differences in illness severity all sit inside that residual.

The large Variant D event count also supports the transport *drop* with an interval (Protocol Amendment 10): both models lose AUROC on transport, and both intervals exclude zero (Table 5.14). This is the study's one demonstrated transport loss. It concerns a deterioration rule, not Sepsis-3, and cannot be pooled

with the anchored arm. It does show that the two databases are not similar enough for a frozen model to move between them at no cost, which is consistent with the Sepsis-3 arm being unable to resolve a degradation of the size the literature anticipates on 29 events.

6.5 Interpreting the Ablation

The ablation is exploratory and run on one cohort, and with 22 covariates the stability counts are small enough that the comparison is descriptive. Its interpretation has three parts.

First, the explicit action-derived features contribute little to AUROC. Removing them changes discrimination by -0.0076 for the primary model under the primary label and by less than 0.01 in all six comparisons. 5 of the 6 paired intervals exclude zero, and no margin of practical equivalence was pre-registered, so this is a statement about magnitude, not absence; a feature that moves AUROC little may still be the reason a particular alert fires.

Second, these features account for most of the cross-label coefficient instability: 27.3% of terms are flagged in the full fit against 7.1% in the ablated one. A published effect estimate for a treatment or order indicator from a model of this kind therefore reflects the labelling regime as much as the patient.

Third, removing them does not remove all clinician-action information. The retained laboratory values are forward-filled from tests somebody chose to order, and vital signs are recorded on schedules that reflect how closely a patient is watched, so the ablation measures the effect of removing the explicit traces only (Limitation 5).

6.6 Implications for How These Studies Are Reported

The findings bear less on which model to build than on what a sepsis prediction study needs to report before its headline number can be interpreted.

Report label agreement, not only performance sensitivity. Performance across label variants should not be reported as the only label-sensitivity analysis. Here, label choice moved AUROC by 0.0123 while the two narrowest readings agreed at $\kappa = 0.267$ despite near-identical prevalence, so every count, rate and denominator changed substantially while discrimination barely moved. Agreement between the variants is where the variation lies, and AUROC does not show it.

State the onset rule and its consequence for the evaluation window. Two studies can share a cohort, a definition and a model and still differ on whether pre-onset prediction is measurable, purely through the choice between t_{susp} and $\min(t_{\text{susp}}, t_{\text{SOFA}})$. That choice is often left implicit in a reference to Sepsis-3. It should be stated in the methods with its consequence for the evaluation window, because under a conjunction rule an empty pre-treatment window is not evidence of anything (Proposition 5.1).

Report discrimination and operating-point burden together. At the event rate here the conventional 0.5 cut-off separates models by calibration rather than clinical value. A threshold-dependent metric is interpretable only as a swept ceiling with the attaining operating point reported beside it, including its alert burden (51.8 false alerts per true alert at the best configuration under the primary label, against the 1.4 benchmark of Moor et al. [21]). Alert burden is rarely reported among the 91 models Wang et al. [4] reviewed.

Audit label-input coverage before external validation. In this study the main constraint on the external estimate was whether the label's inputs existed in the destination database, which neither a reported AUROC nor a reported cohort size reveals. A study transporting a treatment-anchored label should report the coverage of each labelling input in the destination cohort, and the event count an estimate rests on, before reporting performance.

Interpret comparisons between flexible estimators in context. Within this study, comparisons between flexible estimators were less informative than comparisons of label membership, onset timing and alert burden: the two flexible estimators differ by 0.0165 AUROC against 0.1499 across the six specifications. Two estimators are a narrow base for a general claim, since no deep temporal model was fitted and GBT was not tuned (Limitation 6).

Declare subgroup event floors in advance. At a low event rate most demographic strata carry too few events for a subgroup AUROC to be interpretable, and an interval reaching the chance line is easily read as a null result rather than an absent measurement. Declaring the floor before estimation, tabulating below-floor strata rather than suppressing them, and excluding them from the corrected family keeps that distinction visible.

6.7 Limitations

The limitations below are not of equal weight, and three bound the central claims most directly: the discrimination figures measure detection at the recorded onset hour rather than advance warning (Limitation 3); part of the label-variant spread is definitional, because the pre-registered variants share one treatment-based anchor (Limitation 4); and the strongest positive result, H6, is non-inferiority against a single untuned comparator whose boosting-round selection was corrected under Protocol Amendment 8 (Limitation 6). The remainder qualify the scope of particular analyses rather than the direction of the main findings.

1. **Development cohort and population selection.** All models were developed on MIMIC-IV, from a single academic medical centre, and the findings apply to the Sepsis-3 definition on that database; clinical practice, documentation and antibiotic prescribing elsewhere may differ, and SEP-1 and the CDC Adult Sepsis Event were not implemented. The base cohort is also selected: only the first ICU stay of each patient and stays of at least four hours are retained (Section 3.4). Repeat admissions differ systematically from first admissions, and short stays mix rapid deaths, rapid transfers and low-acuity observation admissions; the two restrictions pull acuity in opposite directions, so their net effect is not signed here. The population is the subset of adult ICU admissions for which an hourly real-time task is well posed.

2. **External label transport and external sample size.** The Sepsis-3 anchor requires a culture, which `microLab` documents for only 1.55% of the 181,589 eICU-CRD stays, so the primary external analysis rests on 29 events under the primary label, below the 100-event floor. More data of the same kind would not remove this; it would require a destination database whose microbiology recording matches MIMIC-IV's. The covered sub-cohort is documentation-selected, with a direction of selection that cannot be signed, and the external degradation is not decomposed into case mix, measurement frequency, presentation or residual label imperfection. NEWS2 is also not scored identically in the two cohorts: in MIMIC-IV a person-hour with no recorded Glasgow Coma Scale after forward-fill is charged three points as "not alert", whereas in eICU-CRD, which lacks the variable, it is charged none, so NEWS2 thresholds, specificity and alert burden should be compared within a cohort, not across the pair. No decomposition, verdict or primary-model result depends on NEWS2.

3. **Discrimination is measured on the current-hour hazard, not on a six-hour horizon.** Every AUROC, AUPRC and calibration figure scores the current-hour hazard against an indicator marking the onset hour among

at-risk hours (Section 3.7). The figures are therefore not directly comparable with published AUROCs for binary "onset within k hours" labels, whose positives are more numerous and whose task is easier to score well on, and comparisons with such figures in Chapter 2 are indicative only. Covariates are binned to the hour, so a measurement charted after the onset timestamp within the onset hour still enters that hour's prediction: the figures measure detection at the recorded onset hour, not anticipation, and the lead-time distribution, which speaks to anticipation, is uninformative at these alert rates. A design that lagged every predictor by one hour, or applied a sub-hour timestamp cut-off, would separate the two and is not implemented here. A cumulative six-hour risk, which a deployed instrument would display, is also not reported.

4. **Label construction.** Part of the label-variant spread is definitional: Variants A–C share the culture-plus-antibiotic anchor, so moving the antibiotic–culture windows must move the onsets. The pre-registered onset rule is also not the standard one, so the pre-registered family cannot address onset timing on its own, H3 among it. The alternative labels that address these points are post-hoc (Protocol Amendments 2 and 4) and rest on few variants. Variant F uses the full SOFA score, so its onsets are free of the antibiotic–culture anchor but not of treatment in general, and its t_{SOFA} is searched only inside the 72-hour panel. Variant D is an operational deterioration label built for this comparison, not a validated construct, its onsets have no external adjudication, and its overlap with the model inputs makes its discrimination level uninterpretable.

5. **Feature construction.** The feature set cannot be fully purged of clinician action: the ablation removes the explicit action-derived predictors, but retained laboratory values and vital signs still reflect ordering and monitoring decisions. Clinical notes, procedure codes and ventilator settings were not included. The covariate panel is also single-analyst work with no independent re-implementation; declared-but-absent features and implausible recorded values are guarded against, by an error on any missing feature and a plausibility range on all nineteen physiological variables before forward-fill, but neither guard substitutes for independent replication.

6. **Comparator coverage and the scope of H6.** A deep survival comparator was planned but not built, so H6 establishes non-inferiority against GBT only, not against every flexible estimator. GBT was not tuned, which keeps the search budget equal across the H6 comparison but may place it below its attainable performance. The contrast supports one-sided non-inferiority, not two-sided equivalence (Section 5.10.8). GBT's boosting-round count

is chosen on a holdout from the training stays (Protocol Amendment 8); under the earlier procedure it was chosen on the test set, which favoured the comparator, and the correction moved H6 in the primary model's favour, so H6 is a finding the uncorrected fits already supported and the correction strengthened.

7. **Penalised estimation and H2 uncertainty.** The unpenalised likelihood is completely separated, so the reported fits are ridge-penalised; test AUROC varies by less than 0.02 across a 20-point penalty path, but the coefficients are shrunk estimates and the H2 counts are properties of the penalised solution. The cluster-robust covariance is that of the penalised estimator, so Wald-type interpretation is approximate, and it bears on coefficients only: H2's count has no sampling distribution. The Amendment 7 diagnostic shows that 3 of the 6 flagged covariates move by less than their own estimation noise, so the count is a loose upper bound, but the diagnostic is not a test. Performance intervals come from the cluster bootstrap and are unaffected.

8. **Threshold selection and alarm implementation.** Operating points and maximum Utility Scores are selected on the test set on which they are evaluated, so they are optimistic upper bounds; selecting thresholds on the training years would give a lower deployment estimate. External thresholds are re-derived on eICU-CRD, which hides the further loss a transferred threshold would incur. False-alarm episodes are defined by contiguity rather than acknowledgement; suppression and escalation logic would reduce them, so the alert-hour and episode rates bound the true burden from above and below.

9. **Subgroup analysis.** Only 2 of the racial strata clear the 100-event floor and no contrast in family E1 survives correction, so the data are consistent both with no disparity and with material disparities. The strata are also administrative: the `race` field mixes broad and granular codes from the same population, so the reference level is a residual category, and two levels record the absence of an answer. The label-sensitivity result is better powered but inherits the construct limitation, and its two significant racial strata are the missingness categories. A documentation-based explanation, in which an incomplete registration record accompanies an incomplete clinical record and so makes a label built on two documented clinician actions less stable, is assumed rather than measured; comparing charting density or laboratory coverage between strata would test it.

10. **Literature-search scope.** The search in Section 2.1 is structured and verified but not systematic: no title-and-abstract screening of a full deduplicated yield was performed and no second reviewer screened independently.

The novelty claims for contributions C1 and C5 concern combinations of analyses, each with precedent, and a systematic review could surface a prior study combining them.

11. **Post-hoc amendments and hypothesis evaluability.** Ten protocol amendments were introduced after the initial analyses had been inspected. They are dated and disclosed in the pre-registration, and their analyses are tagged exploratory or descriptive (Appendix C), but the confirmatory interpretation should be read as a pre-specified hypothesis set analysed under a retrospectively specified inference plan (Amendment 1). Amendment 8 corrected test-set use in GBT early stopping and changed the GBT estimates; no amendment altered an H1–H6 estimand. Three others warrant a note. Amendment 6 reduced family E1 from thirty computed contrasts to six, a departure from the rule that the denominator is the number of contrasts computed; the outcome-blind event floor removed the one contrast that had cleared correction, resting on twelve events, so the change runs against this study, and none of the six retained contrasts clears correction under either denominator. Amendment 7's diagnostic favours this thesis's argument and is computed from the same fits as the ablation, so it is not independent evidence. Amendment 10 attached intervals to three quantities outside the corrected families: 5 of the 6 ablation contrasts and one cross-label calibration-intercept contrast exclude zero. Of the hypotheses, H2 has no test statistic, H3 is degenerate by construction, and H5 is tested on a substituted estimand; that substitution is a departure from the protocol, and H5's verdict is weaker evidence than H1's or H4's for that reason.

Chapter 7

Conclusion

7.1 Summary of Findings

This thesis asked how much of the reported performance of real-time sepsis prediction on MIMIC-IV comes from the choice of model, and how much from the researcher's choices about the label, the timing of prediction relative to treatment, the data split and the evaluation metric. Across the six evaluated model specifications, model choice affected AUROC more than the three Sepsis-3 operationalisations did. External transport has the second-largest point estimate, but its interval is too wide to establish that rank, and the label and the split moved AUROC least. The main effect of label choice was on cohort membership and onset timing rather than on AUROC. The six pre-registered hypotheses were resolved as follows.

1. **H1 was not confirmed.** The label spread was 0.0123 AUROC against 0.1499 across the six model specifications, and the interval on the difference, $[-0.1623, -0.1083]$, excludes zero. Label choice nevertheless changed cohort membership substantially: the two narrowest readings agree at only $\kappa = 0.267$ despite near-identical prevalence.

2. **H2 was met descriptively.** 6 of the 22 shared covariates met the pre-registered instability criterion, but only 3 of the changes, all in indicators of whether a test was ordered, exceeded their estimated noise.

3. **H3 was not evaluable as pre-registered.** The pre-treatment event count was fixed at zero by the onset rule (Proposition 5.1); standard re-timing (Variant F, post-hoc) recovered 71 pre-treatment onsets on the same cohort.

4. **H4 was not confirmed.** Random splitting did not produce higher AUROC: the random-minus-temporal difference was -0.0165 (95% CI -0.0372–

0.0040), and every model fitted under both splits scored lower on the random split, although the two test sets also differ in composition.

5. **H5 was not evaluable as pre-registered.** The internal Utility Score was too close to zero for a proportional decline to be defined, and the substituted external AUROC contrast, 0.0630 (95% CI −0.0215–0.1563) on 29 events, was imprecise. Every external point estimate was below its internal counterpart, and the post-hoc anchor-independent analysis showed a transport loss for the deterioration target, but the Sepsis-3 arm establishes neither the existence nor the magnitude of a degradation.

6. **H6 supported one-sided non-inferiority, not equivalence.** GBT did not outperform the primary model by the 0.02 margin: the difference was −0.0165 [−0.0308, −0.0012], with the lower limit outside the margin. The results do not support preferring GBT over the primary model on predictive performance, and the small AUROC difference was not accompanied by a meaningful utility difference (0.031 against 0.025 at each model's best threshold).

7.2 The Central Lesson

The main consequence of label choice was not a large change in AUROC but a change in who was labelled septic and when onset was assigned. Under the pre-registered timing rule, pre-treatment prediction was structurally impossible; standard re-timing recovered pre-treatment events on the same cohort. Across all pre-registered labels, no evaluated operating point approached the alert-burden benchmark: the best configuration under the primary label reached a Utility Score of 0.031 at 51.8 false alerts per true alert, against a benchmark of 1.4, while AUROC remained in the high 0.7s. These findings indicate that improving discrimination alone cannot resolve limitations introduced by outcome membership, onset timing and alert burden. They do not isolate the label as the sole cause of the gap between discrimination and usable alerting: the estimand, the threshold, calibration, prevalence and the Utility Score's own penalties all bear on it.

Splitting the Sepsis-3 conjunction (Section 5.6) both qualifies and sharpens this. Under every pre-registered variant the onset time is fixed by the treatment anchor, so part of the variation between variants follows from the definition. An anchor-independent deterioration label agrees with Sepsis-3 at $\kappa = 0.057$, close to chance, so the disagreement is not confined to readings of Sepsis-3. The treatment anchor alone is about as predictable as the conjunction built on it ($\phi_{\text{anchor}} = 1.041$ [0.990, 1.097] for the primary model, an interval including 1, and 1.099 [1.037, 1.164] for GBT), whereas Sepsis-3 under standard onset timing produced higher

discrimination for the primary model (0.7951 against 0.7590), on an in-horizon event set slightly larger than Variant B's. Reduced discrimination is therefore associated more with an onset placed at the clinician's action than with the conjunction itself.

7.3 Contributions Revisited

1. **C1 (Label variation, separated from discrimination):** Three Sepsis-3 operationalisations were applied to a discrete-time competing-risks model on 65,078 MIMIC-IV stays, with the label and specification spreads estimated on the same test rows, each with an interval; to the best of our knowledge this combination is not previously reported. Label choice moved cohort membership substantially while moving AUROC by 0.0123.

2. **C2 (Treatment leakage: membership and onset timing):** The study proves that the conjunction-based onset rule leaves the pre-treatment window empty, and measures 71 pre-treatment onsets on the same cohort under standard timing. Kamran et al.'s [7] concern therefore holds for label membership, while for onset timing it depends on the onset rule.

3. **C3 (Utility ceiling):** The Utility Score was swept across the full threshold grid for every model and label. Under the primary label its ceiling lies within a few per cent of the no-alert strategy and is reached only at 51.8 false alerts per true alert, extending Wang et al.'s [4] meta-analytic finding to a controlled single-study comparison that does not depend on one operating point.

4. **C4 (Coefficient instability):** 6 of the 22 shared covariates were unstable across labels under the pre-registered criterion; referred to their cluster-robust standard errors, 3 remain, all order indicators. This supports Lauritsen et al.'s [10] framing-instability concern for specific effect estimates on MIMIC-IV.

5. **C5 (Equity by label):** Subgroup discrimination and subgroup label sensitivity were measured on the same strata of one onset-prediction cohort under false-discovery control (Section 2.1). The label-sensitivity contrasts survive correction and the discrimination contrasts do not, and the surviving racial contrasts are missingness categories, so the finding concerns record completeness rather than race.

6. **C6 (Reproducible pipeline):** A reproducible, seed-fixed pipeline built around a pre-registered protocol, with every amendment disclosed and re-

porting following TRIPOD+AI. Within the pinned environment it reproduces the reported estimates from the raw databases, and it writes the numbers in this document as generated constants (Section 4.7.2, Appendix A).

7. **C7 (Definitional vs. empirical label variation separated):** An anchor-independent deterioration label agrees with Sepsis-3 at $\kappa = 0.057$ and matches its onset hour in 6.5% of shared stays, and records 1,362 onsets in the pre-treatment window that is empty of Sepsis-3 onsets under the pre-registered rule. These results show that label membership and onset timing shape the prediction task; they do not show that the physiology is easy to predict, since Variant D's discrimination level is set in part by its construction.

7.4 Future Work

1. **Better sepsis labels.** Future work should evaluate outcome definitions that reduce direct dependence on treatment timestamps and are adjudicated independently of the model inputs, for example definitions built on lactate trajectories and organ-function biomarkers. The deterioration label of Section 5.6 is an operational construct, not a validated definition.

2. **Lagged predictors.** A design that lags every predictor by one hour, or applies a sub-hour timestamp cut-off, would show whether the information associated with the recorded onset hour supports actionable advance warning (Limitation 3).

3. **Non-Sepsis-3 consensus definitions.** SEP-1 and the CDC Adult Sepsis Event were not implemented. Both are defined on antibiotic days, so neither tests treatment-independence, but each yields a different cohort, and Dutta et al. [9] report that performance varies between them.

4. **Prospective validation.** A new label should be tested where sepsis onset can be determined by a protocol-blinded assessor, independently of treatment decisions.

5. **Utility-optimised training.** The models here are fitted to a likelihood or surrogate loss and only afterwards read at the utility-maximising threshold. Training against the Utility Score, through a differentiable surrogate or cost-sensitive reweighting, would change what the model learns. Such optimisation should use clinically justified costs and validation thresholds defined independently of the test set; otherwise it may improve the selected utility score without improving deployability.

6. **Multi-hospital evaluation.** Whether any model can be deployed at an acceptable alert burden requires prospective multi-centre testing with clinical-staff feedback.

7.5 Closing Remark

The central hypothesis was not supported: across the evaluated specifications, model choice affected AUROC more than the three Sepsis-3 operationalisations did. The smaller AUROC effect should not, however, be interpreted as evidence that label choice is unimportant. The variants selected substantially different cohorts, and the onset rule determined whether a pre-treatment prediction task could be defined at all.

No evaluated configuration approached the pre-specified alert-burden benchmark. In addition, the reported discrimination concerns detection of the recorded onset hour and does not establish advance warning. Future studies should therefore report label agreement, onset timing, calibration, clinical utility and alert burden alongside discrimination.

The principal contribution of this study is to show that outcome construction changes the population and timing of the prediction task in ways that AUROC alone does not reveal.

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Appendix A

Source Code and Reproduction

A.1 Repository

The complete implementation of this study is held in a version-controlled public repository:

Source code repository

https://github.com/ahadarif007/sepsis-variance-decomposition

Source code licensed under the Apache License 2.0.
No patient data is included.

The Apache-2.0 grant covers the pipeline source written for this study. It does not extend to material this work does not own, and the repository's `NOTICE` file draws that boundary explicitly: the institutional thesis template and its brand assets are reproduced under their owner's rights and are excluded; the two clinical databases are excluded absolutely, being redistributed by neither this repository nor its licence; and the thesis text and its figures, being scholarly prose rather than software, are made available for citation and academic reuse with attribution rather than under a software licence.

Where a routine implements a published method (the SOFA score, the Sepsis-3 operational criteria, NEWS2, qSOFA, SIRS, the adapted PhysioNet-style Utility Score, the sample-size criteria, pooled logistic regression, decision-curve analysis) the implementation is original to this work and written from the published definition, and the method is attributed to its authors both in `NOTICE` and at the point of use in this thesis. No implementation was copied or adapted from a third party's source.

The repository contains **code and aggregate outputs only**. No MIMIC-IV or eICU-CRD record, derived person-hour table, or fitted model object is included:

the `data/` and `processed_data/` directories are excluded from version control by design, because the PhysioNet Credentialed Data Use Agreement (Section 3.3) prohibits redistribution of the underlying data. Exclusion is enforced by `.gitignore` and was verified against the tracked file list before release. The `output/` directory is tracked only for the frozen run's rendered stage reports, run logs and figures, which report aggregate results and no patient-level record.

A.2 The Frozen Run

Every quantity in this thesis is written mechanically from the pipeline's result tables by `thesis_constants.R`, so no current estimate is transcribed by hand and every such value in the text agrees with the table it is drawn from. The one deliberate exception is named below: the two *superseded* figures this section prints in order to compare them against the current ones. That is a guarantee about transcription and about nothing else: it cannot establish that the right output cell was selected for a sentence, that the sentence still describes what the value shows, or that a claim has not outlived the analysis it was written for. Those remain human obligations, and Section A.4 describes the checks that close part of the gap. Whether the document is consistent across *runs* is a separate question again, and it is worth answering with a measurement rather than an assumption.

The run this document is bound to was executed end to end for that purpose, and it reproduced *every* reported quantity exactly: of the macros `thesis_constants.R` emits, the only ones that changed were the provenance stamp and the generation timestamp. On a fixed machine, R version and library build, with the seeds in `config.R` and a partition defined by admission year rather than by sampling, this pipeline is reproducible to the last digit it prints.

Two qualifications keep that from being a stronger claim than it is. The gradient-boosted comparator is the one component with no guarantee of reproducibility across *differing* library builds or architectures, so the statement above is about re-running this environment and not about rebuilding it elsewhere. And one intermediate artefact is known to vary between runs of identical source: the per-hour external prediction files differ below the fourth decimal place, which is why the result tables rather than those files are used as the reproducibility check. No reported quantity has been observed to move because of it.

Two quantities in this thesis did move during its development, and neither movement is attributable to that instability. The utility ceiling moved from 0.039 to its present 0.031 and the model-class gap from 0.0129 to 0.0165. That was not run-to-run noise. It was Protocol Amendment 8 correcting a defect: the boosted comparator's early stopping had been selecting its number of boosting rounds against the temporal test set, so a model-selection choice was being made on the

rows the model was then scored on. Removing that optimistic bias lowered GBT's discrimination, which moved every GBT-derived quantity at once and widened H6's gap in the interpretable model's favour. Section 6.7 states the amendment and the direction of its effect; naming a leakage correction as though it were benign non-determinism would understate it. The two superseded figures are typed here rather than generated, because they are the only quantities in this document that must *not* track the current run: replacing them with macros would print the present values twice and destroy the comparison the sentence exists to make.

The freeze therefore exists for a narrower reason than run-to-run drift, and it is still a good one: a number is only meaningful when paired with the run that produced it, and anything built outside the pipeline (a slide, a screencast, a figure recalled in a viva) can drift from the document without anything in the document detecting it. This document is therefore bound to one run, and that run is identified here so the pairing can be checked rather than assumed:

Frozen run

tag `frozen-2026-08-31` · commit `f345a92cdb6b`

constants generated 31 August 2026

Two repository states are involved and they answer different questions, so the distinction is drawn here rather than left to be discovered. The commit named above is the **analysis provenance**: it is the state of the pipeline source that produced every quantity this document reports, recorded by the pipeline itself in the generated constants file, and it is what to re-run to reproduce them. The repository's history was rewritten once before publication, which changes commit identifiers but not file contents, so the published history carries this state as commit `34863fcbd248`, with the identical source tree `5919f2b139c7`. The tag `frozen-2026-08-31` is placed on that published commit. The constants file the run generated was committed immediately after it, with the PDF built from it, at `3678120abfa9`.

The **document provenance** is a later commit on the same branch, tagged `submission-2026-10-16`. Text was revised after the analysis commit without re-running the pipeline, so the submitted sources and this PDF are carried by that commit. The revisions were to the thesis; to the narrative text of the twelve pipeline notebooks, including their report titles and inline expressions that print configuration values into that text; to the wording of the pre-registration document, without change to its hypotheses or decision rules; and to the tools that check the thesis or draw its diagrams. The rendered stage reports, run logs and figures of the frozen run were also committed. No code that computes or writes a reported quantity changed. That is a checkable claim rather than an assurance,

and the commands below check it. A `-dirty` suffix on the commit would indicate that the constants were generated from an uncommitted working tree, and must not appear in a submitted document.

```
git grep pcFrozenCommit submission-2026-10-16 \
    -- sepsis/thesis/generated/pipeline_constants.tex
git log -1 --format=%T frozen-2026-08-31
Rscript sepsis/project/check_frozen_code.R \
    frozen-2026-08-31 submission-2026-10-16
```

Run from the repository root, the first prints the line of the generated constants file that records the analysis commit named above: every number in this thesis is expanded from that one generated file, and the file records which run wrote it. The second prints the source tree of the tagged analysis state, which begins `5919f2b139c7`. The third compares the two tagged states: every R code chunk of the twelve pipeline notebooks as text, and every other file that computes or writes a reported quantity, or fixes the software environment, byte for byte. It prints `OK` only if all are identical, which is what establishes that no code producing a reported quantity changed between the run and the submitted document; narrative text outside the code chunks is outside its scope by design. Checking out `submission-2026-10-16` then yields the submitted sources and this PDF alongside the code that produced its numbers. Re-deriving the file from the databases additionally requires credentialed PhysioNet access, since no derived table is redistributed here (Section A.1); the reported quantities themselves are aggregate results and are disclosed in this document rather than as data.

Everything produced after the freeze (the presentation, the screencast and the viva preparation) reads from the frozen constants rather than from a fresh run, so that no figure quoted aloud can disagree with the figure on the page.

A.3 Repository Layout

Table A.1: Repository layout.

Path	Contents
`00_preregistration.md`	The pre-registration, including the G0 analysis gate and Protocol Amendments 1–10, each dated and disclosed as post-hoc

Path	Contents
`config.R`	All constants: label-variant parameters, feature lists, plausibility ranges, seeds, bootstrap sizes, multiplicity families, event floors. No pipeline configuration constant is defined in another analysis script
`utils.R`	Shared helpers, including `require_features()` (errors on a declared-but-absent predictor) and `check_multinom_fit()`
`clinical_scores.R`	SOFA, NEWS2, qSOFA and SIRS implementations
`01_setup.Rmd` … `12_inference.Rmd`	The twelve pipeline stages described in Chapter 4, in execution order
`run_pipeline.sh`, `run_pipeline.R`	Orchestration. Runs all stages, or a named subset, and regenerates the thesis constants on success
`thesis_constants.R`	Reads the result tables and emits every numerical value used in this document as a LaTeX macro (Section A.4)
`utility_score.py`	Reference implementation of the Utility Score, maintained as a mirror of the R scoring rule and cross-checked against it
`check_consistency.R`	Cross-file assertions over the result tables: arithmetic identities that must hold between quantities written by different stages, and predicates for the comparative claims this document states in words
`check_thesis_numbers.R`	Checks that every result-shaped number in the thesis sources is either generated, cited to prior work, or recorded as a design constant
`check_thesis_margins.R`	Measures the built PDF for content passing the text block, which LaTeX does not report for a table
`check_frozen_code.R`	Confirms that no code producing a reported quantity changed between two repository states (Section A.2)
`output/`	Created at run time: `processed_data/`, `figures/`, `logs/`, `pdf/`. Only `processed_data/` is untracked

A.4 Reproducing the Results

Prerequisites. R 4.5 or later with the packages listed in the repository, Python 3.14 with `pandas` 3.0.4 for the Utility Score mirror, and a TEX distribution providing `xelatex` and `pdflatex`/`biber`. The two source databases must be obtained independently through PhysioNet credentialing: MIMIC-IV v3.1 and eICU-CRD v2.0. They are placed under `data/` and located automatically, or pointed at with the `MIMIC_RESEARCH_ROOT` environment variable.

Full run. A complete execution is a single command,

```
./run_pipeline.sh
```

which runs stages 01–12 in order and, on success, regenerates the thesis constants and figures. Individual stages or phases can be run by number. Every stochastic step is seeded, so within the pinned analysis environment a repeated run reproduces the reported estimates; the gradient-boosted comparator may vary slightly across library builds or architectures, as noted below.

Partial runs. Three environment variables scope a run without changing any reported result. `V2_VARIANTS` restricts the per-variant loops in Stages 02–07 to a named subset, so an alternative label can be added without refitting Variants A–C; `V2_ARMS` restricts Stages 05 and 06 to the full or the ablated feature arm (Section 3.14.5); and `V2_BOOT_B` overrides the bootstrap replicate count of Stage 12 for a fast smoke test. Reported results always use the values fixed in `config.R`.

How the thesis stays aligned with the pipeline. No quantity estimated by this study is typed by hand into this document; the exceptions are figures quoted from prior work, constants fixed by the pre-registration, and prose roundings of a macro stated exactly nearby. On completion, `thesis_constants.R` reads the result tables and writes every reported quantity as a macro into `pipeline_constants.tex`, which the thesis includes; it also copies the generated figures into the document's image directory. A quantity the pipeline did not produce typesets as a conspicuous marker rather than as a stale number, so a partial run cannot silently leave an outdated value in the text. This is the mechanism by which the numbers in Chapters 5 and 6 are guaranteed to be the numbers the code produced.

Determinism. All seeds are fixed in `config.R` and the train/test partition is defined by admission year rather than by sampling, so it is invariant. Gradient-boosted tree fitting is the one step that is not bit-reproducible across differing

library builds; the pipeline therefore reports the gradient-boosted results as an explicit consistency check on any re-run, and every other estimate is invariant to it.

A.5 Ethical and Data-Access Statement

Both databases are de-identified and publicly available to credentialed researchers under the PhysioNet Credentialed Health Data Use Agreement. The analysis is retrospective and observational and involves no patient contact and no intervention, and the terms of the data use agreement, including the prohibition on redistribution and on attempted re-identification, were observed throughout. Section 3.3 sets out the full statement.

Appendix B

Pipeline Stage Reference

Chapter 4 describes what each stage does and why the architecture takes the shape it does. This appendix records the file-level interface: what each stage reads and what it writes; the mapping from an analysis to its inference status is in the analysis register (Appendix C) rather than repeated here. Paths are relative to `sepsis/project/`; all intermediate tables are Parquet, with a CSV mirror written for any result table small enough to read directly.

Stage	Reads	Writes
01	MIMIC-IV source tables	n/a (validation only; halts on a missing table)
02	`hosp/`, `icu/` source tables	`02_cohort_{A..F}`, `02_label_variance_table`, `02_alt_label_variance_table`
03	`02_cohort_*`; `chartevents`, `labevents`, `inputevents` (streamed)	`03_person_hours_{A..F}`, `03_early_onset_shift`
04	`03_person_hours_*` (three columns)	`04_sample_size`
05	`03_person_hours_*`	`05_model_primary_*`, `05_coef_all_variants`, `05_predictions_primary_*`, `05_h2_stability_*`; `*_ablated` counterparts
06	`03_person_hours_*`	`06_model_gbt_*`, `06_predictions_comparators_*`, `06_gbt_importance_*`, `06_schoenfeld_*`; ablated GBT counterparts
07	`05_*`, `06_*` predictions	`07_metric_results`, `_random`, `_altlabel`, `07_utility_sweep`, `07_utility_max`, `07_ablation_metrics`
08	eICU-CRD source tables; frozen `05_/06_` models (including the Variant D fits, for the Amendment 9 arm)	`08_external_validation_results`, `_abxonly`, `_altlabel`, `08_eicu_predictions_*` (A–C on the microbiology-covered sub-cohort, D on the full cohort), `08_eicu_coverage`, `08_event_rate_reconciliation`, `_altlabel`, `08_h5_internal_external`
09	`07_metric_results*`	`09_variance_decomposition_table`, `09_hypothesis_verdicts`, `09_label_anchor_attribution`, `09_label_agreement`, `09_pretreatment_onsets`
10	`03_person_hours_*`, `05_/06_` predictions	Per-stratum performance and label-sensitivity tables
11	`05_/06_/08_` predictions	`11_calibration_summary`, `11_operating_points`, `11_net_benefit_at_prevalence`, `11_patient_level_metrics`, `11_lead_time_summary`, external counterparts
12	Stored predictions only (refits nothing)	`12_confirmatory_tests`, `12_internal_ci`, `12_subgroup_auroc_ci`, `_contrasts`, `12_subgroup_labelsens_*`, `12_altlabel_*` (including `_transport_ci`), `12_ablation_ci`, `12_calibration_ci`, `_contrasts`, `12_h2_*`, `12_analysis_register`

Table B.1: Stage inputs and outputs. Which result each stage supports is recorded once, in the analysis register of Appendix C, and is not repeated here. Stage 03 is the only stage that reads the full-size source tables, and Stage 12 is the only one that reads nothing but stored predictions, which is what allows the inference layer to be re-run and checked without refitting any model.

Naming conventions. A file's numeric prefix is the stage that wrote it. A `_{A..F}` suffix denotes the label variant; `_ablated` denotes the Protocol Amendment 5 feature arm; `_random` denotes the random-split arm used only for H4. Files without a variant suffix are pooled across variants by construction. No stage overwrites a file written by another stage.

Appendix C

Analysis Register

Table C.1 records the status of every analysis reported in Chapter 5 and the multiplicity correction applied to it. Its confirmatory and exploratory rows are transcribed from `12_analysis_register.parquet`, which the inference stage writes from the same objects it corrects. Two rows marked ¶ have no counterpart in that file, because they are scored in stages 07 and 08 and enter no corrected family; they are included so that the exploratory record is complete.

Table C.1: Analysis register. [C] confirmatory, [E] exploratory, [D] descriptive. **H5's pre-registered estimand is not evaluable (Section 5.10.7); the slot is retained and the contrast reported in it is a post-hoc substitute, so the [C] tag marks the slot rather than the estimand actually reported. Section numbers refer to `00_preregistration.md`. *§9 pre-registered the equity analysis in scope only: neither the subgroup levels, the reference level, nor the direction of any comparison was fixed in advance, so these analyses are exploratory. †A2 = Protocol Amendment 2 (§§16–17), written after notebooks 02–12 had been run; these analyses are post-hoc and change no pre-registered estimand. ‡A4 = Protocol Amendment 4 (§18), which adds the standard onset timing as Variant F; also post-hoc, and it enlarges family E3, so every BH-adjusted p-value in that family is stricter than it would be without the amendment. §A6 = Protocol Amendment 6 (§20), the subgroup interpretability floor; post-hoc, and it *reduces* family E1, which the inference plan's own denominator rule prohibits on its face; Section 6.7 states the tension and §20.4 of the pre-registration gives the argument. ‖A9 = Protocol Amendment 9 (§23), the full-cohort external arm under Variant D; post-hoc and registered as descriptive, so it carries no p-value and is excluded from H5 on the terms that exclude Variant D from H1. #A10 = Protocol Amendment 10 (§24), which attaches intervals to three quantities otherwise reported as point estimates and tests H6's interval null at its second boundary; also post-hoc, and it changes no verdict. ¶Scored outside the inference stage; no row in `12_analysis_register`.

Analysis	**Status**	**Pre-reg.**	**Correction**
Cohort description by label variant	[D]	§2	none
Sample size adequacy	[C]	§10	none (single criterion)
Internal temporal performance	[C]	§5–6	none (estimation)
Calibration intercept and slope	[C]	§5	none (estimation)
Decision-curve net benefit	[C]	§6	none (estimation)
Operating points at fixed sensitivity	[E]	n/a	none (descriptive)
Per-patient detection and lead time	[C]	§6	none (estimation)
H1 label dominance	[C]	§7	Holm ($m = 6$)
H2 coefficient instability	[C]	§7	slot reserved; no test statistic
H2 instability vs. estimation noise	[E]	Amd. 7	none; ratio to pooled SE, not a test
H3 treatment leakage	[C]	§7	slot reserved; exact bound
H4 split optimism	[C]	§7	Holm ($m = 6$)
H5 slot: metric divergence not evaluable; substituted AUROC contrast**	[C]	§7	Holm ($m = 6$)

Analysis	**Status**	**Pre-reg.**	**Correction**
H6 model-class null	[C]	§7	Holm ($m = 6$)
External validation on eICU-CRD	[C]	§4	none (estimation); feeds H5
antibiotic-only anchor (sensitivity)¶	[E]	n/a	none; different target, not pooled
Variant D on the full external cohort	[D]	A9‖	none; joins no family, excluded from H5
its transport interval	[D]	A10#	none (estimation); excluded from H5
Action-derived feature ablation	[E]	A5	none (estimation); enters no family
its contrast intervals	[E]	A10#	none (estimation); enters no family
instability by feature kind¶	[E]	n/a	none (descriptive split of H2)
Cross-label calibration contrasts	[E]	A10#	none (estimation); enters no family
Variance decomposition table	[C]	§1	none (assembly of H1–H6)
Proportional-hazards testing	[C]	§8	none (single check)
GBT feature importance	[E]	n/a	none (ranking, no inference)
Subgroup AUROC by demographic group	[E]	§9*	Benjamini–Hochberg, family E1
interpretability floor on E1	[E]	A6§	restricts E1 to 6 contrasts
Subgroup label sensitivity	[E]	§9*	Benjamini–Hochberg, family E2
Label sensitivity vs. AUROC association	[E]	n/a	none (visual, no test)
Alternative-label AUROC vs. Sepsis-3	[E]	A2†	Benjamini–Hochberg, family E3
Anchor-attributable fraction ϕ	[E]	A2†	none (ratio, no null at zero)
Onset agreement between definitions	[D]	A2†	none (descriptive)
Standard onset timing $\min(t_s, t_{\text{SOFA}})$	[E]	A4‡	Benjamini–Hochberg, family E3
Pre-treatment onsets under standard timing	[D]	A4‡	none (event count, not a test)